Northern Nut Growers Association Report of the Proceedings at the 13th Annual Meeting

Northern Nut Growers Association

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NORTHERN NUT GROWERS ASSOCIATION

REPORT

OF THE PROCEEDINGS AT THE

THIRTEENTH ANNUAL MEETING

ROCHESTER, NEW YORK

September 7, 8 and 9, 1922

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CONSTITUTION

ARTICLE I

Name. This society shall be known as the NORTHERN NUT GROWERS ASSOCIATION.

ARTICLE II

Object. Its object shall be the promotion of interest in nut-bearing plants, their products and their culture.

ARTICLE III

Membership. Membership in the society shall be open to all persons who desire to further nut culture, without reference to place of residence or nationality, subject to the rules and regulations of the committee on membership.

ARTICLE IV

Officers. There shall be a president, a vice-president, a secretary and a treasurer, who shall be elected by ballot at the annual meeting; and an executive committee of six persons, of which the president, the two last retiring presidents, the vice-president, the secretary and the treasurer shall be members. There shall be a state vice-president from each state, dependency, or country represented in the membership of the association, who shall be appointed by the president.

ARTICLE V

Election of Officers. A committee of five members shall be elected at the annual meeting for the purpose of nominating officers for the following year.

ARTICLE VI

Meetings.—The place and time of the annual meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the executive committee shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the president and executive committee.

ARTICLE VII

Quorum. Ten members of the association shall constitute a quorum, but must include two of the four elected officers.

ARTICLE VIII

Amendments. This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or a copy of the proposed amendment having been mailed by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

ARTICLE I

Committees. The association shall appoint standing committees as follows: On membership, on finance, on programme, on press and publication, on nomenclature, on promising seedlings, on hybrids, and an auditing committee. The committee on membership may make recommendations to the association as to the discipline or expulsion of any member.

ARTICLE II

Fees. Annual members shall pay two dollars annually, or three dollars and twenty-five cents, including a year's subscription to the American Nut Journal. Contributing members shall pay five dollars annually, this membership including a year's subscription to the American Nut Journal. Life members shall make one payment of fifty dollars, and shall be exempt from further dues. Honorary members shall be exempt from dues.

ARTICLE III

Membership. All annual memberships shall begin either with the first day of the calendar quarter following the date of joining the Association, or with the first day of the calendar quarter preceding that date as may be arranged between the new member and the Treasurer.

ARTICLE IV

Amendments. By-laws may be amended by a two-thirds vote of members present at any annual meeting.

PROCEEDINGS

at the

THIRTEENTH ANNUAL CONVENTION

of the

NORTHERN NUT GROWERS ASSOCIATION

Rochester, N. Y., September 7, 8 and 9, 1922

The convention was called to order at 9:40 A. M., Thursday, September 7, 1922, by the President, Mr. James S. McGlennon, of Rochester, New York, at the Osburn House, Rochester, N. Y.

THE PRESIDENT: This is the thirteenth annual convention of the Northern Nut Growers' Association. We have been favored by Rev. Dr. Cushman in consenting to give the invocation.

Invocation by Rev. Ralph S. Cushman.

THE PRESIDENT: I believe I voice the sentiment of all present in saying that we are grateful to Dr. Cushman for his prayer and I personally extend to him my sincere thanks and on behalf of the association.

I have the great honor and the rare privilege of introducing to you our Mayor. He has very kindly consented to come here and make an address of welcome to this association.

MAYOR VAN ZANDT: Mr. President and ladies and gentlemen, members of the Nut Growers' Association: Your President has said I was going to make an address; I never did such a thing in my life. I am glad to welcome you to the city of Rochester; I hope your meeting will be profitable and so pleasant that you will want to come again. I believe there are very few people in Rochester who know anything about nut growing. We have a splendid exhibit here from our parks and one that I am very proud of and we have a man here, Mr. Dunbar, that we are very proud of; he is a wonder; I confess that I didn't know there were so many nuts to be found in the parks myself—that is no joke. It is a wonderful thing, it is a revelation to me, I never dreamed that you could find such things growing around this part of the country at all. I fancy that most people don't know anything about nuts at all, except the five-cent bag of peanuts. I certainly wish you success in every way and particularly with reference to the plantation that I understand has been started here close to Rochester where they are doing some wonderful work. Most of us have the idea that nuts are used by people to put on the table for dessert at Christmas time and but little appreciate their true food value.

I sincerely trust that you will all come again, that you will have pleasant weather and that you will have time after work to see something of our beautiful city. We think it is the most beautiful one in the country. Thank you. (Applause.)

THE PRESIDENT: If you can wait just a minute, I am going to ask for a reply to your address of welcome. Mr. Patterson comes from Albany, Georgia, and is probably the biggest producer of pecans in the world. Mr.

Patterson is a member of this association and has very kindly consented to come all the way from Georgia to be with us.

MR. PATTERSON: Mr. President, Mr. Mayor, ladies and gentlemen: I wonder if the President in saying I was the biggest nut grower in the world had any reference to my physical proportions. You have certainly a wonderful exhibit here, Mr. Mayor, of the products of your parks and you have reason to be proud of it, as you have for many other things in the city of Rochester. It has been my privilege to make short visits to the city, my wife having some relatives here. I said to my cousin this morning, if there is any place outside of the South where I would rather live, it would be Rochester.

The nut proposition is in its infancy and we all believe, those of us who are wholly nuts, that it will grow into a giant. We have a little giant in the south in the shape of the paper-shell pecan and we are expecting that this Northern Nut Growers' Association will, within the next few years, develop some varieties of nuts, or discover some varieties of nuts, that are adapted to this northern climate and will do for the northern states, the northern, eastern and western, what the pecan is promising to do and really is doing for the South. While not a native of the South I think I may extend the cordial greeting of the South to you in the North. There was a time when a northerner like myself who moved into the South had just one name and that was a "damned Yankee", and a good many people through the South thought that was one word, but that time has passed and they are welcoming in the South today the northerner who comes with an honest purpose of helping develop that wonderful country. The day of bitterness is fast passing away, so I bring to you not only the greetings of the southern nut growers, but of the South and I bring to the Mayor, and through the Mayor to the citizens of this beautiful city, the greetings of the membership of this association. (Applause.)

THE PRESIDENT: I am very grateful to you for your consideration of my impromptu request.

THE MAYOR: I will promise to give an order to the policemen to crack no nuts while the nut growers' association is in town. As to the 18th amendment, I think that nuts are about the only vegetable that I know of that they are not making hootch out of at the present time.

THE PRESIDENT: I feel that we have been particularly favored not only in receiving an address of welcome from our Mayor, but also in having with us the President of our Chamber of Commerce, who has kindly consented to come and welcome us also. It gives me distinct pleasure to call upon the president of our Chamber of Commerce, Mr. James W. Gleason.

MR. GLEASON: Mr. President and ladies and gentlemen: On behalf of the Rochester Chamber of Commerce, I certainly want you to know that we appreciate the honor and privilege of having this convention held in Rochester. I don't know of a convention that has come to Rochester that should be more welcome on account of the scientific nature of your work and the magnificent aims and purposes of your organization in extending the planting of trees and the culture of your product. I know the Mayor has extended to you a welcome for the city but we have one citizen here in Rochester, Mr. George Eastman, of whom we are very proud because of the unselfish work that he has done, and in the work that you are doing you can appreciate what he is doing in a larger way than is given to most of us to be able to do. This week saw the opening of the famous new five million dollar Eastman Theater, dedicated to the public, and I believe the motto over the door is "For the enlargement of community life". Now, Mr. Eastman wants the people to consider this theater as their own, and that means you, that means all of us here. He would like to have the people from Rochester and the people from out of town take advantage of this magnificent structure, the wonderful orchestra, probably the finest thing of its kind in the world.

I won't make an extended address but I can promise that if you can come to the Chamber of Commerce we will make you all welcome. Thank you. (Applause.)

THE PRESIDENT: Mr. Weber of Cincinnati has kindly consented to make a reply to your address.

MR. WEBER: Mr. President and Mr. Gleason: We really know each other as old friends, for some years ago we had our convention here and we are very glad to have it in your city again. Such bodies as yours, the Chamber of Commerce, can further the activities of the Northern Nut Growers' Association and what it stands for in the North; which is demonstrated by the exhibits shown on the table. I see at both ends of the table exhibits that show what can be done in this community in particular in the way of nut growing. Right out behind us there is one of the largest English walnut groves in this part of the country. I think it has 228 trees. The mistake the gentleman made who planted them was that he didn't plant grafted trees. Had he planted grafted trees he would have had a gold mine right there on his farm; Mr. Vollertsen, one of your citizens, has begun an industry which in time may become another one for your Chamber of Commerce to look after. We appreciate your very fine exhibits, we are glad to be here with you and thank you for your address of welcome. (Applause.)

THE PRESIDENT: According to the program the next feature is your president's address. I feel that it is unnecessary for me to even attempt to add anything to what His Honor, the Mayor and President Gleason have said relative to our wonderful city. It is one of the great cities of the world.

THE SECRETARY: What is the population of Rochester?

THE PRESIDENT: Over 300,000.

To Members of The Northern Nut Growers' Association:

GREETINGS:

Your President recommends that definite action be taken to the end of increasing our membership, to the still further end of exemplifying the truth of the old saying that "in union there is strength." More members mean the spreading of our gospel over greatly increased areas that should be interested in nut culture. The present membership is approximately 250, an increase of only 24 since the Lancaster Convention in October last year. And while it is also an old and true saying that "self praise is no recommendation," the fact remains that 18 of these new members were secured through my office.

It has been suggested at previous conventions that a systematic campaign for members can be perfected through organized co-operation by our State Vice-Presidents. I believe this to be the most efficacious medium through which the greatly desired results can be obtained. Of many, I am sure, systems that can be employed to such end there are two that always appeal to me as most desirable. Doubtless you all have thought of them at some time or other; in fact I have heard at previous conventions casual mention of the second. But the first I have heard little if anything of, and it is that effort should be exerted to interest women more actively in nut culture. We have a few women members. Why shouldn't there be as many women as men? I can think of no reason why there shouldn't. I believe that women are just as competent as men to conduct any feature of nut culture, with the possible exception of specific manual labor. And I can think of no more delightful vocation for women who love the great and wonderful outdoors—and where is the woman who does not?—than nut culture, the cultivation of nut trees and bushes, beautiful things not only for the grace and beauty of trunk and limb, foliage and flower, but for their real substance, their fruit, nuts, one of the most nutritious foods for human beings. More and more nuts are being consumed every day, and I venture to say that their consumption as a leading item in our dietary is only in its infancy. So I feel that here is another opportunity for our women to demonstrate the justice of her recent acquired suffrage in our national affairs.

The other possible source of membership I have in mind is a systematic campaign to enlist the interest and co-operation of school teachers. Just think of the possibilities of such a campaign. School teachers, every one, being the high-class people they necessarily are, would respond finely, I'm sure, and serve as a most desirable medium through which that very potent additional force can be reached, namely, the pupil. What parent would refuse a child's request to enable him or her to participate in the planting of a tree! Recently I cut out the following little poem, by Charles A. Heath, from my old-home-town Canadian paper:

THE MAN WHO LIKES A TREE

I like a man who likes a tree,
He's so much more of a man to me;
For when he sees his blessing there,
In some way, too, he wants to share
Whatever gifts his own may be,
In helping others, like a tree.

For trees, you know, are friends indeed,
They satisfy such human need;
In summer shade, in winter fire,
With flower and fruit meet all desire,

And if a friend to man you'd be,
You must befriend him like a tree.

A beautiful sentiment, I know you will agree, and applicable to any tree, but especially so to nut trees, for the reason that they combine all the laudable qualities enumerated plus that of food—food for man—one of the very finest of foods for man.

There are, of course, numerous other ways that can be employed to get new members. Another I might mention is that of offering suitable prizes; but I urge you to action, definite and specific, along this line, that our Association may better ably execute the worthy ambitions in which it was founded in 1910.

Then, again, more members mean more money. With more money we can get along faster. "Procrastination is the thief of time," you know. I trust that real action will be taken at this convention to the end of increasing our membership to at least one thousand by the time of the 1923 convention. It can be done—yes, easily. If only each member would pledge himself or herself to get three new members during the year the 1923 convention would find us with the desired membership; and I am sure that a considerable excess would be found on the roll at that time.

Also, increased membership is desirable to the end of increasing subscriptions to, and widening the scope of our official organ, The American Nut Journal, the only publication of the kind in the country. Under the able editorship of that Roman, one of our most earnest and intelligent members, Mr. Ralph T. Olcott, it is a power for good in the interests of nut culture. It can be made an even greater power with a materially increased subscription list, and I know that I speak for my friend, Olcott, when I say that he is ready and willing to expand the Journal's columns as will be required, of course, by the expansion of nut culture—I believe I voice the general sentiment of our membership when I say that no more welcome messenger comes to us each month than the American Nut Journal.

Another recommendation I am going to offer is, that the association consider the advisability of establishing a nursery at a point agreed on as best adapted for the propagating and nursing of such nut trees and bushes as it endorses as suitable and desirable for the area of country naturally governing the origin of our title—Northern Nut Growers' Association. This recommendation germinated in my thought from a casual remark made to me recently by our esteemed member, Mrs. W. D. Ellwanger, while I was a visitor at her charming summer home, Brooks Grove. Viewing her nursery of several thousand black walnut seedlings she casually mentioned that she would be very happy to present to any one desirous of planting such trees any consistent number he or she desired. As my thought dwelt on the expression of such a splendidly magnanimous nature I began to wonder, if a lady was willing to perform such a noble act, why should not the association elaborate on the worthy plan along the lines I have suggested. And with more members, and, thereby, more money, we can do it. Then The Northern Nut Growers' Association will be doing a real thing, something tangible, something that will attract new members in a way nothing else would, because people would then be able to see the living evidence of the practicability of our ideals. We could start in a small way, and grow. After long and earnest thought on the subject I came to the conclusion that it was worthy of our consideration.

From Mrs. Ellwanger's reference to "Johnny Appleseed" I believe that she found precedent for her nut tree nursery initiative in the work of inestimable value to posterity done by that same worthy. If the legend be true, he worked with much happiness of heart, but not more so than that of Mrs. Ellwanger, I am sure you will agree, when I tell you that many of her nursery trees are growing from nuts she garnered from roadside and field trees manifesting some exceptional trait, or indicating rare strain.

And I cannot refrain from urging action to the end of influencing our other states to pattern after good old Michigan in our effort to enact legislation, as she has done, providing for planting our roadsides with nut-bearing trees. It is something tangible, like this, that really counts. I believe that it is a fundamental of life, and living, that precedent, pro or con, is invaluable as governing subsequent action along similar lines. Here we have, in Michigan's action, a most worthy precedent, and I can think of no good reason why OUR other states should not do likewise. And I believe that this association, functioning efficiently, can exert the necessary influence to bring about a similar condition in OUR other states. My emphasis of the word OUR means The Northern Nut Growers' Association's states, you know.

I just wish to mention in passing that the author and collaborator, respectively, of the Michigan roadside planting of nut trees legislation are our esteemed members, Senator Harvey A. Penny and the Hon. William S. Linton, both of Saginaw, Mich.

In closing I desire to refer to our wealth, as an association, in scientific lore. The association is particularly well equipped in having a faculty, so to speak, than which there is none better in the country—yes, the world—in whose hands our recommendations, to the planter of nut trees, can be entrusted with absolute safety. For genuine scientific research in nut culture of the northern states this association stands singly and alone. This tribute is born of vivid remembrance of the really scientific work done by several of our worthy members, notably, Jones, Bixby, Morris, Deming and Vollertsen. Them, especially, I salute. (Applause.)

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MR. OLCOTT: With reference to the suggestions in the President's address, why wouldn't it be desirable to refer them to a committee to report upon and take any action that may be desired?

THE PRESIDENT: I believe, Mr. Olcott, that is a good suggestion.

MR. OLCOTT: I move that the President's address be referred to a special committee to consider and report at a later meeting in respect to the suggestions made and the plans for carrying them out. Motion seconded by the Secretary and carried.

Committee appointed: The President, Mr. Olcott, Dr. Deming, Mr. Bixby and Mr. Jones, to report Friday evening.

THE PRESIDENT: The next feature of our proceeding is the report of our secretary, Dr. William C. Deming of Hartford, Conn.

THE SECRETARY: Mr. President, I beg to say that the secretary has no formal report; but I have a number of items that will be of interest to the association which we can take up at this time if you think best. I think first should be taken up the notices of two members who have died this year, both of whom were very prominently connected with nut growing, Dr. Walter Van Fleet and Col. C. K. Sober. I will read a notice of Dr. Van Fleet's death which has been especially prepared for us by Mr. Mulford of the United States Department of Agriculture.

DR. WALTER VAN FLEET

In the death of Dr. Walter Van Fleet on January 26, 1922, the United States has lost one of the greatest plant breeders in its history, and garden rose growers an ardent advocate and sincere friend. Since a lad he had been interested in these lines of work and the products of his unremitting and painstaking energy, combined with unlimited patience, are known by garden lovers all over the country, as well as in Europe.

Rosarians naturally know him best by his roses, of which there were many, among them that splendid variety that bears his name, as well as such others as Silver Moon, American Pillar, and Alida Lovett. Many more are still in the trial grounds of the United States Department of Agriculture at Bell Station, one of which, christened Miss Mary Wallace, will be available in two or three years.

The ideal rose for which he was striving, in all his later work at least, was a garden rose with foliage that would compare in healthfulness and disease resistance with the best of the rose species, that would be hardy under ordinary garden culture, and that would be a continuous bloomer. His experience taught him what would be likely to give the desired results, but often he could not come directly to the ends sought. For example, when he wanted to combine the characters of some newly found species with the Hybrid Tea roses, he would often find the two could not be crossed directly with one another. He would then seek some other rose that would combine with the new species, without changing the characteristics which he wished to preserve, after which he would grow the resulting hybrids and cross them with the hybrid tea. Sometimes he would need to make another cross before he could get the seedlings for which he was striving. When it is realized that each cross of this kind would take from three to five years before he could take the next step an idea is gained of the patience required. Sometimes the results of these crosses would be infertile, producing neither perfect pistil nor viable pollen, as in the case of a handsome scarlet rugosa growing in the National Rose Test Garden which he was unable to use for further breeding on this account.

His great love of his work is shown in his having given up a successful medical practice in 1891 to devote all his time to plant breeding. He did this, even though he had taken a post graduate course in medicine at the Jefferson Medical College in Philadelphia in 1886-7, after having graduated at the Hahneman Medical College in the same city in 1880. His first work after this change was primarily with the gladiolus on a farm between Alexandria and Mount Vernon, Va. The soil was not adapted to his purpose so he abandoned it and went from there about 1892 to the Conard and Jones Company of West Grove, Pa., then to Little Silver, N. J., and in 1897 to the Ruskin Colony in western Tennessee as the colony physician.

In 1899 he became associated with the Rural New Yorker and lived at Little Silver, N. J., where he continued his breeding work on his own place. As associate editor for the following ten years and as writer of the column of "Ruralisms" in this paper he has left much valuable information on plant life and plant growing. From 1902 to 1910 he was also Vice-President of the Rural Publishing Company. While at Little Silver he was breeding fruits, roses, chesnuts, lilies, freesias, azaleas, and other ornamentals.

In 1909 he went to the Plant Introduction Gardens of the United States Department of Agriculture, at Chico, Cal. As the climate did not agree with his wife, he remained at Chico but a year and moved to Washington, D. C., where his official work was with drug plants and chestnuts, but his own time was largely devoted to breeding work with a wide range of other plants, a continuation of much of the work he had been doing at Little Silver. The move to Chico, Cal., resulted in a great loss to his breeding work. Some of his material was left at Little Silver, much of it died in the uncongenial climate at Chico, and other promising plants were lost in the long shipment across the continent, both going and coming.

In 1916 he was transferred to the office of Horticultural and Pomological Investigations where he was permitted to devote himself to plant breeding along such lines as looked promising to him, while at the same time he continued his work with chestnuts and chinquapins and a few drug plants.

Dr. Van Fleet was born at Piermont, N. Y., June 18, 1857. His early years were spent on a farm but later he lived at Williamsport, Pa. In early life he made a study of birds, his first book being "Bird Portraits," published in 1888, apparently being a reprint of magazine articles, one of which dates back to 1876. He was also a successful taxidermist, having studied under Maynard, and trained several of the leading taxidermists of his generation, including Charles H. Eldon of Williamsport, Pa. At nineteen he spent a year in Brazil, first connected with a party constructing a railroad around some of the rapids of the upper Amazon, and later in connection with the Thomas scientific expedition collecting birds and plants.

August 7, 1883, he married Sarah C. Heilman of Watsontown, Pa., who was associated with him in his medical practice and in his breeding work, and has been a sympathetic and helpful companion, and who survives him.

His was a most lovable personality. Those who came into contact with him day after day appreciated best his sterling qualities. He was kindly and considerate and nothing was too much trouble, and yet he had an intolerance of hypocrisy and cant that was almost violent. He was steadfast of purpose and there is nothing that shows this better than his lifelong work in plant breeding and the ruthless manner in which he rooted out his inferior seedlings as soon as he felt them to be valueless. His likes and dislikes were strong. Above all, he was modest and retiring in the extreme. He not only avoided, but shunned publicity. He avoided the outdoor meetings of the American Rose Society in the National Rose Test Garden as much from the fear of publicity that we, his friends, could not refrain from giving him, as for any other reason. He regretted in his later years that he had given up, during his editorial career, the little public speaking that he had previously done and had gotten so out of practice that, with his disposition, he could not again take it up.

He was an amateur musician with a thorough knowledge of orchestral and band instruments, harmony, theory, and orchestration but during the last few years none but intimate frequenters of his home had the privilege of hearing him, although until within the last two or three years he often played the violin.

In 1918 he was awarded the George Robert White Medal of Honor for eminent services in horticulture by the Massachusetts Horticultural Society, probably the greatest honor that can come to a horticulturist in this country. He had also been awarded three medals for the rose Miss Mary Wallace, a gold medal by the American Rose Society, a gold medal by the City of Portland, Oregon, and a silver trophy by the Portland (Oregon) Rose Society. He was associate editor of the magazine "Genetics" at the time of his death.

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Although he was an honorary member of the association I think very few of us knew that he had such varied activities in his life as this little biography tells us he had. The death of Dr. Van Fleet has been a great loss to American horticulture and nut growing.

Also during the year Colonel Sober has died. Colonel Sober, as you know, was a man who had made a very great success of growing the Paragon chestnut. His was the first commercial success in nut growing in the North. Then the blight came along and wiped out his industry. The Colonel was loath to admit for a long time that he had the blight or that his trees were not immune and that his nut growing was going to be a failure on account of the blight. I have no biography of Colonel Sober to read but one was published in the American Nut Journal for August.

THE PRESIDENT: I feel that we ought to make some record here of our feeling for these two men. I knew them both personally. I met Dr. Van Fleet at Washington two years ago and Colonel Sober seven years ago when the convention was held here. I had a great deal of correspondence with Colonel Sober. I think that we should adopt a resolution now and send copies of it to the families of these two deceased gentlemen to let them know the high regard in which this association held them as members and men.

MR. O'CONNOR: I make that motion.

THE SECRETARY: I second that motion and ask that the President appoint a committee on resolutions, which will also cover any other resolutions that may be necessary during the course of the meeting.

(See Appendix for Report of Committee on Resolutions.)

THE PRESIDENT: I will appoint on that committee Dr. Morris, Mr. Patterson, Dr. Deming, Mr. Jones and Mr. Rick.

THE SECRETARY: I have still a number of things here that will take up a good deal of time. I don't know that it is particularly interesting to any one outside of the association but I have a letter that I think is interesting to the members, especially those who have attempted chestnut culture, from Mr. G. F. Gravatt, assistant pathologist, United States Department of Agriculture, in which he says as follows:

As you may be asked questions at the Northern Nut Growers' Association meeting at Rochester regarding chestnut blight work of the Office of Forest Pathology I am sending the following letter:

By means of short field trips and correspondents I am keeping up in a general way with the spread of the chestnut blight. The disease is steadily spreading southward and westward. Infections are now known in seven counties in Ohio and thirteen counties in North Carolina. There is every reason to expect that the disease will ultimately cover the range of the native chestnut and chinquapin.

In Ohio several orchards have been reported as infected by State authorities. The blight is now present on native and planted chestnut in a number of localities in the Northwest quarter of that state. State authorities have reported one orchard in Indiana as infected.

It is evident that chestnut orchards located in the middle west are in danger of becoming infected with the blight. The most important means of spread to localities outside of the range of native chestnut are by chestnut poles and lumber products, and by infected chestnut nursery trees. Owners of chestnut orchards should keep on the watch for the disease and any suspicious specimens will be gladly identified.

There is some disagreement among pathologists as to the practicability of controlling chestnut blight in orchards located outside of the range of native chestnut or in localities within the range of the native growth where the native trees are very scattering, such as in many parts of Ohio.

My personal opinion is that the orchardist thoroughly familiar with the disease who will systematically inspect his trees, properly remove any infection as soon as it becomes visible and who has eliminated the sources of new infection in his neighborhood has a good chance of success. Control of the disease in some orchards is being tried out and I am desirous of getting in touch with other chestnut orchardists who have infected trees.

The chestnut breeding work at Bell, Md., started by Dr. Van Fleet, is being continued. Mr. Reed is looking after points relating to culture, quality of nuts, productions, etc., while I am looking after the hybridization and disease work. The Chinese chestnut seems to be the most resistant to the disease though a number of trees of this species have been killed primarily by the blight.

A number of reports of chestnut blight becoming less virulent have been investigated but in all cases the reports were incorrect. Professor Graves is continuing his observations on resistant trees around New York City.

That, I think, summarizes the chestnut blight situation very well.

I have a letter from Mr. Reed from China; it is a long letter and I will only read from it one or two extracts which tell why he was sent to China:

My task is that of obtaining a summary of the so-called "Manchurian" walnut industry of this country. So many walnuts from here are being delivered in the States each year that our own industry is considerably affected. The extent of production, its present rate of growth and its probable character and magnitude ten years hence are things our own people needed to know. So serious is the situation that Thorp, manager of the California Association left San Francisco for over here more than two months ago to get a short general glimpse, then to go to European points for the same purpose.

The consuls here have reported that no walnuts are grown in Manchuria, except in half wild, low-grade, scattered product which is assembled in small quantities only and probably not exported. The exported nuts are mainly from the provinces of Chihli, Shantung, Shansi and Honan. Tientsin and Hankow are the chief points of export.

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Mr. Reed expects to be back about Thanksgiving time. We miss Mr. Reed very much here at the conventions because he is the Government representative of the nut industry. He has a wider general knowledge of the nut industry in the United States than any other man.

In connection with the suggestions that our President has made, I think I ought to call the attention of the association again to the address of Dean Watts that he delivered at the convention last year in Lancaster. (This address, entitled "A National Programme for the Promotion of Nut Culture," will be found on page 80 of the report of the proceedings at the twelfth annual meeting.)

I have brought here a cluster of burrs from some chinkapin bushes that have been growing in Elizabeth Park, Hartford, for 23 years. They are loaded with nuts and although attacked by the blight, the fact of their being there so many years shows how resistant they are. I have also some clusters of burrs from chinkapin bushes in my own garden. They bore a full crop the second year from transplanting.

MR. O'CONNOR: Before I forget it, I want to say a word in regard to chinkapins. Right close to where I live there was a fire swept through the place and burned them down to the roots. But they have come up from the roots and are full of chinkapins at the present time; I have seen where the blight has hit them and they died back to the ground and they have shot up new shoots again and are bearing. The chinkapin is a coming nut; the school children are looking for them like I used to look for the butternuts in the early days.

THE PRESIDENT: That is very interesting information, Mr. O'Connor, and I am very glad you have stated it.

THE SECRETARY: Mr. Wycoff of Aurora, N. Y., has brought here a little branch containing two well developed Indiana pecans grown on a grafted tree. I think that is the first instance in which a grafted pecan tree of the Indiana variety has borne in the North. Mr. Snyder says he has fruited a Witte pecan at his place. A number of us have been striving to make the record for first bearing of a grafted "Indiana" pecan tree in the North. Mr. Wycoff has won it.

Mr. O'Connor, I think, has brought with him a number of branches of pecans grown in Maryland.

MR. O'CONNOR: I have some hazels and also some chinkapins.

THE SECRETARY: Have you any pecans fruiting down there this year?

MR. O'CONNOR: Several nights of frost hurt us pretty bad this spring. We have one tree that has got a few pecans on this year; last year the same tree had over a hundred; this year it hasn't got more than a dozen, but it promises to have a heavy crop next year.

THE PRESIDENT: What variety of pecans?

MR. O'CONNOR: If I am not mistaken, it is the Indiana. There are several trees that promise to bear heavily next year. In the spring we had a severe frost for seven nights in succession and that hurt our trees pretty bad. We are in the frost belt down there. Last year we didn't have any apples or peaches; this year we have some apples and some peaches but the grapes were severely hurt by the frost, also there are very few walnuts on the trees this year.

MR. CORSAN: From traveling around as much as I do I can vouch for that gentleman's statement in regard to the frost. I was up in the extreme northern part of the United States, northern New York, and I never saw such a crop of hickory nuts in my life and I have gathered nuts since I am able to remember. I have also seen more peaches up in Ontario and even north of Ontario. When you talk about frost and the South having such an advantage over the North, it is entirely wrong; I have had that idea knocked out of me for a good many years.

THE SECRETARY: I wish also to say that I brought here a small branch from the Hartford pecan tree bearing two nuts. The Hartford pecan tree is undoubtedly the largest pecan tree in the North. It is about ten feet in circumference, over seventy-five feet high and has a very large spread. I will ask Mr. Weber if he will give us the account again of the finding of that black walnut in the river and tell us the result of his investigation.

MR. WEBER: Whenever I come across a black walnut I want to open it up and see what it looks like inside. Following that custom when I found a walnut that had lodged against the dyke north of the central part of the city, I was surprised when I opened it because the partitions were very thin, like an English walnut. Later on I found another similar nut lodged against the dyke of the river about a quarter of a mile along. Then through a statement in the paper and an advertising campaign we tried to locate the tree. Finally we got the name of a man in Floyd, Va., who said he knew of the existence of such a tree, but a few years previously they had cleared the land and it had been cut down. So that finished that. But he gave me the name of the man who had owned the place and said that there were some other trees that had originated there and that they were bearing. It is down in Virginia at the extreme western end and off the railroad and rather hard to get to. I thought possibly on my way home I would get there this trip.

THE SECRETARY: As an example of nut enthusiasm here is the corporation counsel of the city of Cincinnati, who on his walks abroad picks up nuts that he finds and examines them. He finds one on the dyke of the river that he considers remarkable and in conjunction with the president of this association conducts an advertising campaign in the watershed of the river where that nut was found in order to locate the tree, and succeeds eventually in doing so.

Mr. President, here is a communication which I received in July from the Secretary of the American Pomological Society inviting us to become a member. I didn't feel that I had the authority to send him a check for ten dollars, but I would like to put before the association the question as to whether we ought not to make this association a member of the American Pomological Society. I would ask, Mr. President, that you put that matter up for discussion, if you think it is of sufficient importance.

THE PRESIDENT: I do, Mr. Secretary, and think it would materially help in gaining names in our plans for increasing the membership if we were able to say we were a member of that society. What do you suggest relative to the procedure in that connection?

THE SECRETARY: I think all that is necessary is the motion by some member that the treasurer be authorized to take out a membership for the association in the American Pomological Society.

BY A MEMBER: I so move. They will know we are in existence and if we take an interest in their work they will take an interest in ours.

Motion duly seconded and carried.

THE PRESIDENT: Your reference to Mr. Reed reminds me that prior to his receiving orders to go to China, he and Mrs. Reed both had promised to come and make addresses at this convention; Mrs. Reed on the subject of nuts as a food and Mr. Reed with a fine exhibit and also an illustrated lecture. He wrote me quite fully just before going saying he was awfully sorry that he could not be here. With reference to the Secretary's remarks regarding Dean Watts, I had the privilege of meeting Dean Watts last year at Lancaster and I think his ideas are very much along the same line relative to increasing our membership and improving our financial condition so that we can do real things. I had a letter from Mr. Littlepage early in the season and he expected to be here. Then he finally wrote me and said it would be absolutely impossible for him to come but he was sending his able lieutenant, Mr. O'Connor. I was beginning to feel a little worried this morning that perhaps Dr. Morris might not be able to get here but I was very happy a few minutes ago to see the Doctor come in and now I feel considerably more comfortable because he is a great aid and help at these conventions. Is there anything further, Mr. Secretary, that you have in mind?

THE SECRETARY: I just want to call your attention to the exhibits; they really hardly need any one to call attention to them, but I would like to mention especially the exhibits at the two ends of the table. The one at the further end of the table by Mr. Dunbar of the Department of Parks of Rochester is really a very remarkable exhibit, especially from a scientific point of view. (See list of exhibits in appendix.) At this end of the table is a splendid exhibition of filberts grown in Rochester in Mr. McGlennon's filbert nursery under the direction of Mr. Vollertsen; it needs no word of praise from any one, it speaks for itself. Also I call your attention to these three English walnut trees in pots, each one bearing fully developed nuts, which were grown by Mrs. Ellwanger. Last of all I will mention again the cluster of Indiana pecans brought here by Mr. Wycoff of Aurora.

MR. DUNBAR: Dr. Deming didn't tell us about the Chinese chestnuts that are fruiting—the castanea mollissima.

THE SECRETARY: Dr. Morris has had them fruiting for a number of years. I don't know whether any others have or not.

DR. MORRIS: They fruit very well and are a good hardy nut. They are on limestone land.

THE SECRETARY: It is a very interesting nut.

MR. CORSAN: Out of twelve varieties of chestnuts that I planted on my place it is the only one that died. I got them in Washington. I looked after them probably too well. I will try them again to be certain they had no climatic reason for dying. It is very strange that that chestnut didn't grow. Nobody near me grows chestnuts so I can cultivate them for a good many years without any worry about blight.

DR. MORRIS: I doubt if the blight amounts to much with you. It is carried by migrating birds. Some birds will take the blight north and our friends in Canada will finally have it, so cheer up, the worst is yet to come, but it will be a good many years.

MR. CORSAN: The blight has got to the extreme northern part of the chestnut growth, that is, to the top of Lake George. The chestnut doesn't go a quarter of a mile beyond Silver Bay.

DR. MORRIS: I have found chestnut trees in Quebec.

PROFESSOR NIELSON: Speaking of the range of nut trees, I have seen the hazelnut in the Saskatchewan several hundred miles north of the international boundary and at Edmonton, Alberta, Canada.

THE PRESIDENT: That is very interesting to me for about the time that we started in experimenting with filberts I received a letter from an old friend of mine in Canada, Mr. Edward Kennedy; he stated that he believed the hazelnut or filbert would do very well in the Canadian Northwest. At that time we were in the nursery business and were finding it difficult for our general nursery stock to survive the severe winters in the Canadian Northwest. Mr. Kennedy thought that from his observation of the filbert throughout that country it was the one item in the nurseryman's list that would do very well there.

DR. MORRIS: In that connection I would like to say that I have seen the hazelnut growing as far north as Hudson Bay and it is very hard to distinguish it from the elm. The hazelnuts grow to a height of from twenty to twenty-five feet and the elm comes down to about that height. The leaves look so much alike that I found myself looking for hazelnuts under an elm tree.

THE PRESIDENT: Mr. Patterson told me that while fishing on one of the streams near Albany he had found some of the common hazelnuts in fruit. I have sent down to some of my friends at Albany some of our filbert plants to see how they might do there and the reports up to the present time have been altogether favorable. My thought up to the present time has been that perhaps the climate there is a little too hot.

The next item on our program is the report of the treasurer, Mr. Willard G. Bixby of Baldwin, N. Y.

NORTHERN NUT GROWERS' ASSOCIATION

In Account With

WILLARD G. BIXBY, *Treasurer*

Receipts:

From annual members, including joint subscription
to American Nut Journal $222.25
From contributing members, including joint
subscription to American Nut Journal 80.00
From contributions 357.50
From advertising in report 5.35
From sale of reports 12.00
From sale of Bulletin No. 5 8.58 $685.68

From Life Membership W. L. Linton 50.00

$735.68

Deficit September 1, 1922:
Balance Special Hickory Prize $25.00
Balance Life Memberships 95.00

Deficit for regular expenses 176.87
Net deficit 56.87

$792.55

Expenditures:

American Nut Journal—their portion of joint subscription $74.00
1921 Convention 71.46
Printing report 12th meeting 212.19
Printing and stationery 142.82
Nut contest 111.01
Postage and express 5.00 $616.48

Deficit October 1, 1921:
Balance Special Hickory Prize $25.00
Balance Life Memberships 45.00
Deficit for regular expenses 246.07
Net deficit 176.07

$792.55

The work of the treasurer for the past year has not been satisfactory to him.

The amount of attention he has been able to give it has been much less than he had hoped. While supposed to be retired with nothing to do except just what he wants to this is far from the facts. While it is true that in 1919 he did retire from business, in which he had spent practically all of his time since leaving school, he has never been able to retire entirely and is still president of one corporation and vice-president of two. In the case of one of these the conditions under which it operated have changed so entirely that he has had practically to get back into business and the work of the association has had to be sandwiched in as best it could and at times has had scant attention. Had it not been for Mrs. Bixby's help on the work of the treasurer proper, he would have had to resign.

There is a deficit[1] shown by the treasurer's report although less than that of a year ago. The attempt to induce a rather large proportion of our members to become contributing members, paying $5.00 per year as membership fee, including subscription to the American Nut Journal, has been reasonably successful, about one-quarter of our receipts of membership fees being from this source. The real difficulty, however, is that our total membership is not sufficient to enable receipts from dues to pay expenses. In every year, for a good many years, receipts from contributions have been about equal to those from dues and apparently that condition will have to continue until our membership is doubled, unless the activity of the association is materially reduced, which course seems inadvisable to your treasurer.

[1] This was wiped out at the meeting by contributions and guarantee of new membership which more than equalled the amount of the deficit.

The results of the nut contest the past year have been unsatisfactory. The nut crop was a failure over quite a portion of the country covered by the association. The number of nuts sent in was not over one-tenth of those received in 1920 and no nuts of notable excellence were received. Were it not for the fact that this year promises to be a great year for nuts in the northeastern United States, one might think that the nut contests had outlived their usefulness. They have, however, brought us so many good nuts and are so comparatively inexpensive that your treasurer would not want to give them up yet.

During the past year an earnest effort was made by the treasurer to get new members by getting nurserymen to enclose in their catalogs circulars regarding the association as well as membership application blanks, over $100.00 being expended on this item. The nurserymen on the accredited list responded heartily. The results, however, were far from being as satisfactory as a year ago when the literature sent out by the nurserymen simply called attention to bulletin No. 5. Literature regarding the association and membership application blanks were inserted in bulletin No. 5 and between five and ten per cent. of those who received bulletin No. 5 became members, the number being considerably greater than those from similar efforts this year.

This shows conclusively that direct appeals, unless there is personality behind them, do not have much force. A year ago bulletin No. 5 in the possession of one interested enough to purchase it, supplied the personality and gave force to the appeal that was lacking this year.

Thirty-eight new members have joined the association since the last report, making 561 since organization, of whom we have 249 at present, making 312 who have resigned, or dropped out, or have been removed by death. The additional members obtained this year are largely due to the personal efforts of the president and those in his office.

During the past year we have lost by death our only honorary member, Dr. Walter Van Fleet of the United States Department of Agriculture, and one life member, Col. C. K. Sober of Lewisburg, Penn.

Respectfully submitted,

WILLARD G. BIXBY, *Treas.*

* * * * *

THE PRESIDENT: I feel that we have got to get busy and get some more members and more money. At nearly every convention a deficit is reported; it ought to be the other way, and it can be. We will all agree, I believe, those of us who are in the habit of attending these conventions, that they resolve themselves largely into meetings of a mutual admiration society. Outside of Dr. Deming, Mr. Bixby and one or two others, there is very little thought given to this association during the year except immediately prior to the convention. Of course, we can't get ahead very far that way. Ever since I have been actively connected with this association I have given first thought to the matter of membership and the improvement of our finances. I do hope that at this convention some definite and specific action will be taken so that a year from now there will be a decided increase of members, because I am confident we can do it if we put our shoulders to the wheel. Then we will have a surplus instead of a deficit. As I said in my paper this morning, the association is engaged in scientific work, but we are not going to get very far along unless we have more money, and we can't get more money unless we get more members. We ought to put our shoulders to the wheel and pull this association up to a membership that is worthy of its title. If each member would get from three to five new members during the year we would have a membership in the neighborhood of a thousand another year and that would give us a surplus of money. I hope that definite action will be taken at this convention to stimulate that development of the association. If any of the other members have anything to say on that subject I would be very glad to hear from them.

MR. OLCOTT: I think that the membership is really one of the most important things for this association to consider. But I do not think it would be well to go away from this convention with only the idea that each member should try to get three or four others. That is all very well and it would mean considerable IF they would do it. I think there are enough business men here and brains enough here so that if this matter were referred to a good big committee that would spend some time on it, and before we go would report some definite way of stimulating interest in nut culture and in this association, that it would bring the membership up to a point where it could accomplish something in a business way. It is not a matter for individual action but a matter for association action. It needs publicity and a good comprehensive plan. The money will come as more members come. The wider knowledge of what this association is doing for an active membership would make a bigger membership. If you will remember President Linton suggested that each state should provide twenty-five to fifty members; it does seem as though there should be twenty-five or fifty members, men and women, in each one of the twenty or so northern states. If there were fifty there is a thousand members in the twenty states. He pledged, I believe, twenty-five names from Michigan on his own account; I don't know whether he made good or not but the plan is good to aim at fifty members in each of twenty states.

MR. SPENCER: I am very much interested in the production of nut trees largely as a matter of curiosity. My home is in Decatur, Ill. Illinois has 56,000 square miles, 30,000 square miles of that state are, or were, covered with hard wood timber. In Bureau County the hickory, the hazel, the walnut and butternut grow with a great deal of vigor; less than two blocks from me there is an ordinary sweet chestnut brought from the East by a gentleman a great many years ago. I measured it last fall and it is six feet nine inches in circumference, it has a spread of about sixty feet and it is about seventy-five feet high. The neighbors told me that they got a bushel of chestnuts every year off that one tree. I presume if they took better care of it and gave it some fertilization they would get more than that. I happen to be the chairman of the tree committee of the Bird and Tree Club. The city of Decatur purchased 42 trees and planted them in seven parks of the city of Decatur; members of the Bird and Tree Club came to me for advice and last year I placed 114 trees for them. They placed a number of trees with the

Oberlin Conservatory of Music, chestnut trees, and they planted them on the campus. I believe that persons who are associated with different clubs would take up the matter of nut growing. That means that you can interest the children and if you can interest the children then you get the parents interested. In Macon County alone the county surveyor told me there are 20,000 acres of ground that are absolutely worthless except for pasture because they form bluff land along the Sangamon river. It isn't a large stream, I suppose down here you would call it a creek, but the city has put a dam across the river and trees were planted. I tried to create a sentiment to have that shore planted with nut trees instead of ash and elm and the various trees that can bear nothing but leaves, but the hardest thing in the world is to start a new idea.

An ordinary crop of nuts after a tree commences bearing is worth a great deal more than a crop of wheat or oats and in the meantime you can use the ground under it if you want to.

Now these are simply my individual efforts in Macon County to get people interested in nut-bearing trees. It is a hard road and I am like some other people, I don't like to be pointed out as a crank, but I am pretty near that on this subject. With the co-operation of Mr. Reed a year ago I delivered an address, illustrated with pictures that were supplied by the Bureau of Plant Industry, on the subject of "The Value of the Nut Trees for Shade and Food," with the idea of having farm homes made beautiful by trees and attractive by the fruits thereof to keep the children home. Last year I delivered an address on "Nut Trees and Roadside Planting," also illustrated by pictures sent me by Mr. Reed and through the courtesy of McMillan & Company I reproduced pictures describing Dr. Morris's new way of grafting. If you will take steps along those lines and work through the Bird and Tree Clubs and get the children interested I believe you could do something toward spreading the gospel of nut culture. I thank you for your attention. (Applause.)

MR. CORSAN: As to getting new members, I am ashamed to say that since I joined in 1912, I just got one new member actually into the club and that was Dr. Kellogg. I interested hundreds of people but he was the only person I got in. The only way to do is to step right up and ask a man for his money as soon as you give him the proposition. Now that is where I fail. I struck Mr. MacDonald, the permanent Boy Scout Director, 200 Fifth avenue, New York City. He is very enthusiastic but he hasn't come in as a member. Then the Overseer of the Boy Scouts, a tall young fellow with sandy hair and a good complexion, I have forgotten his name, but he is a splendid fellow. He was enthusiastic but he hasn't come in as a member. I met Mr. McLean of the Orphan's Home and he is going to have the Orphan's Home planted with nut trees, but he didn't join the society. I suppose I didn't beg them enough. I suppose I should say, "Give your money to me right now, immediately, and let me send it over to Mr. Bixby." I think that would be the best method of getting in new members. Then they will read the literature and keep in touch with the association. I must confess downright negligence for not getting members into the association. I thought we were a kind of a rich gang and don't need money. But we have got to have money in order to get people into the idea of growing nut trees.

THE PRESIDENT: What seems to be the objection?

MR. CORSAN: No objection at all except I had that fault of not gathering in their membership while I was speaking to them upon the possibilities of nut culture.

THE PRESIDENT: If you don't get some members in this year there will be trouble!

MR. CORSAN: Why not give a tree with every new membership so that the member can plant a nut tree on his own farm, and the Boy Scouts and also the Girl Scouts would come into this thing, too, as the tall gentleman from Decatur has said.

MR. PATTERSON: I should like to tell you what happens in our association in the south of Georgia. For a number of years our treasurer has come up with a deficit each year. The only practical way that we have found in the southern nut growers' association for increasing our membership and getting additional funds is to do it by subscriptions taken at the meeting. Let each man pledge so many members and turn over the money to the treasurer to pay up for the members that he has pledged. Then let him go out and get the members to reimburse himself. In that way we have increased our membership very much. I do not say that that is the way that it should be handled here but that is the only way we have found of solving the problem.

MR. TAYLOR: I represent the Northern Apple Growers' Exchange. We want to get people who grow apples into our association and the first thing of all is to get them interested. You first have to attract the attention of a man, your prospective member, and then you have to arouse his interest and you have to create a desire. We found that in order to attract his attention a circularization of people who were eligible for membership accomplished a great deal. These people were circularized, given little bits of information here and there, not the information that was given the members as a rule, not to that extent, but they were given a certain lot of information from time to time to let them know that the Apple Growers' Exchange was there. After a while they were approached personally and if they said "No" we continued circularizing them a little while longer along a different line. Finally, when we thought we had gotten them to a point where they were interested, the problem was to get them properly signed up. So we then made a drive for those particular individuals by showing them what they could personally get out of it. After he had joined our problem was to hold him, to keep him interested until he became enthusiastic. Unless you keep them interested they are liable to cool off, and once they are cooled off it is almost impossible to get them interested again. We find the members who have gone out are the hardest to get back. A way of keeping that new member in, and helping him to feel that he is a potent factor in the organization, might be by having some sort of a special communication with him at the time he joins, or at the next meeting of the association. I know that in California that is the way they work it. Keep members informed, not merely with reports of proceedings but with something like an occasional sheet or two on the latest thing that is going on, especially for the new members. (Applause.)

THE PRESIDENT: I would like to have any other suggestions. Dr. Morris, have you anything to say?

DR. MORRIS: No, I have been doing a lot of thinking.

THE PRESIDENT: It seems to me it is the one vital thing for us to consider. We have got to increase our membership.

MR. OLCOTT: Apropos of the remarks of Dr. Taylor comes the question of the desirability of giving a prospective member something for his money. Our first problem is to interest someone to the extent of membership and then to keep him after we get him. Those are problems that require thought. I think the President in his address suggested that the association produce young nut trees to be given away to someone to plant, to interest that someone and others who see it. Would you give him another tree at renewal time?

THE PRESIDENT: That was the idea.

MR. OLCOTT: The renewal proposition with trees selling at $2.50 to $3.00 apiece would be pretty expensive for the association—for a member to pay us $2.00 and get a tree for nothing. My personal idea has been that there should be a state organization in every one of the northern states, subsidiary to this association; that each association have its monthly meeting, or maybe quarterly or annual, taking in those who cannot find it convenient to come to the parent association's convention.

DR. MORRIS: I will pay the dues, and subscription to the Journal, for any Boy Scout for ten years if you will make that the object for striving for a prize in some organization of Boy Scouts.

THE PRESIDENT: I appreciate that very much.

THE SECRETARY: I have two suggestions for ways of drawing attention to our association. The first is lectures. There are a number of our members who have given lectures on the subject of nut growing. Mr. Spencer has just told you that he has and Dr. Morris loses no opportunity to give them. I have given them myself and Mr. Reed of the Department of Agriculture speaks on nut culture. There is hardly a member of this association but belongs to some agricultural society or club. That is one possible place for bringing nut culture to the attention of people who are interested in either agriculture or horticulture. I am sure that Mr. Reed of the Department of Agriculture will send a collection of lantern slides on nut growing to responsible persons. These slides make lecturing much easier. I will undertake to get Mr. Reed to make up a collection of slides to be sent out to members for the purpose of illustrating lectures. My other suggestion is the writing of articles for magazines, horticultural and agricultural, and especially high-class horticultural magazines that reach wealthy people who are interested in new things and in trying experiments, such as the Country Gentleman, Country Life in America and the Garden Magazine. What we really want is some person who will give himself continuously to the promotion of this nut-growing idea. It is a great misfortune that Mr. Bixby has taken up business again because he made a splendid beginning in devoting himself to the interests of nut culture. I did a great deal more myself in the earlier days of this society but circumstances have been such that lately I have not given it much attention. I feel that there must be members who are all ready to do work, members who would like to jump in and take a hand. I would be very glad to share my work as secretary. I would be glad to hand over the entire work of secretary to some member who feels an itch to get in and do this sort of work.

THE PRESIDENT: You are very liberal in your service but I think others ought to take a bigger share so that your duties will be easier and also Mr. Bixby's. Now that we have this thing going I hope we will stick to it until we get something concrete because I can't see that we are going to make much progress just meeting from year to year with an increase of twenty to twenty-five members. I personally will guarantee a hundred members for this year for this association. I speak advisedly because I know what we have been doing in our office this last couple of months. I am satisfied that I can bring to the association a hundred new members this year if the rest will bring ten each. We have got to get more members and more money; let's get down to bed rock and look the thing squarely in the face and make up our minds to go to it and do it.

MR. CORSAN: Where can these slides be got?

THE SECRETARY: I will undertake to furnish them through Mr. Reed of the Department of Agriculture. There is also a good moving picture film of Colonel Sober's chestnut grove that I think can be had. I have used it myself two or three times.

MR. KAINS: Rochester, as a good many of you know, is the center of the fruit industry in western New York. Right here is also the scene of one of the greatest fights to get an association on a paying basis that ever occurred. Some of you probably know that away back in the fifties Patrick Barry and Mr. Worter and several others of the fruit growers got together and formed the Western New York Horticultural Society. Gradually people came in and took an interest in the work but, as always in the beginning, there was trouble to make ends meet and Mr. Barry and some of the others put their hands in their pockets to keep the association going. At last it got so bad and the amount of the deficit was so great that it was decided to have a closed meeting, no one to be admitted except those who had actually paid their one dollar membership fee. The year that it was announced that this would be put into effect the following year there was all kinds of a fuss at the meeting. The next year the people came there in a crowd to see if the rule was going to be put in effect and the result was the largest meeting the association had ever had. The only men and women who got inside the door had paid their dollar. That was the first year that the association got on its feet. One other method that could be used to spread the love of nut growing would be to have the association offer a nut tree to different schools where they would plant it as an Arbor Day tree. In that way the children would learn the value of the grafted nut tree and the value of real first-class nuts. The result would be that other people would become interested in grafted nuts and thus extend the interest in the whole nut-growing proposition, and your membership would most likely increase. (Applause.)

THE PRESIDENT: I will ask for nominations from the floor for the nominating committee.

Mr. Pomeroy, Dr. Morris, Mr. Olcott, Mr. Rick and Mr. Patterson nominated and elected.

THE PRESIDENT: The next order of business is to call for the reports of any of the standing committees.

THE SECRETARY: The chairman of the committee on incorporation, Mr. Littlepage, wrote me not long ago that he was taking active steps to incorporate the association. I don't know whether Mr. O'Connor may know if Mr. Littlepage has done anything about it or not.

MR. O'CONNOR: I can't say about that.

THURSDAY EVENING, SEPTEMBER 7, 1922

THE PRESIDENT: I am going to ask Dr. Taylor to present his paper now, if he will please.

PROF. RALPH H. TAYLOR: Through a previous arrangement with our secretary I had assigned to me an entirely different subject from that on the printed programme, "The Use of Nuts the Year Around." I have prepared a paper on the original subject and so I will proceed to deliver it in accordance with my arrangement with him.

I do, however, want to say first, in connection with the use of nuts the year around, that we from California are vitally interested in that problem. I know of no problem that faces us more at the present time than the one of marketing the product that we grow in competition with the tremendously increasing imports from abroad, brought in from countries where labor costs anywhere from twenty to fifty cents a day, and at the highest a dollar a day for what they call skilled labor, most of it twenty to fifty cents, and with freight rates across the Atlantic that amount to less than half of our freight rates, or one-quarter of them. With the commodity in the hands of speculators who are able in various ways to make tremendous profits, and giving the public none of the benefit of these conditions, we find it almost impossible to market our product at a profit. We must get it into the hands of the consumer cheaply. We are endeavoring to do it. One of the plans is to encourage the use of nuts the year around, and the California Almond

Growers' Association, whom I represent, are planning now to shell their own almonds and put the kernels up in vacuum packages, both tin and glass, and make it possible for the housewife, instead of going to the candy stores and buying salted almonds for a dollar to a dollar and a quarter once or twice a year, to secure her own almonds, blanch them herself and use them considerably more often because she can get them cheaper. We believe it is going to be worth while for us to go into the business the year around. The demand at the present time is for almonds for a brief period up to the first of January. Thereafter there is no sale until the following November. Under those conditions you can see that with increasing crops we are facing difficulties that are almost insurmountable. Therefore we are changing the form in which we are marketing part of our crop. I want to say to those people who do recognize the value of almonds for food that it is going to be possible for you to secure them in a most desirable form, clean, wholesome and absolutely fresh, as almonds packed in vacuum. They will be just as fresh as when they are put in from the orchards of California.

ALMOND POSSIBILITIES IN THE EASTERN STATES

BY R. H. TAYLOR[2]

There is probably no better way to open a discussion of this kind than by asking a question and then using it as a text. The future possibilities for almond production in the eastern states can not be stated any other way than as a question. For my text I am indebted to your secretary, Dr. W. C. Deming. It is taken from a letter written by him under date of June 22nd to Mr. T. C. Tucker, the manager of the California Almond Growers' Exchange, and is as follows:

> "Why can't we breed an almond that will do in the East what its sister, the peach, does?"

Any answer we might give must be, of necessity, more or less empirical in nature.

[2] In charge Field Department, California Almond Growers' Exchange.

In order properly to understand that answer, and I shall attempt to give one later, certain fundamental relations and limitations must first be considered; then the possibilities of any given line of procedure may be more clearly understood.

Botanically the almond is very closely related to the peach, both belonging to the genus *Prunus*, sub-genus *Amygdalus*. The species of the peach being *persica*, and of the almond, *communis*. In fact the two trees are in many respects so much alike that it is possible to select twigs and leaves from each which cannot be distinguished except by an expert, and even he may be misled at times. Ordinarily, however, they are of sufficient difference to be readily distinguished.

In the fruit the principal difference is that the fleshy portion of the peach becomes in the almond a leathery hull which splits at maturity revealing a seed or nut, the shell of which is generally softer than that of the peach pit. The kernel may or may not be bitter, depending upon the characteristics of that particular seedling. If 100 almonds from a sweet almond tree are planted and brought to bearing it is probable that from a third to a half of them would produce bitter almonds. As a matter of fact, we have had by actual tests as high as 50 per cent. bitter. The peach, on the other hand, will, probably in 99-1/2 per cent. of the cases, produce a seed with a bitter kernel, only very rarely a seed developing which will produce edible kernels. The same is true of the apricot, the Smyrna variety being an edible apricot with an edible kernel.

The almond is normally the first of the stone fruits to begin growth and come into blossom in the spring and is also normally the last tree to become dormant in the fall. It is evident, therefore, that its normal winter resting period is comparatively short. The peach has a much longer resting period than the almond although less than the apple, pear and other similar fruits, and it is for this reason that peach production is possible in a commercial way in many sections of the East.

In California, where almonds and peaches are very often planted in close proximity, many seedlings are known which are very evidently natural crosses between the peach and the almond. In addition many artificial crosses have been made with no difficulty and have been planted and brought to maturity. The products of these crosses have shown the same general characteristics as those found naturally.

We are familiar with a peach-almond growing on the edge of a large almond orchard in California which produces good crops of fruit quite regularly. The fleshy portion or hull is almost edible, being much drier than the flesh of an ordinary peach and yet much more fleshy than the hull of the ordinary almond. It has a slight amount of astringency, a characteristic of the almond hull, but not sufficient to prevent its being eaten. Upon maturity this fleshy portion or pericarp splits but does not open as is usually the case with almond hulls. Inside this the pit, stone, seed or nut, or by whatever name it may be called, exhibits characteristics of both the peach and the almond. It does not have the deep corrugations of the peach pit nor does it have the comparatively smooth shell with small pores of the almond. In this particular variety the kernel is mildly bitter. In almost every respect this cross exhibits characteristics of both the peach and the almond. In other cases this is not true, some approaching more nearly the almond type while others are almost indistinguishable from peaches. In other words, the variations are those naturally to be expected in hybrids.

Now to return to the almond again. We find that for best results in production the almond must be grown in a climate where the winters are comparatively short and yet where there is sufficient cold weather to force the trees into complete dormancy. Where the winters are long or the summers are so dry as to force the trees to come dormant too early in the fall there is a great tendency to premature blossoming in the spring. In other words, the first warm weather in the late winter will bring the trees into bloom because of the fact that they have completed their normal rest period. This same condition has been found to be true of certain varieties of peaches which can be grown in the South but do not do well when planted in the North. It is for this reason primarily, in our judgment, that almonds do not produce under eastern conditions. There are other factors, such as extreme humidity, which may have a bearing, and undoubtedly would in the maturing of these nuts, but this should not prevent them bearing provided they could escape the adverse weather of late winter and early spring.

A mistaken notion has been given considerable credence that the almond is much more tender to frost or cold than the peach. Our experience, where the two have been grown side by side under identical conditions, is that the almond will stand fully as much cold as the peach and in some cases even more. The reason why almond crops are lost oftentimes when peach crops are not is due to their earlier blossoming and consequent subjection to the more severe weather of early spring which the peaches avoid.

It is evident, therefore, that the principal problem in producing almonds in regions of long winters, as compared with those localities where almonds can be produced, is to secure an almond which naturally has a long resting period, resulting in late blossoming, and yet one which will mature its fruit reasonably early. An almond tree beginning to blossom about the first of February will usually ripen its crop between the first and middle of August, though sometimes later. Those beginning to blossom about the first of March or later ripen their crops during September usually and often extend into October.

The question of soils and stocks is too broad to discuss here, except to dismiss it with the statement that the soils that will successfully produce peaches should also prove reasonably satisfactory for almonds through the use of peach rootstocks. These are commonly and successfully used in commercial almond orchards in the West.

Whether it will ever be possible to produce commercial almonds will depend upon whether an almond can be bred which will fulfill the requirements of late blossoming and early ripening and at the same time answer the requirements of a commercial nut. We should judge that it is possible, although we believe it is a big problem. Our reason for thinking so is that the Ridenhauer almond under eastern conditions will often produce nuts and it is recognized as doing quite well. We have never had an opportunity of tasting this nut but have seen photographs of the tree and have examined personally the nuts. Without any knowledge as to the actual ancestry of this nut we are very much inclined to the belief that it is a peach-almond. If this is so it opens up a line of breeding possibilities which should not be overlooked.

The procedure which should be followed will depend necessarily upon the conditions under which breeding experiments may be carried on. We believe that under eastern conditions the only opportunities for outdoor breeding work will lie along the line of interbreeding with peaches and almonds. The feasibility of indoor breeding with almonds is questionable in view of the difficulty of properly hardening for winter and yet affording protection during blossoming and providing at the same time for conditions which will favor the setting of the fruit. We do believe that there is abundant opportunity for experimentation, with the possibility that valuable results may be secured by systematic breeding along the line just mentioned.

Along with this cross breeding simple almond breeding experiments should be carried on, but these must be done in a locality where almonds can be brought to fruitage. Of course, the ideal place for this would be in California in a known almond district, and it is hoped that as time goes on experiments along this line will be conducted in an effort to secure later blossoming varieties and earlier ripening varieties. Our guess is that it would not be possible, at least within the lifetime of one man, to lengthen the normal resting period of any strain of pure bred almonds to the point where they would be able to withstand the long eastern winters and at the same time shorten the ripening period to practical limits. The development of this work, as far as it can be practically carried, should result in relatively late blossoming almonds which could then be used as a basis for breeding with peaches in an effort to still further approach the desired results and yet maintain the desirable characteristics of the almond. This simply involves the application of known breeding methods to these fruits.

To accomplish anything of this kind involves the development of a long-time plan which must be consistently followed. We would not look for any results to speak of before ten years, and would not expect any definite worthwhile results short of twenty years. It appears, however, that the possibilities are great and well worth striving for, and it is our sincere hope that some day a variety may be developed which will prove adaptable to eastern conditions.

The usual summer climatic conditions which prevail in the eastern states are not favorable to the economical production of almonds in a commercial way but we see no reason why they should not be eventually developed to the point where they may prove of considerable value and satisfaction for home orchards. The very fact that thus far no varieties of peaches have been developed which are immune year after year to spring frosts would indicate that it would probably be impossible to secure an almond which would be better than any peaches now known. On the other hand, one never knows until he tries and we believe that out of the effort much good could be accomplished, not only in the possible production of satisfactory varieties of almonds, but possibly in the accidental development of new and highly desirable peach varieties.

The possible development of a desirable table or canning peach variety with a sweet kernel would in itself be well worth the effort.

I had occasion to examine those Illinois almonds on the table here. It is quite evident that even though dried out somewhat they have some of the characteristics of the peach. The hull itself is fleshy even though thin. That is a characteristic that does not appear in the normal, pure bred almond hull.

I was just talking with Dr. Morris about some efforts he made at Stamford, Connecticut, to grow almonds. He stated to me, what was a very great surprise, that almonds there are afflicted with peach leaf curl and other diseases to which, under our weather conditions, they are not subject at all. There are undoubtedly other conditions here, due to a different climate, which we of California do not recognize at all.

I have endeavored to make this paper just as short as I could. I think that after it comes out in the proceedings there may be opportunity to study a few of the suggestions made here, and I want to express, on the part of the people in California, our desire to co-operate with those of you from the other sections of the country in every way possible for the development of varieties of almonds, or peach almonds. I can see that it will be difficult to compete with the sections in which almonds are naturally produced under semi-arid conditions. But I do believe in being close to your market if it is possible and in developing an almond which will be worth while for local consumption, especially for home use.

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THE PRESIDENT: Dr. Taylor, we thank you for the good advice and suggestions offered in your paper. I believe some attempt has been made to study the almond here in this vicinity. I know of one instance down in Forest Lawn by Mr. Baker. I believe that some years ago Mr. Wile attempted to do something in a commercial way with the almond, but I have since learned it proved a failure.

As Mrs. Ellwanger was very gracious in giving up her place I am going to call upon her now to read her paper.

OPPORTUNITIES FOR A WOMAN IN NUT CULTURE

MRS. W. D. ELLWANGER, Rochester

When at Mr. McGlennon's request I agreed to give some of my experiences in nut growing at this meeting I had no idea such a large and comprehensive title was to be given to my brief remarks.

Are not such opportunities wide and open to all? Women are now taking up so many branches of agriculture, gardening, farming, landscaping, that specializing in nuts is but one more. A real love for growing things, perseverance in face of many discouragements, and incidentally a place to grow the trees, are all that is necessary. I hope before long there will be classes in nut culture in the women's horticultural schools.

What is more delightful than to plant a tree? Planting flowers is a pleasure of the present but a tree is a link with the future.

My interest in growing Persian Walnuts in this region was started in 1912 by reading in a newspaper that these nuts could be grown in any climate suitable for peaches. Then I remembered that when a child I had picked walnuts from a tree on our lawn here in Rochester. Having a farm on the shore of Lake Ontario, part of which was a peach orchard, it seemed worth while to experiment with walnuts. Needless to say I am still experimenting!

The first trees planted were about one hundred Pomeroy seedlings and some fifty grafted trees, of the Rush variety. Dynamite was used at this time with such success that we have used it ever since. The seedlings are now quite large trees but not over half a dozen of them have borne any nuts. I early learned from growers in California that seedlings are a waste of time and money. I own a few acres of land in Southern California and of course have planted walnuts there. A few years ago I received word that the crop from my trees was being shipped to me. They arrived. There were six nuts. If I were a Californian I might say six bushels.

Three years ago the trees here bore quite a crop and no squirrel ever hoarded his winter supply with more satisfaction than I had with that first peck or so of nuts. Last year promised well, and many trees had nuts set for the first time, but owing to the intensely hot summer, or some other reason they did not mature.

There is a question as to the adaptability of Persian walnuts to this climate. The severe winter of 1917-18 with its sudden and extreme changes of temperature killed scores of my peach trees, while the established walnuts came through practically uninjured by a temperature of twenty-three below zero.

The World War did not take all the black walnuts in the country for gun stocks, for there are many fine trees still in the Genesee Valley. Every fall I am on the watch for trees bearing an abundance of large nuts which we use for parent stock.

It would be quite out of place for me to discuss the various methods of grafting before this audience all of whom know so much more about it than I do. But after many trials we have had the best results from grafting in the greenhouse. The black walnut stock is about four years old when potted, and the scions are cut in January or February and used immediately. Fifty per cent. is our average of success by this method, and some of the trees not two years old are bearing nuts.

I have tried planting pecan trees, but so far they have always been winter killed. Some Indiana trees planted this spring are growing and I am hoping they may prove hardy.

The Sober Paragon chestnuts have shown wonderful growth and bear nuts most abundantly. Each year, however, a tree or two is killed by the blight and I suppose soon my orchard will meet the fate of all the other chestnuts in the East. It seems as if someone ought to discover a remedy for this destructive pest. Tomorrow I hope to see you all at my farm where you can see what use one woman has made of her opportunities for nut culture.

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THE PRESIDENT: On behalf of the association I am certainly very grateful to you for your paper which contains some very valuable information.

Last week I went up to East avenue here to see the Thompson walnut grove and met Mr. Thompson and talked with him. The grove is in a very much run down condition. In fact he is thinking of using dynamite to blow it up and market the wood in Batavia for gunstocks at the gun factory there. He told us that in the thirty-six years that he has had it, he has had only three crops of nuts. One of the crops was an especially good one, I have forgotten the number of bushels he had, but he sold one hundred bushels, he said, to Sibley. Lindsay & Curr at nine dollars a bushel. If he could get a crop every year at that price I think he would be making pretty good money. I would class that orchard as a failure.

Last week, however, I had the privilege of seeing a walnut orchard that certainly surprised me greatly. I went to Lockport at the invitation of our very enthusiastic member, Mr. Pomeroy, to see the Pomeroy orchard, and I saw several trees heavily loaded with good sized nuts. Mr. Pomeroy estimates that he will have in the neighborhood of six or seven thousand pounds of nuts. The trees look healthy and show no evidence of disease. As I understand some of the trees are fifty years of age and there have been only two crop failures in that time. My idea is that the Pomeroy walnut is very hardy and of unusually fine strain. I believe that there is little hope for the commercial development of the English walnut much north of the fortieth parallel. I believe there will be some instances found, like that of the Pomeroy nut, where the seedling will do very well. It certainly has done very well with him. The Avon orchards are seedling trees, of course, the nuts having been gotten from a residence on Lake avenue, Mrs. Cramer's, at the corner of Emerson street. Evidently that strain is entirely different from the strain of nuts represented by the Pomeroy orchard which were brought from Philadelphia by Mr. Pomeroy's father.

I am going to ask Dr. Morris if he will present his paper and make his demonstration at this time.

DR. ROBERT T. MORRIS: I have had a good many experiences in grafting for a number of years. I have finally discarded most methods and have gotten down to rather simple principles. As a matter of fact this is the last word from my own point of view. During the past thirty or forty years I have changed my mind so many times on so many subjects that I have no confidence at all in anybody who puts any trust in me.

I am getting down to the splice graft. The reason why I didn't try it before was because it didn't seem reasonable to believe that the simple splice would hold. It was because I was so busy with many other responsibilities that on one occasion I neglected to brace some large splice grafts. Thus I learned that the splice graft would hold even through the very severe storms in our vicinity of Stamford, Connecticut. We have violent thunder storms and sometimes for a few minutes in advance of a storm we have a wind velocity of sixty or seventy miles an hour. If at the time the leaves happen to be wet the battering power of a seventy-mile wind is so tremendous that it will break out almost any form of graft. But my splice grafts during the past two years, simple splice grafts, subjected to this sort of storm, have not given way on a single occasion so far as I know, much to my surprise.

I will pass about some examples of the simple splice graft first and then show how we do it.

Here is a Stabler black walnut graft on common black walnut stock last year. For years I had been in the habit of cutting my scions and throwing the stubs away. I had a nice lot of hardy looking stubs in the grass and I said to myself "Why not try some of the stubs?" They made a very fine growth. I didn't lose one of them. Here is one of the big stub grafts and here is the growth it made last year. Here is another plain splice and the growth it made last year. This tree was killed by the ice in the river on my place last year. Sometimes in the spring we have great masses of ice come down that run through the orchard and kill some of my trees. That is the reason I cut off this one. I have only brought specimens that were injured but they show perfectly well. In this smaller splice you see I fitted the scion to the diameter of the stock. In the larger one I took no pains to do that. Furthermore the paraffin method was used. The scion is covered entirely with paraffin and I think you will notice, by rubbing your fingers over this stock, that the paraffin, although two years have elapsed, is all there. It is because I put it on in such a fine layer that it expanded with the growth of the scion.

Not always, but in order to make sure that my simple splice graft would hold, I have sometimes put in screws. I use flat-head, brass, wood screws, seven-eighths inch long.

I will put in some screws for you. So, if any of you fear that the simple splice grafts may not hold, put in screws and study Basil King's book on the "Conquest of Fear." This is a black walnut graft that I put in late this year with screws. You can see the screws projecting from the paraffin cover. I do not care if the screw sticks out quite a little distance. It is covered with a thin layer of paraffin. This graft caught and started to grow but was killed off by sprouts springing from the butternut in great masses before it had a chance to assert its own individuality. The graft, however, is all complete. Here is another one, where the screws are projecting, which was killed off by the stock sprouts below, with the repair all complete. In fact it would have gone on well enough to a successful growth if I hadn't been away and allowed the stock sprouts to grow. This shows, incidentally, the thin layer of paraffin. If we use a thick layer of paraffin it will crack and not be successful.

The simple splice graft is a very simple affair. In the first place it is well to have a knife with which you can shave. I think, Mr. Chairman, you could shave with that (handing knife to the President). That is the sort of edge to use in all our grafting work, the sort of edge that will bring terror to the heart of the mother of boys. I find very few people who really can sharpen a knife. I have been surprised at the small proportion of people who are really able to do it. They put on a feather edge, or they leave a round edge, or at any rate they are unable apparently to use the little finesse required to put the finishing touch on a really good knife. Above all other essentials is this little piece of carborundum made at Niagara Falls, F F Fine. Moisten it, hold it in the fingers this way, and then by simply rubbing it back and forth in this way you can put on the very finest edge. Do not use a knife unless you can shave with it because it is quite essential to have the cambium layer very nicely kept.

A couple of years ago hearing of Mr. Biederman's work in the use of the plane for grafting with his Persian walnuts, it occurred to me to try it with shagbark hickories. I went out in the barn to look for a block plane and I found three or four rusty ones. I wondered where they came from and then it occurred to me that about eight years ago I had thought to try the plane, and did try the plane, but it was not a success. That was before we had any success in grafting hickories. Now we may use the plane almost to the exclusion of the knife in cutting our scions of hard wood trees. Perhaps the majority of scions are shaped with the plane rather than with the knife because it gives a much truer surface. The block plane, then, I believe, is to be used more and more instead of the knife because of the very true surface that we make on the scion and on the stock and very quickly.

Of course with a small scion of this sort that would be about the slope that I would use for my ordinary splice. Fasten the splice together and simply wrap it with raffia. There is an ordinary splice graft fastened with raffia. That is the simple form that has given me the best results and I have tried out all the fantastic forms of grafting.

Now I am going to use the plane on a little larger scion. That is about the slope that I would use ordinarily. We will say this is to be the scion and this the stock. In order to make them fit perfectly I will use a smaller block plane. Now I will pass this about. You see with what absolute perfection those surfaces fit. You can get absolute perfection of fit by trimming a scion with a plane instead of with a knife. Even the best experts, like Mr. Jones, who make beautiful free-hand cuts, will find that with a plane they may make still better ones. That is one of the grafts that I would ordinarily fasten just with raffia, but I will fasten one together with screws to show how it is done. Now we will say that this is the stock and this is the scion. I am going to prepare them to fit each other. Some will ask if I ever use a scion as large as that. Sometimes I use a scion two or three feet long and as large as that in diameter. They are full of vitality and make wonderful growth. In order to do this I trim it down roughly with the knife to the general shape before I use the plane. I will cut as true as possible with the knife in order to simplify my work later.

MR. WEBER: In a large scion don't you have to have a larger exposed surface?

DR. MORRIS: I do not think that really counts.

MR. SMEDLEY: Isn't the tree in the ground when you graft it?

DR. MORRIS: This is supposed to be in the ground.

MR. JONES: You couldn't do a thousand of those a day?

DR. MORRIS: If you have something special, where you want to use up some big scions. But you can use the plane on little grafts just as well. Now this is the stock and Dr. Deming is going to represent Mother Earth.

MR. SMEDLEY: Are the scion and stock necessarily of the same diameter?

DR. MORRIS: Not necessarily, but preferably so. One's sense of nicety might demand that they be just alike, but you will find it doesn't make any difference. It takes a little longer to put in a big scion of his sort but it is very sure to grow. Your tree is already made by the time you have done this.

MR. SMEDLEY: Should you have bark contact all around?

DR. MORRIS: I could do it with contact on one side.

MR. CORSAN: What time of the year do you do it?

DR. MORRIS: Almost any time of the year, preferably May or June.

MR. CORSAN: Do you wax that before you put the raffia on?

DR. MORRIS: After everything is all complete that is my final touch.

MR. WEBER: When the stock is sappy wouldn't the sap jam the edges of the plane and roughen the bark?

DR. MORRIS: Not if you make it shave. I get the edge of my plane so it will shave. Then it will not roughen it. I can screw in a scion two feet long. I have tried it and had it start into growth. Thus I have got half my tree under way. Now I cover the whole thing with melted paraffin.

MR. CORSAN; How do you apply the paraffin, paint it on?

DR. MORRIS: Yes, with a soft brush.

MR. CORSAN: Do you use the stuff you buy at Woolworth's by the pound?

DR. MORRIS: Yes, I buy what they call parawax.

QUESTION: It is not necessary to wrap a scion with raffia if it is fastened with screws?

DR. MORRIS: No. After it is screwed you don't have to use raffia. I use either screws or raffia. In a large one like this the screw is preferable. In a smaller one the raffia would suffice. It is the plain splice graft that I use almost to the exclusion of anything else.

MR. WEBER: Wouldn't it assist the union, if the graft didn't make a perfect fit, to wrap it with raffia to hold it together?

DR. MORRIS: Possibly, but I think with the plane one can make a perfect fit. That is the idea at any rate. After three weeks of growth that will stand any storm.

QUESTION: How do you tell when the paraffin is the right temperature?

DR. MORRIS: That is very much as a woman does in cooking. You put in so much of everything. It is a matter of experience. I get it very hot but not hot enough to scald. The idea is to have it hot enough and to have it very thin. On one occasion my light went out when I was grafting walnut trees. It went out when I was grafting the very last tree. I put in perhaps twenty or thirty grafts in all. All the other grafts caught but on that tree, after my light went out, only one caught. In examining into the philosophy of it a week later I found that the paraffin, being a little too thick, had cracked.

QUESTION: When is the best time to do the grafting?

DR. MORRIS: I think the best time is after the sap season in the spring; all through the latter part of May and in June and the first half of July.

QUESTION: Do you use paraffin of a particular melting point?

DR. MORRIS: I have tried many but the one I use the most is the commonest one. You can buy parawax in all groceries. If you wish to make the parawax harder for the southern sun put in stearic acid. It may be bought at any drugstore. Melt it with the paraffin and that will harden it very much.

QUESTION: What proportion do you use?

DR. MORRIS: It would depend on the degree of heat to be resisted. I suppose you might use it in the proportion of one to four of parawax, but very little stearic acid will harden it.

QUESTION: Isn't there a tendency to melt under the high temperature of the sun?

DR. MORRIS: As a matter of fact I pay no attention to that in the North. Although we have very hot days and the paraffin does soften, it does not seem to interfere with the repair on the part of the tree.

QUESTION: In the case of smaller grafts, what would be your objection to the use of the ordinary whip graft?

DR. MORRIS: It makes one more motion.

QUESTION: It seems to me that it is more quickly done?

DR. MORRIS: It may be; that is a matter of individual technic. My idea is to do the thing the quickest way. If a man has found that he can put on one graft more quickly, that he has a technic that gives him speed, which is one of the essentials of grafting, if you can put on the whip graft quicker than I can put the other on, do it.

QUESTION: Do you have any trouble with the oxidizing of the cambium?

DR. MORRIS: Yes and no. Of course you free a certain number of enzymes. I haven't thought of it as an oxidizing process so much as an enzymic injury, where enzymes are freed from an organic solution.

QUESTION: I think that is correct. That is the common method of expressing it.

DR. MORRIS: I use sometimes, when the weather is very hot and I am grafting in the midst of sunshine on a hot day, a solution that I have described containing salts belonging to the salts of trees. I use that to dip my graft in and in that way the enzymes that are freed from the cut surface are removed by the solution in such a way that they do not interfere. Practically we can get almost one hundred per cent. of catches of our grafts now by the paraffin method, that is, with perfect scions, perfect stocks and perfect technic by the operator.

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THE PRESIDENT: Time is pressing and we have with us a member whom I am very anxious to have you all hear. I refer to our beloved member and my warm personal friend, the Pecan King, Mr. J. M. Patterson of Putney, Georgia, who is here this evening with Mrs. Patterson and their two sons. It affords me great pleasure to introduce Mr. J. M. Patterson.

MR. PATTERSON: Ladies and Gentlemen, your distinguished president has set a nice pace for me, introducing me as a king! Of course I am not unmindful of the fact that crowned heads are not any longer in favor in this democratic world of ours.

THE PRESIDENT: When I introduced Mr. Patterson at the Chamber of Commerce yesterday to Secretary Woodward, I introduced him as the Pecan King. He is known as the Pecan King and he is the Pecan King. There is no question about it. Mr. Woodward responded in what I thought was a very gracious way. He said he was much happier in meeting a pecan king than he would be in meeting some of the kings in the old world.

MR. PATTERSON: That is my apology for being here. You have made it easy for me. I have been away from home for nearly five weeks traveling on four wheels, and I received notice from your worthy president just a day or two before leaving my office that he would expect me to read a paper on the Commercial Possibilities of Nuts. At all events I had no time to collect my thoughts or make any preparation, and those of you who have toured through a new country and through some twelve or fifteen states, and passed through eight or ten universities and got your graduation papers each time as you went through, will realize that I have had not much time to compose my thoughts on this subject.

However, I am exceedingly glad to be here and I am going to talk a little like a preacher I heard once in the city of Pittsburgh. He said, "My text will be found in the Gospel of John, 4th Chapter, 15th verse, which reads as follows:" and he read the text. Then he proceeded without a lapse of breath and said, "From which we now take our departure." My subject is the Commercial Possibilities of Nuts, "from which we now take our departure."

California, or the Pacific Coast, has found the commercial nuts, the almond and the walnut. The Southland has found the commercial nut in the pecan. You good people of the effete and frozen East are still looking for the commercial nut. That is how it comes that we are here. It looked to me very much this afternoon when we were out at Mr. McGlennon's nursery that he had helped you very materially to answer that question, that he had discovered for you one commercial nut.

We have in the South two pecan organizations, one of which we call the National Nut Growers' Association. You will notice the word *National* Nut Growers' Association. The association is composed wholly of pecan growers. Many of us recognize that the name is a misnomer. We have been hoping that the time would come when we could have the name of that organization changed to the Southern Pecan Growers' Association, but we have one old member who has one English walnut tree in his orchard, who says we are a national nut growers' association and he objects. Some time that English walnut tree will die or he will die, and then we will be able to change the name. Then we have the Georgia-Florida Pecan Growers' Association. There is a California Walnut Growers' Exchange and a California Almond Growers' Exchange, and I am hoping to see a time when this Northern Nut Growers' Association will have discovered some real commercial nut, and then we will have complete the organization of the nuts of this country, the Almond Association, the Walnut Association, the Pecan Association and then the filbert, or whatever nut you discover here. We will bring them all together in one great national organization, and we will have an organization of real nuts. I am expecting to see that day. (Applause.)

I read a criticism the other day of a book that was published in which the reviewer said: "It is well for a man when he sits down to write a book that he know something of the topic on which he is going to write." I know very little about the possible nuts that may become commercially important in this section of the world. If it wasn't for the fact that when I come North here I like to meet some fellow nut, we wouldn't care very much whether you fellows ever discover a commercial nut in this part of the world or not, because the Lord has been so generous to you. The Lord has not given us a perfect climate. He gives one climatic feature here and another one there and another one some place else. He distributes his benefactions. It seems to me he has been lavish with you people, especially in New York and all through the middle West and the East. You have so many things. Why should you want to grab off the nut business? But just for the sake of letting you have a little variety and having some real good things to eat, I am willing to have you discover some real good commercial nut and then the time will come when we will have this national organization.

I am going to tell you a little bit about the history of the pecan. I think you would be interested in that. The cultivated pecan is of comparatively recent history. It is not so long since those who were in the South dreaming of a commercial nut were in very much the same position as this association is here, although the South seemed to be the natural place for the pecan. There were no commercial pecan orchards twenty years ago. There were wild groves in the river bottoms of Texas which there are today, but there were practically no cultivated pecans. There were actually no bearing groves of cultivated pecans. It is only a matter of fifteen or eighteen years that the cultivated pecan has been commercially planted.

I think our concern was among the earliest. I think we may claim to be the very first who, in a large way, planted pecans. We did not start with the intention of planting them in a large way. It was a sort of natural growth. It was only sixteen years ago this month, sixteen years ago, that I first heard of the paper shell pecan from John Craig of Cornell University; right under the shade of where we are meeting tonight I first heard of the paper shell pecan and was induced to put a little money in planting groves. I think I may say that New York State, through the instrumentality of old John Craig, can take credit for the start of the great commercial pecan groves of the South. Since that time pecan groves have been planted very extensively. I don't think that any accurate statistics are obtainable of the acreage planted to pecan groves in the district in which we are located in southwest Georgia, but in an area of probably forty or fifty miles I imagine there are seventy-five thousand acres of pecan groves. They have not all proven successful. Some have been planted on soil that was not adapted and there are some cases of insufficient or unwise care, and some of not having the proper stock to plant. For one reason or another a good many groves have not proven successful today. Others have proven quite successful. There is no question but what that which was a hope fifteen years ago is today a reality and that the cultivated pecan is today an established industry. I do not mean by that that we have reached the stage which our friend Mr. Taylor has reached with his almonds or which the almond growers have reached. We are still in our infancy and have many problems and the problems multiply as days and years go by. Fifteen years ago we would have said there were no insect pests nor any diseases of the pecan. They have certainly made themselves known in the last few years. We have a good many insect pests and we have some fungus. We do not believe that any of these will be beyond the skill of scientific investigation and that they will ultimately be brought into subjection.

As an indication of the growth of the industry, eight years ago the association of which I chance to be president gathered their first crop of nuts of something like six thousand pounds. Last year we harvested over four hundred thousand pounds of nuts. In eight years of course there was an increased acreage but they were all young groves. I tell you that fact just to show you that when you do find a nut that is adapted to your soil and to your climate, as the pecan is adapted to the climate and soil of the South, it will not take many years to develop such a nut into a commercial proposition.

I had the pleasure last fall of entertaining Mr. Pierce, the president of the California Almond Growers' Association. Mr. Pierce was very much interested in this young giant of the South in the nut world. He had had a very unfortunate experience in the use of pecans. He had passed through Chicago a short time before and a friend of mine, an officer of our association, happened to be a friend of his, and gave him some pecans, and he liked them so well that as he started from Chicago on the way to Washington he indulged too freely, and by the time he got to Washington he had to go to the hospital for repairs. Mr. Pierce wrote me a letter after that and said that he didn't know why the Lord permitted trees to grow such nuts until he created a new race of human beings with gizzards in place of stomachs. That is because California men were not used to eating good, rich nuts. We claim for the pecan that it is about the best nut there is. We don't claim the earth but if you people can develop or discover any nut that is better in quality and more tasty and more alluring than the pecan, we shall be mighty glad to have you discover it, and we hope it will be adaptable to the South. You know the Buick automobile says, "When better cars are made, Buick will make them." "When better nuts are made, we will make them." We know that all people can't have the best. We know that some people have to eat cheaper steaks. The trouble with this country today is that everybody wishes the very best. The packers tell us they have great trouble in disposing of the cheaper cuts of the meat. I do not imagine that the nut growers are going to have much trouble in disposing of the round steaks, but we are going to furnish the best nuts. The market for cultivated pecans has developed in a most marvelous way. There has never been any advertising, except in a very small way, and yet the demand has always exceeded the supply. It has grown just naturally. People learn of a good nut and they spread the good news to their friends so that the demand increases. Customers in New York but four or five years ago would order eight or ten barrels of nuts; they are ordering 150 barrels now.

I want to say to you, find a nut like that that you can grow in New York State or that you can grow down in Connecticut, or in any of this part of the world, and we will be mightily glad to see what you can do, and we will try to steal it and grow it in the South. It has been said that every great institution is only the shade of some great man. If you can build up a great institution of a great commercial nut here in the North let it be the shade of the Northern Nut Growers' Association.

I am not going to keep you longer because this rambling talk is not prepared. I have been interested as I drove through New England in seeing great groves along the public highways of maples and elms, and I have thought how wonderful it would be if those were all pecans or walnuts or almonds or some tree that would bear nuts instead of furnishing shade.

There is a world of opportunity in this country for a commercial nut. They are used as delicacies now, most of these nuts, but they are food, and they are food of the very highest type. I expect to see the day when all our best hotels and restaurants will have on their menus nut steaks, almond and pecan steaks, and when a great many of their guests will order these steaks in place of the beef steaks that they are ordering now.

I want to say that we are glad to have your distinguished president as a fellow pecan nut. He is largely interested in Georgia and we see his smiling face frequently in that section of the world. We are interested to see him succeed there and I am sure the members of this association are all interested and pleased to see what he has accomplished in developing the filbert right here in the shade of Rochester. (Applause.)

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THE PRESIDENT: Mr. Patterson, I thank you. I feel that I cannot let this opportunity pass to correct an impression that might have gotten over from one remark of Mr. Patterson's about the filbert nurseries being the result of my efforts. That is a long way from being so. In every successful operation I believe the master hand can be traced. In this operation of ours here the master hand has been that of my esteemed friend of long standing and very close cooperation covering a period of over a decade, Mr. Conrad Vollertsen. Mr. Vollertsen is entitled to the full credit for the success of our industry. I feel that I am justified in claiming for myself in connection with it the credit for the enterprise. Each of us in life has our particular place to fill. Mr. Vollertsen brought to me the idea of this filbert operation some years ago, over a decade, especially the idea of propagating the filbert from the layer instead of from the bud or graft, it being my belief up to that time that it could be propagated only by budding and grafting. He had worked in the nurseries in Germany as a young man and had told me of his experiences. So I sent to Germany and got five plants of twenty varieties, leaving to the nurseries from which I purchased them the selection of the varieties. I think the plants were six to twelve inches in size. From these, under the ability and knowledge of my friend, Conrad Vollertsen, has been developed what you saw this afternoon. I am mighty proud of it and so is he because he and I alone know what we have had to buck these last ten or eleven years. Speaking frankly, it has been pretty hard going sometimes, but personally I feel tonight, after what has been said to me by many of our members at our place this afternoon, especially the praise of our faculty to which I referred in my paper, that we have accomplished something really worth while, and it is my ambition and Mr. Vollertsen's, too, I know, to prove that we have a really worthwhile thing for the people. The pecan is the highest in food value of any nut known to the world today. The filbert is the second highest in food value and I believe it is a nut adapted for a wider range of soils and climates in the North than any other nut. I know this may sound a little like blowing my own horn, but I want you to understand that I am chuck full of filbert as well as pecan. I am certainly mighty happy for my pecan association in southwest Georgia, and I am feeling pretty happy tonight in connection with the filbert also.

I am met with a disappointment this evening. Mrs. Patterson tentatively promised to favour us with a paper on the use of nuts as foods. But I regret to say that she is somewhat indisposed and unable to favor us with a paper as promised. So I am going to ask another member, a new member, to make a few remarks on the subject of nuts as food. I know that he knows what he is talking about when it comes to a discussion of the subject of nuts as food, because I come in rather vigorous contact with him twice a week, and he talks nuts as food to me on those occasions. I am endeavoring to follow out his suggestions as closely as possible and I know that I am benefiting in health by so doing. I refer to James B. Rawnsley, the noted physical culturist who lives in this city. I have great pleasure in introducing to you Professor James B. Rawnsley.

MR. RAWNSLEY: Mr. Chairman and ladies and gentlemen: The gentleman that Mr. Patterson referred to as going to the hospital for repairs was not taken there because of eating nuts. The cause of the need for repairs was good food going into that man's stomach and mixing up with a lot of refuse matter that he had been eating at some previous time.

MR. PATTERSON: Almonds!

MR. RAWNSLEY: I hope that there are no medical doctors in the place or any butchers because if there are I am liable to go through the door or window. The nuts that you people are growing I hope will be the only thing, along with fruits and vegetables, that will be eaten in the future. As Mr. Patterson said tonight, since God put nuts and fruits and vegetables on this earth, those are what we ought to use from the commencement of life. The nut is one of the cleanest and most wholesome foods that is grown. I have tried it a good many years and I want to tell you, ladies and gentlemen, that there is nothing so sweet, so good or so substantial. It does not take much of a meal of nuts mixed with fruits to keep a person alive and well and strong. The sooner you people that are growing nuts get that into your minds and use it the sooner you will find it the best advertisement by which to get new members into the association. Show it yourself by using them.

THE PRESIDENT: I am mighty grateful to you for your words. We are going to try and get through one more paper this evening. It is by Mr. John Dunbar, Assistant Superintendent of Parks, Rochester, N. Y., on the subject, Nut Trees in Rochester Parks. I have great pleasure in introducing Mr. Dunbar.

MR. DUNBAR: Mr. President, and ladies and gentlemen: I picked up the program this morning and looking it over I was quite surprised to see that I was down there for a paper. We have given much attention for possibly twenty-five or thirty years to the establishment of an arboretum in the parks of Rochester of all the trees that are hardy in the north temperate zone. I think that perhaps the Rochester parks today stand next to the arboretum at Harvard University in the number of species and variety of trees from all parts of the north temperate zone.

We are studying trees generally from the ornamental point of view and to educate the people in the value of trees. Of course we have a large number of nut trees, hickories, walnuts and hazels, and incidentally we are interested in their food value.

In listening to Mr. Rawnsley tonight I was much interested in what he said because he is a neighbor of mine and lives across the street. I remember seeing him on a cold winter day when I was walking down street in a big overcoat, five below zero. Across the street there was Mr. Rawnsley shoveling snow and all he had on was trousers and a shirt. I have found out tonight how he could do it, by eating nuts. I said to my wife that I didn't see how he could stand it but now I shall tell her that I have found out.

Of course there are some nuts that are commercially of no use here. The pecan is the nut of the South. Mr. McGlennon and Mr. Vollertsen are doing great things with the filbert here. I think there is a great future here in the North for the hazels and king nuts. Other nuts that are very important here because they are hardy are the black walnut and the butternut. If walnuts and hickories can be grafted in tens of thousands like apples and peaches, all right, go ahead, but in the meantime raise all the seedlings you can. I am surprised that so far nothing has been said here about the king nut. There are only two places in New York State where the king nut grows. It grows in the Genesee Valley from Rochester up to Mt. Morris quite abundantly and it grows around Albany and Central New York. There are no other places in New York State where it grows. It is a larger nut than the common shell bark. It makes a magnificent tree. I think the king nut should be planted. We are growing it ourselves in the park. The tree itself grows fifteen miles from here. We have it in the park today and I have planted a good many of these nuts. I think the big shell bark or king nut and the shell barks should be planted quite extensively. Put them in the ground and let them come up. They will come up. Another good tree we have here with great possibilities in it is the Japanese butternut. It is hardy and I understand it is growing at Lockport. These are a few rambling ideas. Incidentally we are doing all we can to spread the gospel of nut culture and the growing of nut trees. If people could see them in the parks it would help along their education.

MORNING SESSION, SEPTEMBER 8th, 1922

The Convention was called to order by the President at 9:30 o'clock A. M.

THE PRESIDENT: After a night of good rest we are ready to proceed with our deliberations and as we have a lot to do we are going to try to push things along fast this morning.

Some of the papers have not arrived and some of the speakers will not be here. Senator Penney of Michigan wrote me that he was not only in rather poor health but he was in the midst of an election primary and that it would be impossible for him to be here but that he would endeavor to send a paper. I am sorry to say that it has not arrived.

I was pretty sure that ex-President Linton would be here. But I have a telegram from him this morning saying it is absolutely impossible and that he, too, hasn't had any time to prepare a paper. Mr. Linton is a very busy man and about the only way to get a rise out of him is by wire. I have written him three times and wired him five times. Finally I succeeded in getting a telegram from him this morning. I was particularly anxious that he and Senator Penney be here to discuss the roadside planting of nut trees and the legislation of Michigan in that regard, believing that such aid would materially help us in getting other states interested along the same line. I'm sorry, therefore, that they are not here.

This telegram from Mr. Linton, received this morning, reads as follows:

"Expected until yesterday that I would get to Rochester convention but am bitterly disappointed in being unable to do so owing to fatal illness of chairman of our state commission, whose called meetings and pendent duties have fallen upon me. Senator Penney is in midst of strenuous primary campaign closing Monday and can not leave and Mr. Beck is in hospital recovering from operation. So your Saginaw trio, positively with you in spirit and good wishes, is held here this time absolutely and all regret the situation beyond measure. I expressed to you yesterday, prepaid, the Washington walnuts, fine young trees only eighteen months old, and will replace them next spring if necessary. Penney and Beck join me in sincerely desiring the success of your convention and extending kind regards to you and those present, all of whom we hope to meet another year.

WM. S. LINTON.

The trees we are going to plant tomorrow morning, if these seedlings get here, are grown from nuts furnished Mr. Linton by the superintendent of Mount Vernon. Last year we planted some in one of the parks at Lancaster.

I will ask Mr. Vollertsen to read his paper now.

MR. CONRAD VOLLERTSEN: Ladies and gentlemen: My paper this morning will necessarily be very short as the subject assigned to me is one of which I so far have not had any practical experience and therefore am unable to say much about.

According to our program I have been assigned to make a few remarks on "The Blight-proof Propagated Filbert," a subject I think rather hard to discuss as we have so far no positive proof that blight, if it at all exists on the improved filbert, will not eventually appear on varieties we are now growing. I therefore believe the subject, "Blight-proof Propagated Filbert," should have been worded somewhat differently, as we have no assurance when blight may appear, nor any guarantee against its appearance. It may fall on our plants over night or at any time. That we can not prevent nor control.

In the nursery of improved European filberts which we have maintained for ten years, blight is so far not known and has never made its appearance. We know of other filbert plants, several varieties, all of German origin, in this, our home city, from thirty to forty years old, never affected by blight, bearing nuts today. But all this will not guarantee the improved propagated filbert to be blight-proof. We certainly do not claim our propagated improved filbert plants are blight-proof. In fact to our knowledge there is no such thing as blight-proof filberts no more than there are blight-proof pears, quinces or other fruits. But we do claim that our improved filbert varieties, imported from Germany, will stand our climatic changes very much better and will resist the attack of blight to a greater extent than any other variety imported from France or Italy.

We really do not fear blight. We have heard very much about it and have so far seen nothing of it. But should it eventually appear in our nursery I am fully convinced we can easily control it and prevent its spreading by cutting the affected parts thoroughly away, removing the diseased twigs or branches so low as to make the cut in entirely sound wood. Through such an operation I am fully convinced the disease can be completely eliminated in a comparatively short time, should it ever appear.

We have been repeatedly told blight will not only attack small parts or branches of the improved filberts but will kill them entirely. Such a thought I can never entertain, not for a moment. I have had too many years' practical experience with the growing and cultivating of improved hazel or filbert plants, and have never seen anything of the kind. It would be very interesting if members of this association who have observed blight on the improved hazels and seen plants actually killed by that disease would relate their experiences and the real facts so as to enlighten the public on the subject. For instance:

Where did it happen that blight killed the plants entirely?

What varieties were attacked and killed?

And was it genuine blight that killed them?

These questions should be well considered, particularly the last one, as it is a well-known fact that in a general way the term blight is frequently used for various injuries or diseases of plants causing the whole or parts to wither and die, whether occasioned by insects, fungi, or atmospheric influences.

We will, in the early summer, occasionally see on various shrubs or trees numerous little twigs and branches dead and decaying and the general saying then will most assuredly be, the shrub or tree is blighted, where a close and thorough investigation will not reveal the slightest sign of blight, merely injuries by frequent climatic changes in the late winter or early spring months.

I also have observed the same thing where insects were the cause of all the trouble. A little downy species of the aphis, or plant louse, had completely overrun a Stump apple tree and really caused it to die. The owner told me that tree was blighted. But here also no sign of blight could be detected. Nothing but insects caused the tree to die, not blight.

I merely mention these instances to show how thoroughly and readily a disease or ailment of a tree or shrub is called blight where in reality not the slightest sign of it can be discovered.

If our people had the understanding and would take the time to investigate the cause of their diseased trees I am fairly satisfied the complaining of trees or shrubs being killed by blight would not be heard as freely as it is today.

Now under no circumstances should this be construed as meaning that I dispute or doubt the existence of blight among our filbert plants. Not at all. Quite the contrary. We have, as stated above, so far no blight-proof filberts and no guarantee that blight will not eventually attack our plants. We therefore will have to be more or less on the alert, will have to watch our filbert plants as we do our pear or quince orchards or other fruit trees more or less inclined to blight. By no means let blight discourage the planting of filbert or hazel nuts, as I am fully convinced should it eventually appear it will not kill our plants. In fact it will not harm them as much as it will our pear trees, our quinces or other varieties of fruit inclined to that disease, of which we, in spite of blight, plant and maintain large orchards.

My advice would be to stop all talk on blight and wait until it appears. Do not let us cross the bridge before we come to it but let us watch our trees inclined to blight, particularly our hazel and filbert plants, as they are not blight-proof, but eventually should blight make its appearance let us be ready for it, fully prepared to receive it, not to welcome but to eliminate it. That we can do, that we can accomplish very thoroughly through the operation set forth in the beginning of this paper.

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THE PRESIDENT: That is a subject that I feel we ought to have a little discussion on and I would like to hear from Mr. Jones, Doctor Morris, Mr. Bixby, Doctor Deming, for a brief discussion on the points just touched on by Mr. Vollertsen.

THE SECRETARY: I have had very little experience with the blight. Two years ago Mr. Bixby and I visited the very large hazels in Bethel, Connecticut, seedlings raised from grocery store nuts, and we saw there the blight on some of the largest trees, on the large limbs, unquestionable blight with sunken areas covered with pustules. I didn't see the trees last year, but on Wednesday, just before taking the train to come here, I ran in to this place to get a bunch of hazels to bring here, and I saw the tree on which Mr. Bixby and I had found the blight looking as well as ever. In a hasty examination of the tree I saw one or two stubs where large limbs had been cut off. I presume that the owner had followed our advice and had cut off the blighted limb and, apparently, the tree itself was none the worse for the blight.

I have had hazels planted and neglected for twelve or thirteen years and this is the first year in which I have found the blight. I have found before other causes of death of parts of the shrubs, girdling by insects and apparent winter killing, but this year I found several of my trees on which were undoubted patches of cryptosporella. That is the extent of my experience with the blight.

MR. JONES: I have not had any actual experience with the blight but I have seen it in Connecticut. I have not found it on any of the wild hazels of Pennsylvania. Therefore we do not have it at Lancaster. I have not regarded it as nearly as serious as pear blight and some other blights that attack fruit trees.

THE PRESIDENT: What is that, Mr. Jones?

MR. JONES: I say I have not regarded the filbert blight as nearly as deadly as some of the blights that attack the fruit trees, because of the fact that it works very slowly, and it takes, I understand, about two years to girdle a limb of any size; therefore, it is easily cut out and controlled.

MR. CORSAN: Could it be that the blight would be very much more active in a tree growing in the shade than on one growing out in the strong sunlight and well nourished?

MR. VOLLERTSEN: I know of some trees that were for at least ten or eleven years practically overgrown by butternut trees. I have known the trees for more than thirty years. I visited the place about a week ago and found a tree doing fairly well under the circumstances. That tree is between thirty and forty years old and has grown steadily for the last five or six years entirely in the shade and is bearing fruit fairly well. There were quite a few nuts on it although there were more over the top than on the lower branches; but I did not notice any dead limbs or anything of that kind.

THE PRESIDENT: Do you refer to Doctor Mandel's plant?

MR. VOLLERTSEN: No.

DOCTOR MORRIS: Stamford is a natural home of the hazel. Wild hazels fill the fields to such an extent that they destroy pastures very often.

Hazel blight, therefore, is to be found there as an indigenous organism or parasite. Among the native hazels it apparently attacks only those that have been injured or are weakened by age or otherwise. That is the common history where a plant has existed along with a parasite for centuries or ages, a certain amount of tolerance is established by the resistance of a few individual plants and the elimination of the others. By natural selection the best survives.

Now when I brought some European hazels to this place a little over twenty years ago they made a good start. In two or three years all were attacked with blight and at the end of four or five years all were dead. I spoke to Mr. Henry Hicks about it. He has a place on Long Island. Mr. Hicks said, "I have given up foreign hazels. They are no use. They all die. I don't try them." Whenever anybody says that to me it starts me right off doing it. When they said we couldn't graft hickories I said, "Well, here is something to do," and I did it. They said, "Well, we couldn't raise hazels; we might as well give up." I said, "Well, here is the best thing for us to do then." So again I got a small lot and observed them day by day. Very soon the blight began to attack them. I found it grew slowly and gave me plenty of time to cut it out. I neglected some purposely to see how long it would take the blight to girdle a limb and some of the larger limbs took two years. In all of the limbs that were affected, in the hazels which I wished to save, I simply cut out the blight with a sharp jackknife, painted the spot with a little paint, an antiseptic or something of the sort, and had complete control. In fact I found that I needed to go over my hazel bushes not more than once a year to look after the blight, and in one day, or part of a day, with a sharp jackknife I had absolute control of the blight.

There are some large European hazels that I have neglected and have allowed the blight to get under way. Some of them are so resistant that they bear very good crops notwithstanding the fact that they are neglected and have the blight. Others have died. Therefore it is a relative question, a question of relative immunity to the blight. My belief is that the blight will not be any more injurious to our hazels than the San Jose scale has been to the peaches. We have complete control of the San Jose scale because we know the habits of the scale insect. I believe we have complete control of the hazel blight because we know the habits of that particular sporella.

As to the question of growing in the shade or in the sunshine, on the Palmer property not very far from me, there are some very large bushes of red and white avellana and of the purple hazel that have been overshadowed by other trees because they haven't been looked after. Those are all very large bushes, in fact they have grown to be small trees and they are completely overshadowed by other things. They have some blight but continue year after year to bear heavy crops of nuts.

THE PRESIDENT: Mr. Bartlett, have you any remarks on the subject?

MR. BARTLETT: My experience has been very similar to that of Doctor Morris. I have visited possibly a hundred places and have seen hazels growing, some of which have probably been there seventy-five years. In talking with the people connected with the place I have often heard said, "Why, years ago we used to have hazels, a great many hazels here, picked maybe a bushel at a time, but the best varieties have died, and what we have left are worthless." Or perhaps, "There is only one bush left and we don't get any hazels now." Apparently the purple hazel is freer from blight than most of the other imported varieties. I have seen the blight in these places. I have seen branches from three to four inches in diameter that were attacked with blight and were still growing but were not fruiting very much. I know a very few places where hazels are grown within fifty miles of New York, and I know of some places where they are getting some nuts. But the general impression is that the European varieties will be attacked with blight and killed.

I have seen bushes that have been attacked by blight where the roots are alive but sending up very weak shoots. That is probably through neglect of stocks. Certain of those that I have raised, five or six years old, are absolutely free from blight. Most of the older trees that I have seen around have blight in some form or other.

MR. BIXBY: Doctor Morris' remark as to what Mr. Hicks says of giving up attempting to grow hazels because the blight would take them, seemed to me very appropriate in view of an observation I made on Mr. Hicks' place last fall. I found there a large hazel which was probably twenty-five feet high and bearing a fair crop of nuts. Mr. Hicks told me that he had brought that tree from Germany many years ago—I think it was over twenty years ago—and that that was the only one left out of a lot. Now if other European hazels had been killed there with the blight and this one was left there was apparently a blight-proof hazel in that lot.

I have seen a good many hazel bushes affected with blight, but I have not seen any since I went with Doctor Deming up to Bethel. I have seen no blight since then though I have looked for it whenever I have been where there were European hazels. I examined that tree in Mr. Hicks' nursery very carefully and found there was no evidence of blight. I feel as the other speakers do who have expressed themselves, that we have little to fear from the hazel blight; that if it does appear in the nurseries we can control it by cutting out the blighted portions.

MR. PIERCE: In northern Utah I have a number of bushes of the foreign and the American hazel and they are ten years old. So far I have not seen any evidence of blight.

I would like to ask a question. What form does this blight take, and is it deadly? In other words, will it kill the bush? Is it good to cut out the affected parts?

DOCTOR MORRIS: You find a depression of the bark over a small area, gradually increasing, and around the part that is depressed you will find a little swelling of the healthy part that is trying to grow over the blight area. This also contains the roots, if you can call them that, of the blight. You can recognize it everywhere on the hazel by the distinctly depressed area of bark, which should be cut out before it gets to be the size of a quarter.

In other cases the blight will encircle a small branch and cause a swelling instead of depression that looks very much like the swollen area around the depressed bark. There may be depression in the branch parts but the swelling blocks that so you can see only the swelling. These branches may be very easily removed, with as much ease as a boy would steal the nuts, so there is nothing to be feared on that score. If the blight is left uncared for it will kill some of the plants and it will not kill others. It will injure some also without killing them, so that we have to consider the question of what we call relative immunity. In the case quoted by Mr. Bixby we have a case of relative immunity of a hazel which has grown to be twenty-five feet high and bearing crops in the midst of the blight area on Long Island, while others have disappeared from the vicinity.

MR. BIXBY: I would say in connection with that hazel that Dr. Deming and I visited in Bethel that I took a blighted branch away with me and it was such an excellent example of a blighted area that I had a photograph made and it was printed in the Nut Journal.

THE PRESIDENT: This discussion on the blight of the filbert is of intense interest to me. It is a considerable relief to us to hear these encouraging statements, because during our experiments, covering the past decade, the bugbear of all of our deliberations has been the possibility of blight wiping us out, it having been suggested at the time we imported plants that we would never get anywhere with them. I think we have little cause to feel very much worried on the subject of the blight.

It now gives me real pleasure to introduce to you our friend, Mr. Pomeroy.

MR. POMEROY: Mr. President, ladies and gentlemen: Josephus says that he has set down various things according to his opinion and if anybody holds to another opinion he will not object. That's my position in regard to nut growing. I will tell you a few things that I believe and if you hold another opinion you are entitled to keep it.

Professor John Craig once referred to a thing that surprised me very much. We Americans believe we are a very energetic, smart people not to be fooled much in a trade. Well, he had statistics which showed that after we have shipped millions of dollars worth of wheat and cotton and various other products to Europe we receive our pay in the form of great quantities of nuts which we use for food, holiday nuts, all-year-round nuts. Now I believe that those nuts can be raised right here and we can pocket that money instead of leaving it in Europe.

I was a very small child when my father went to Philadelphia to visit the Exposition in 1876. While he was there he picked up a few walnuts which had dropped from a tree in front of his lodging house and brought them home and planted them. A very few years after he amazed us all by taking a load of nuts to Buffalo and obtaining more money for them than the hired man and I did for a large load of fruit.

At one time I put out some English walnuts in a cemetery as a memorial orchard and the trees are now doing fine. The other night my wife and I strolled over and looked at them and when we were on our way back we passed a neighbor's house where there were a number of maple trees on the lawn. I said to my wife, "Those maple trees are fifty years old, and there by the side of his lawn is a chestnut tree that is forty-four or five years old." She made the remark, "Those English walnut trees over there cast a much more beautiful shade than those maples," and it was true. I think Mr. McGlennon saw them.

THE PRESIDENT: Yes; that's so. I thank you very much, Mr. Pomeroy.

Mr. G. H. Corsan, of Toronto, Canada, is known as the "Canadian Johnny Appleseed." Mr. Corsan goes about the country and when he can find nuts and seeds of what he thinks are good trees and plants he gathers them up and arranges to distribute them. If Mr. Corsan will give us about ten or fifteen minutes I should certainly appreciate it very much.

MR. CORSAN: Mr. Chairman, ladies and gentlemen: Our friend here called me Johnny Appleseed, I suppose because I went around among my friends who had gardens and said, "Let me plant this," and I would plant a nut tree. I said, "Why don't you plant something with a utility value as well as a thing of beauty?" I said, "Why not plant something that will not only grow rapidly and cast a splendid shade but that will also return you something in the way of food?"

I first devoted twelve acres to the culture of nut trees. I afterwards added four more. I just planted seedlings. In the year 1912 I joined the Nut Growers' Association and I set out a hundred chestnut trees. When I found the blight was in them and I cut them all down but two. I have those two now and last year I gathered a peck of very large chestnuts from them which caused the Ontario government to take notice of what I was doing.

I bought a great many other trees, among them some of Mr. Pomeroy's. I had a hard fight with Pomeroy's trees. They would die down one year and grow a foot or a foot and a half the next and then die down again. But each year they increased a little in size and now they are over my head and are not dying down at all.

I tried a lot of others, among which were seedling English walnuts from St. Catherine's. They did not freeze down at all, but whether they will throw as good a nut as Mr. Pomeroy's I don't know. They are certainly a different nut.

Then I got a Chinese walnut of Black's nursery, Hightstown, New Jersey, and it is growing remarkably well. All three types of trees are doing very well and are all over my head, sometimes growing three or four feet a year, very rarely less than a yard from each terminal branch, and I have had no winter killing.

It may be interesting to recount a few other things about my place. I had an awful fight with mice. My land is in a valley and the spring floods come down and I can't plow the land or it would all be washed away. I put a tree in and protect it with a certain amount of space around it. I found that the mice would chew down the trees almost as fast as I could get them in, so I got some cats. The cats soon learned to prefer birds to mice so I killed the cats. Then I bought a flock of geese and the geese cropped the grass short and prevented it from growing so powerfully as to smother out the trees. But the geese had hard bills and when the trees were small they clipped off pieces of bark with their bills, so I traded the geese for wild geese. I learned that they are more discriminating in their choice of food and that though their wings are more powerful their bills are not as strong. They have kept the grass down for me and destroyed the homes of the mice. Then I got pheasants in order to rid myself of the insect pests. I feel that in another ten or twenty years we will have a very beautiful place.

I need not refer to the fact that nuts are very valuable for food. Dentists would all go out of business if we ate nuts.

Pennsylvania is a state which should certainly take up with its agricultural authorities the possibilities of nut growing because that is a state that can be ruined utterly by trying to grow grain on the hillsides. The water comes down and washes all the rich top soil off into the creeks and it is lost to us and our children.

* * * * *

THE PRESIDENT: Will the secretary please read Doctor Kellogg's paper?

THE SECRETARY: Mr. President, this is a very long paper and I have not read it over. It seems to me that perhaps we have devoted so much time to genealogy and reminiscences that the time is short for the papers which are to be read by members present. Would it not be well to defer the reading of this paper of Doctor Kellogg's to a later time, or, possibly, merely print it in the proceedings?

DOCTOR MORRIS: I move it be laid upon the table and printed in the proceedings.

The motion was duly seconded and carried.

(See Appendix for Dr. Kellogg's paper.)

THE PRESIDENT: One of our important visitors is Professor James A. Neilson, Guelph, Canada. The title of Professor Neilson's paper is, "Nut

Culture in Canada." This is an especially interesting subject to me.

PROFESSOR NEILSON: Mr. President, ladies and gentlemen: I want to express my appreciation of your kind invitation to attend your convention and for the opportunity of talking to you for a while on the subject that is more interesting to me than any other branch of horticulture. I had looked forward to coming to this convention but wasn't just sure that I would be able to be here. Therefore when I got a wire from your president I immediately got busy and pulled the wires at the college and asked them to authorize me to come here at college expense. I am very glad to be here. It has been most interesting to me, and I am very pleased, indeed, to meet so many whom I knew already by reputation.

NUT CULTURE IN CANADA

J. A. NEILSON, B. S. A.

Lecturer in Horticulture, Ontario Agricultural College Guelph, Canada

The conservation and improvement of our native nut trees and the introduction of suitable species from foreign countries has not received much attention by horticulturists in Canada, except in British Columbia and in Ontario. In British Columbia, Persian walnuts, Japanese walnuts, filberts, almonds and European varieties of chestnuts have been planted to a limited extent in the fruit districts and small plantings have been made at the Dominion Experimental Farms located at Aggaziz on the mainland and at Sidney on Vancouver Island.

In Ontario very little had been done by the Provincial Experiment Stations to test the different varieties of nut trees until about one year ago when the Vineland Station undertook to establish experimental plantings. A few enthusiasts like G. H. Corsan of Toronto, Dr. Sager of Brantford, Dr. McWilliams of London and William Corcoran of Port Dalhousie are about the only parties who have attempted anything along the line of nut growing. These remarks of course do not apply to those people who have planted a few black walnuts or Japanese walnuts on the home grounds or along the roadsides. Of such plantings there are a few here and there in the older settled parts of the province.

For some years the writer has felt that something should be done by the Horticultural Department of the College to interest the people of Canada in planting more and better nut trees and in conserving the remnant of the many fine nut trees which formerly grew so abundantly in parts of Ontario and elsewhere. Therefore an attempt was made during the spring of 1921 to interest the public in the possibilities of nut culture. A letter and questionnaire asking for information on nut trees were prepared and sent to officers of horticultural and agricultural societies, agricultural representatives, agricultural and horticultural magazines, daily and weekly newspapers, school inspectors and other interested parties.

The following is a copy of the letter and questionnaire which were sent out:

"Dear Sir:

"We are investigating the possibilities of nut culture in Ontario and would be pleased to have you assist us by reporting the occurrence and distribution of the various species of native and introduced nut trees growing in your locality. We are particularly anxious to learn of the exact location of superior trees and if any such are found we plan to have these propagated and distributed for test purposes. We would also like to secure the names of people who are interested in nut culture. Please fill out the enclosed questionnaire and return to the undersigned at your earliest convenience.

"Thanking you in anticipation of receiving some interesting information on nut trees, I am, yours sincerely, (signed) James A. Neilson, Lecturer in Horticulture."

Questionnaire:

Q. 1. Are any of the following kinds of trees growing in your locality:

American Black Walnut European Chestnut
Japanese Walnut Japanese Chestnut
English Walnut Chinese Walnut
Butternut Beechnut
Hickory nut Hazel nut
Pecans Filbert
Sweet Chestnut

Q. 2. Do you know of any individual trees of the above mentioned kinds that are superior because of large size of nuts, excellent flavour of kernel, thin shells, rapid growth or high yields? Please give exact location of such trees.

Q. 3. Is any one in your section making a special effort to grow any native or foreign species of nuts? If so please give their name and address.

Name of correspondent
Post Office
Township
County
Province

I am delighted to say that I never did anything in my life that met with such hearty and general approval as this venture. From almost every quarter of Canada I received commendatory letters and offers of assistance. One encouraging feature was the keen interest shown by wealthy business and professional men in our larger centres and by some of our more progressive fruit growers and farmers. Inasmuch as my venture was an innovation there were of course some humorous comments to the effect that we had enough "nuts" in the country now without encouraging any more. I replied to my humorous friends that the "nuts" they had in mind did not grow on trees whereas the kind I had in mind did.

The information I received in answer to my questionnaire was very interesting and instructive and confirmed some of my impressions regarding the occurrence of nut trees in our province. More important still it showed that there were several superior trees of various species growing here and there throughout the country.

Geographical Distribution of Nut Trees in Canada

The chief native nut trees are the black walnut, the butternut or white walnut, the hickory, of which there are four species—the chestnut, the beechnut and the hazelnut. Of introduced nut trees there are the Persian, Japanese and Chinese walnuts, the European, Japanese and Chinese chestnuts, the pecan and the European filberts.

THE BLACK WALNUT (*Juglans nigra*).

The black walnut is one of our finest native nut trees and is found growing naturally along the north shore of Lake Erie and Lake Ontario and around Lake St. Clair. It has been planted in many other parts of Ontario and does well where protected from cold winds. The tree grows to a large size, sometimes attaining a height of 90 feet and a trunk diameter of 5 feet. When grown in the open it makes a beautiful symmetrical tree, having a large, rounded crown with drooping lower branches.

The black walnut is not found growing naturally outside of Ontario. It has been planted in Manitoba but does not do well there because of the cold winter. In 1917 the writer observed a few specimens near Portage la Prairie which were about five feet tall. These trees made a fair annual growth but most of this froze back each winter.

Many people in Canada believe that the black walnut is a slow grower. This impression is not correct as some trees grow very rapidly. About eighteen years ago I planted a number of nuts along the line fence and along the roadside on my father's farm near Simcoe, Ontario. Most of these nuts sprouted and grew and some have done exceptionally well. One of these trees is now thirty-seven feet tall and has a trunk circumference of forty-one inches at the ground. It has borne nuts since it was six years of age and this year has a very heavy crop. Some of the first crop of nuts were planted and these in turn have developed into trees which have produced nuts. Nuts from the second generation have been planted and will likely make trees which will yield nuts in a few years. An interesting feature of the original planting is the great variation in the size, shape of nut, thickness of shell and yield. Some are large, some are small, some are round and others are pear-shaped. The majority of the trees yield well but a few, however, are light croppers.

THE BUTTERNUT (*Juglans cinerea*)

The butternut is much hardier than the black walnut and has a much wider distribution in Canada. It occurs throughout New Brunswick, in Quebec, along the St. Lawrence basin and in Ontario from the shore of Lakes Erie and Ontario to the Georgian Bay and Ottawa River. It has been planted in Manitoba and does fairly well there when protected from cold winds. West of Portage la Prairie the writer observed a grove of seventy-seven trees. Some of these trees were about thirty-five feet tall with a trunk diameter of ten inches and had borne several crops of good nuts.

The butternut in Ontario sometimes attains a height of seventy feet and a trunk diameter of three feet.

THE ENGLISH OR PERSIAN WALNUT (*Juglans regia*).

The English walnut, or the Persian walnut, as it should be called, is found growing in the Niagara district and to a lesser extent in the Lake Erie counties. It is stated on good authority that there are about 100 of these trees growing in the fruit belt between Hamilton and Niagara Falls. There are several quite large trees in the vicinity of St. Catharines, which have borne good crops of nuts. One of these trees produced nuts of sufficient merit to be included in the list of desirable nuts prepared by C. A. Reed, Nut Culturist of the United States Department of Agriculture. This variety has been named the "Ontario" and is now being propagated, experimentally, in the United States. In the vicinity of St. Davids, on the farm of Mr. James Woodruff, there is a fine English walnut tree which produced ten bushels of shelled nuts in one season. This tree is one of the largest of its kind in Ontario, being about sixty feet tall with a trunk diameter of three feet at one foot above the ground and a spread of branches equal to its height.

The English walnut is not as hardy as the black walnut and is adapted only to those sections where the peach can be grown successfully. At present this tree cannot be recommended for any part of Ontario except the Niagara district and the Lake Erie counties and even in these areas it should not be planted unless it has been grafted or budded on the hardier black walnut.

JAPANESE WALNUTS.

The Japanese Walnut is known to occur in Canada in three different forms—Juglans cordiformis; Juglans Sieboldiana; Juglans mandschurica.

Juglans Cordiformis.

This species is cultivated extensively in Japan and is the most valuable one for Ontario. The tree is very beautiful, comes into bearing early, bears heavily, grows rapidly and is reported to live to a great age. It is believed to be as hardy as the black walnut and ought to do well wherever the native walnut grows satisfactorily. In the best types the nuts are distinctly heart-shaped, have a thin shell, crack easily and contain a large kernel of good quality which can often be removed almost entire from the shell with a light tap from a hammer.

There are two fine heartnut trees growing near Aldershot which is near Hamilton on the road to Toronto. These trees are eight years of age and are about twenty-eight feet tall with a trunk diameter of eight to nine inches. In the seventh year one tree produced about a bushel of fine nuts with thin shells.

Juglans Sieboldiana.

This type was first introduced into the United States about 1860 by a Mr. Towerhouse in Shasta County, California. Since then it has been widely distributed and is now found in many parts of the United States and Canada. It is much the same in appearance as the one first described and grows just as rapidly and bears just as early but does not produce so valuable a nut. The nut has a smooth shell of medium thickness with a kernel of good quality. It does not usually crack easily and the kernel cannot be taken out entire, therefore, is not so desirable as the cordiformis type. In rapidity of growth the Japanese walnut is only excelled by the willows and poplars. In the vicinity of Grimsby there is a tree eight years of age which is about twenty-five feet high and has trunk diameter of seven inches at the base. It began to bear nuts in the third year and in the sixth year produced one bushel.

Juglans Mandschurica.

The general growth characteristics of this species are somewhat similar to the other two types but the nut, however, is quite different, being somewhat like a butternut. Because of this it is sometimes called the Japanese butternut. It is the least desirable of the Japanese group and should not be planted except where the cordiformis type will not grow.

CHINESE WALNUTS (*Juglans regia sinensis*).

The Chinese walnut is being grown experimentally in the northern part of the United States and has been tried at only one place in Canada, e. g., in the grounds of G. H. Corsan, Islington, Ont. The tree is reported to be fairly hardy at the Arnold Arboretum, Jamaica Plains, Mass., and should be sufficiently hardy for southern Ontario. It is believed that the Chinese walnut will prove to be hardier than the English walnut and it may have an important place amongst the trees in the northern part of the United States and in Southern Canada. The nuts are quite large and have a shell which is thicker than the English walnut but not nearly as thick or hard as the native black walnut. The kernel generally has a fine flavour, being almost as good as the English walnut. Nuts of this species have been planted at the Ontario Agricultural College, Guelph, and at the Experiment Station, Vineland, Ont., and it is expected that trees will be hardy enough for our climate and produce nuts which will be as good as the Persian walnut.

THE SWEET CHESTNUT (*Castanea dentata*)

The sweet chestnut is found growing naturally on sandy ridges in that part of Ontario extending from Toronto to Sarnia and southward to Lake Erie. At the Central Experimental Farm, Ottawa, there is a fair sized tree and near Newcastle there are a few fine specimens.

It grows to a large size, sometimes reaching a height of one hundred feet and a diameter of five feet at the base. When grown in the open it forms several heavy branches and makes a broad rounded crown, but when grown in a dense stand it makes a tall, straight tree.

The native chestnut is subject to a fatal disease called chestnut bark disease. This disease is not known to occur in Ontario, but there is no assurance that it will not appear and, therefore, the planting of this tree is attended with some risk.

A dwarf type of chestnut has been reported from east of Ottawa in the Ottawa valley. The tree is about fifteen feet tall and produces a small burr containing only one nut. I have not seen this tree so cannot vouch for the accuracy of the above statement.

EXOTIC SPECIES OF CHESTNUTS.

Inasmuch as very few of the Chinese, Japanese and European chestnuts have been planted in Ontario very little can be said regarding their behaviour. Dr. Sargeant reports the Chinese chestnut (Castanea Mollissima) as being hardy at the Arnold Arboretum and therefore it should be adapted to southern Ontario. The Japanese chestnut is also quite hardy but is susceptible to chestnut bark disease. A few Japanese chestnut trees are growing near Fonthill, Ontario, and have borne some good crops. The tree is a small, spreading grower, comes into bearing fairly early and bears quite heavily.

THE HICKORIES.

There are four species of Hickory native to Canada. The shagbark, the bitternut, the pignut and mockernut.

The shagbark hickory (Carya ovata) is the chief one of value for the production of edible nuts. It is confined to the St. Lawrence valley from Montreal westward and along Lakes Erie and Ontario for a distance of 40-50 miles back from the shore. It reaches a height of fifty to ninety feet and a trunk diameter of one to three feet and grows best on deep, fertile loams.

BITTERNUT HICKORY (*Carya cordiformis*)

This species has a wider range than the shellbark and is found in southwestern Quebec and throughout Ontario from the Quebec border to the Georgian Bay district. It grows best on low wet soils near streams but is also found on higher well-drained sorts. There are two fair sized trees on such a soil on the O. A. C. campus. This species may prove to be of value as a stock for grafting with the shellbark kingnut and some of the good hybrid hickories.

The mockernut (Carya alba) and the pignut (Carya glabra) occur along the north shore of Lake Erie and along Lake St. Clair.

The mockernut is not of much value as a nut tree but the wood is considered to be superior to other species of hickory.

The pignut is generally a small tree which produces nuts of variable size, form and flavor. The kernel may be bitter or it may be sweet and the nuts vary from round to pear-shape.

THE HAZELS.

There are two species of hazels native to Canada—the common hazel (Corylus Americana) and the beaked hazel (Corylus cornuta). The hazels have a wider range than other nut-bearing plants in Canada, being found in almost every province from Nova Scotia westward to British Columbia and as far north as Edmonton in Alberta and Prince Alberta in Saskatchewan. In Ontario the beaked hazel grows as far north as Hudson Bay and in many other parts the common hazel grows very abundantly and bears heavily. In Norfolk County it is very common and in places almost covers the roadside in the little traveled sections. Dr. N. E. Hansen of Brookings, South Dakota, has made a few selections of the common hazelnut found in Manitoba and is now propagating the best of these for distribution.

A few filberts have been planted in Ontario but have not done very well. The growth of wood has been good but little or no nuts have been produced. In Guelph there is a filbert about fifteen feet tall growing on the grounds of J. W. Bell, but like most other filberts in this province it has not yielded nuts.

THE BEECH (*Fagus grandiflora*)

This tree grows in the hardwood region from Nova Scotia westward to the western end of Lake Superior.

On suitable soils it attains a height of eighty or ninety feet and a diameter of four feet. The nuts are much appreciated by old and young, but on account of the slow rate of growth and the irregularity of bearing very little has been done to plant this tree.

THE ALMOND (*Prunus amygdalus*).

Almonds have been tried to a limited extent in the warmer parts of Canada but only with indifferent success except on Vancouver Island. It is possible that a satisfactory strain will eventually be found that will extend the range of this desirable nut-bearing tree.

Introduction of New Species

In addition to the efforts to gather data regarding nut trees I decided to introduce some good exotic species for trial with the hope that some of these might prove hardy enough for our climatic conditions. I thought that northeastern Asia would be the most promising region from which to obtain nuts for planting. Therefore, I wrote to the Mission Boards of the Methodist, Presbyterian and Anglican Churches and obtained the names of their missionaries in those fields. I then wrote to several of these missionaries and outlined my programme and asked them to send me samples of the best nuts growing in their respective sections. Here again I received great encouragement and assistance. Several packages of fine chestnuts and walnuts were received from China and Japan. Some of these nuts were planted at the College and the remainder were distributed throughout the province to interested parties. Owing to the length of the period between the gathering of the nuts and their arrival at Guelph many had lost their germinating power, consequently I only succeeded in getting ten walnuts to grow and failed entirely with the chestnuts. However, we may succeed in germinating a few more walnuts after a winter's frost.

I am aware that we might not get as good nuts from these plantings as the parents were, but it is also possible to get a real good tree which would be hardy enough for our climatic conditions. Should we succeed in this endeavor it would be a desirable acquisition and a great improvement on our native black walnut.

Improvement of Native Trees

Attempts were made to improve ordinary black walnut trees by grafting. Scions of the Persian walnut and the Japanese walnut were obtained and grafted onto some of the seedling black walnuts planted by myself years ago. I regret to state that in this phase of the work I was greatly disappointed. Not one scion grew but the stock trees grew amazingly after being cut back and would have been fine for budding this summer if I had been able to get the buds at the proper time.

Educational Work

An attempt has been made to bring before all our students at the O. A. C. the advantages of paying more attention to nut culture. These lectures have always been well received and almost invariably have aroused special interest in the minds of those who are horticulturally inclined.

Addresses on nut culture have been given to Kiwanis Clubs and Horticultural Societies and articles have been written for the agricultural and horticultural press.

A small bulletin is being written and it is hoped that it will be available for distribution in a short time.

Plans for the Future

The activities outlined above will be continued on a larger scale and in a more thorough manner, provided I can get the necessary funds to carry on the work. The search for superior trees and bushes will be continued and nuts from good trees in China and Japan will be introduced in much greater quantities for test purposes. The conversion of poor or ordinary native nut trees into superior trees by grafting will receive special attention.

In this way, ladies and gentlemen, I hope to attain the ideal of all true horticulturists, e. g., "To make our country more beautiful and fruitful and thereby help to serve the aesthetic and physical needs of our people."

* * * * *

DOCTOR MORRIS: Mr. Chairman: Canada is the next country in which great developments in all of the branches of science will occur. It is to develop, of course, in our present cultural period and I hope this movement for the development of nut culture in Canada will keep pace with the other developments.

I want to speak about one point of Mr. Corsan's. Game breeding can go very well with nut raising. Wild geese will graze like sheep, they will keep the grass and weeds down, and after they are ten days old they need no feeding at all until winter comes. They will graze like sheep, live out of doors like sheep, take the place of sheep, and will return to the land immediately valuable fertilization.

The pheasants Mr. Corsan spoke about are tremendous destroyers of insects. I have had pheasants in my garden this year and the other morning I looked out of the window and saw a pheasant in the midst of a nest of fall web worms. The pheasants will destroy insects of every sort. The only difficulty is that where there are rosebugs in abundance they will kill young pheasants.

I hope every one will take a copy of this "Game Breeder" that Mr. Corsan has left on the table. The subscription price is very small and we may profitably add game breeding of certain kinds to our nut breeding with benefit all around.

MR. BIXBY: Mr. President: There are some points brought out upon which I could throw some light. I have some specimens of Juglans mandschurica which were sent by E. H. Wilson from Korea. I also have a young tree growing that is apparently larger leafed and with thicker shoots than even Juglans cordiformis. The nut is rougher than the other.

I had the privilege of talking to Doctor Wilson regarding his travels in Japan, particularly in relation to the Japanese walnuts. He tells me that Juglans sieboldiana is a wild tree he has found all through the Japanese islands, from the southern part of the northern island Yezo to the mountains of Kyushu, the southern island. He says that Juglans cordiformis is a cultivated tree found in only three or four provinces in central Japan where the walnuts are cultivated. He also tells me he has never seen any of the so-called Japanese butternut type with the rough shell.

I devoted some time three or four years ago to finding out what this so-called Japanese butternut really was. I could never find any instance of where Japanese walnuts, either cordiformis or sieboldiana, had been imported from Japan and planted here and trees grown from them, where those trees had borne rough-shelled nuts like butternuts. In every case where I found any trees bearing those so-called Japanese butternuts they were grown from nuts, Japanese walnuts, which had been grown in this country. In a number of instances I was able to find that the nuts which were planted were smooth-shelled nuts, either sieboldiana or cordiformis.

When they were planted and the trees grew they bore these rough shelled Japan nuts. In a number of instances I was able to find native butternut trees not far away.

The other question was about the varieties of the American hazel. We have here specimens of the best variety which we have found, the Rush hazel. The gentleman who asked about it may see specimens on the table. I believe that will be commercially valuable.

THE PRESIDENT: I think you have all enjoyed Professor Neilson's address quite as much as I have. I wonder, Professor, if it would be agreeable to you that we, as an association, should communicate with these people who answered your questionnaire, inviting them to membership in this association.

PROFESSOR NEILSON: Mr. President, I think that would be an excellent suggestion, and I would be very glad indeed to prepare a list of those that I know are interested in nut growing, and also give you a list of the names of people who gave me exceptionally good replies.

THE PRESIDENT: That's fine. That's perfectly fine.

PROFESSOR NEILSON: Yesterday when you were talking about a membership campaign it occurred to me that it might be well for me to write personally to several people whom I know are interested in nut growing, asking them to join.

As a matter of fact there is one gentleman in southwestern Ontario who suggested to me that we form a Canadian branch of the Northern Nut Growers' Association.

THE PRESIDENT: Don't do it. Just let us all be one.

PROFESSOR NEILSON: I think that's the better way to do it.

THE PRESIDENT: Thank you very much. Is Mr. John Watson here?

MR. OLCOTT: He asked me to state in his behalf that he really didn't have much to say, he noticed your program was pretty well filled up, and he asked to be excused. I hoped Mr. Watson would say something here, but what would be more important would be for him to speak before the nurserymen and induce them to take more interest in our work. Mr. Jones is here and Mr. Watson was here. Of all the nurserymen in this nursery center here that is the only representation.

Nursery catalogues list seedling trees for the most part. One nurseryman wrote me the other day saying he was continually receiving requests for nut trees but he couldn't supply them and knew nothing about them. He asked me for a list of nurseries growing them. Nursery nut trees are not being produced in very great quantities except by Mr. Jones, and they are unlisted in the nursery catalogues, or only listed in an incidental way, very much as though they were tacking on something in the way of citrus fruit, or something of that kind.

A subject that this association might well take up in the enlisting of the nurserymen's interest in this work. Mr. Brown, by the way, of Queens, New York, was here last night. There was a third one here, the head of a very large nursery down there. I talked with him. He was here with Mr. Dunbar. He was interested mildly but not from a practical point of view. I don't know what is the reason for this lack of interest. I thought maybe Mr. Watson could tell us.

THE PRESIDENT: This thought occurs to me in connection with Mr. Olcott's remarks, that it might be desirable for us to send a representative from this association to the annual meeting of the national nurserymen, and let such representative put before the nurserymen the possibilities of making the growing of nut trees in their nurseries a real feature.

MR. SPENCER: Mr. President, several years ago when I first became interested in nut raising I wrote to the University of Illinois which has really one of the great agricultural schools. It is especially famed for its soil fertility studies and for engineering. I asked them what they were doing in the way of spreading information in regard to nut trees, and if they could give me a list of persons from whom I could purchase reliable stock. To my amusement they said they had no list of nurserymen who produced nut trees. I wrote back to them and said that it seemed to me that in a country which is a nut country they ought to know the products of their own state, and I sent them a list of the people from whom they could get trees.

Now I think it would be good policy to send information to the various agriculture schools, giving them what we know of their particular territory based on our experiences, and also send this information to the farm bureaus.

THE PRESIDENT: Mr. Olcott, what do you think about the suggestion to send a delegate to the nurserymen's convention. You are familiar with the nursery trade.

MR. OLCOTT: That's a good suggestion, Mr. President. I don't know—I had thought of Mr. Jones, who is in the nursery business. It might mean competition for him but I didn't think he would be able to supply all the trees that might be needed. Mr. Jones, by the way, is a regular attendant at the nurserymen's association.

THE PRESIDENT: He would be the man of all men to carry the message and I am sure that he would be very glad to.

MR. CORSAN: Mr. Chairman, I have an idea that the best thing we can do is carry on a magazine campaign this winter. Now my wife is a very good magazine writer and can fix up anything in good shape. Send me along all the photographs you can to the Brooklyn Central Y. M. C. A., where I will be located this winter, and on cold, wet days and odd days I don't work, why, we can get up some magazine articles on nut growing.

THE PRESIDENT: It affords me great pleasure to introduce Mr. Bixby.

THE EXPERIMENTAL NUT ORCHARD

WILLARD G. BIXBY, Baldwin, N. Y.

We have heard much about the desirability of the experimental nut orchard and the association has repeatedly urged the planting of such by each one of the agricultural experiment stations in the country. These have been advocated in order that we might learn of the behavior of the fine varieties of nuts that we now have under varying conditions of soil and climate, and in this way accumulate the experience out of which to make positive recommendations as to the species and varieties that might be planted in any given section with reasonably assured prospects of success.

The association has been criticised, sometimes a little harshly I have thought, for the lack of specific planting recommendations, for, as a general rule, that was what those interested have wanted. They did not want to be experimenters; they wanted to plant varieties and get reliable estimates of the returns that might be expected and information as to the returns that similar plantings have shown. Indeed the statement has been made that, unless the association could give this, it could not hold its members and would largely fail in its mission.

That it has not until recently made any very specific recommendations of this character is to my mind an evidence of wisdom. There is a legend told of King Canute whose courtiers flattered him by telling of his power, not differentiating between the immense power he did possess from that which he did not, and who persuaded him to try it on the rising tide. The King learned a lesson by the test that he never forgot. Had the association attempted to make very definite recommendations before it could point to specific instances where things had been done it would almost certainly have failed as signally as did King Canute.

It is not because it did not realize the value that such recommendations would have, but because it did realize that the experience necessary had not been accumulated before it could safely make them. It is only through experience that recommendations worth while can be made, and it is because of the need of accumulating this for the various sections that the association has advocated the planting of experimental orchards.

It is encouraging to note that while these are not being planted as rapidly as we would wish, the work is going on steadily and we are continually learning of new plantings. Some of the older orchards are now giving us their experience. The oldest plantings are those of Mr. John G. Rush, West Willow, Pa., consisting largely of Persian walnuts, and of Mr. E. A. Riehl, Alton, Ill., consisting of chestnuts and black walnuts.

Mr. Rush's orchard has given us an American hazel, the Rush, the best native variety that we have and which seemingly has commercial value. It has also shown us that the nuts on a young grafted hickory tree, a Weiker, are considerably larger and crack easier than the nuts from the parent tree, and that the English walnut will grow and bear when grafted on practically every species of walnut, black walnut, butternut, and Japan walnut, and it seems likely that this orchard will be a source of knowledge for us for many years to come.

A number of others have been started some of which are beginning to give us evidence of value. Probably more problems have been solved, particularly those relating to propagation on Dr. Morris's and Mr. Jones's than any others so far. Dr. Deming is giving us evidence on grafted hickories of a large number of varieties and Mr. Littlepage's and Mr. Wilkinson's orchards are giving us evidence on pecans. There are also a number of others still too young to give us much information. Mr. Riehl's orchard of chestnuts and black walnuts has gotten beyond the experimental stage and is now a commercial success.

I had a desire to establish an experimental orchard when living in Brooklyn, before I owned any land on which to plant trees, and I bought and set out trees on the land of three relatives before it was possible to set any on my own land. The principal thing gained from these early plantings was experience and the principal things learned were things not to do, for none of the trees then planted are alive today. Buying my present place in Baldwin, at the close of 1916 gave me about three acres available land and since then I have been gathering grafted, budded or otherwise asexually propagated trees of all the fine varieties that we have. At present there are on my place some

14 varieties of black walnuts
2 " " butternuts
12 " " Persian walnuts
4 " " Japan walnuts
14 " " chestnuts
20 " " pecans
25 " " hickories
23 " " hazels
4 " " almonds

The only nut tree, native in the northeastern United States of which I have no named variety is the Beech.

In addition there are seedling trees of four additional species of walnuts, seedlings from several hybrid walnut and hickory trees, besides some thousands of seedling nut trees of practically all species for use as stocks.

I have for the past two years been gathering selected native hazels from the various sections of the United States taking care to select bushes that bore nuts that were relatively large, thin shelled and fine flavored.

Inasmuch as the hazel is native all over the country, and just how to get bushes that bear the best nuts is not generally known, I will tell how I do it, hoping that many others will seek out the best hazels in their section and get them into cultivation. I provide myself with a cloth about as large as a large handkerchief, a number of wooden labels, some paper bags, a hand vise, a pair of calipers, a scale and tools for digging plants. A spade or round-nose shovel is about the best tool for digging the plant and frequently a hatchet, axe, mattock, or bar is required in addition in case the hazels have to be dug away from among the roots of large trees or from among stones of considerable size.

When a plant is found where the nuts look promising the branch on which nuts are to be examined is marked temporarily by throwing the cloth over it. A nut is then carefully cracked in the hand vise, taking pains to extract the kernel whole. This is then calipered with the calipers, set at a minimum size desired. If it is undersize the bush is rejected and another sought. In measuring the longest dimension is the one considered. The minimum size depends on the section from which the hazels are being taken, no kernel which is less than 3/8" in its longest dimensions being considered. While sometimes it requires a good deal of hunting to accomplish it, I have never had to take bushes where the kernel was smaller than this and it is seldom that it is necessary to take those where the kernel is as small as this. In many instances it is very much larger. If the size is satisfactory the kernel is then eaten, only those bushes having well flavored kernels being taken. If all tests are satisfactory the cloth is removed and a wooden label put on the bush which is then dug. The nuts are removed from the bush and put in a paper bag labeled the same as the bush; the bush is cut back to about 6" in height and then put in a sack or other convenient means for keeping moist till it can be put into the ground.

The gathering of the above mentioned trees in a small compass and closely observing them have enabled me to make a number of observations which may be of interest.

Fertility of Soil: The importance of this was shown strikingly in the case of a lot of Japan walnuts received in the spring of 1918. They were quite large and seemingly never had been transplanted and were dug with small roots. For lack of a better place they were set in sod ground which had not been cultivated or fertilized for many years. They eked out a miserable existence during the years 1918 and 1919. During the spring of 1920, I put chickens in that patch and an improvement was noted that year but this year practically every tree has grown six feet or more. The manure of the chickens and the thorough cultivation of the soil caused by their scratching have certainly worked wonders. While I do not minimize the effect of clean cultivation, I am inclined to believe that abundant plant food is the really important thing, for a goose watering pan under a tree pushes the tree along at a remarkable rate, and geese never scratch. They do keep the grass closely cropped, supply an abundance of manure, and the watering pan puts the plant food where the trees can get it.

Pruning: The importance of severely cutting back was strikingly shown this spring. A butternut raised from a nut in a lot of "Virginia" butternuts, bought in a nut store and which had outgrown every other tree in that lot and which I believe to be a Japan walnut butternut hybrid was transplanted this spring. Care was taken to get as much of the roots as possible and practically all were obtained; good soil was taken to fill in around the roots. Over the half of the branches were removed but the five highest ones were not shortened. This tree has not grown as well this year as some others not as vigorous and set in poorer soil but where all branches were cut back severely. Were this the first time I had noticed this, I might have considered it an isolated case, but the need of severe pruning was emphasized even in this case where I hardly expected it to show on account of the tremendous natural vigor of the tree which was transplanted, and the ideal conditions under which the transplanting was done.

Varieties: I get frequent requests from persons who want to know the best variety of this nut or that nut with the idea of planting only the best. The thought behind the request is one with which I heartily sympathize, but the method of accomplishing it that the enquirer has in mind will not accomplish it. The failure of most plantings of European hazels has, it has been thought, been due more to lack of proper pollination than to any other one reason. This year several varieties showed abundant pistillate flowers but there was but one European variety where it was not evident that the staminate flowers had suffered greater or less winter injury. This variety, Grosse Kugelnuss, shed an abundance of pollen when pistillate flowers of several of the others were receptive and there are nuts on three or four varieties for the first time. I believe that the success of Messrs. McGlennon and Vollertsen in fruiting the European hazels would have been but a fraction of what it has been had they not set out the large number of varieties that they did. In setting out nut trees at the present time as large a number of varieties as practicable should be planted. Later we will have the accurate observations that will enable us to select a few and feel sure of getting good crops of nuts, but we cannot do this now.

Chestnuts: While the blight is all around me and several of my trees have been killed by it, there are enough left to produce nuts of nearly every variety and I see no reason yet to change my belief that, by watching, cutting out blight and occasionally setting out new trees, chestnuts of nearly every variety can be grown and fruited in the blight area.

Age of Bearing: My experience would seem to show that grafted or budded nut trees are as a class not slow in coming into bearing provided they have had good care. I have had Lancaster heart nut trees set out in the fall bear next spring and have had hand-pollinated English walnuts bear the third year. Apparently a year or two longer will be required before they bear staminate flowers. Walnut trees certainly appear to bear fully as young as apple trees, in fact sooner, as a class, than apple trees which I set out at the same time that I did walnut trees. Pecan trees appear to take about two or three years longer than walnuts and hickories several years longer than pecans. On the other hand top-worked hickory trees bear about as soon as young transplanted Persian walnuts. Hazels with me have taken about as long as Persian walnuts but I think that they are more rapid in most instances. The soil of most of my place is quite heavy, walnuts, pecans and hickories doing finely. I am inclined to believe that a lighter soil would be fully as good if not better for hazels.

Stocks: The varying rapidity of growth of trees of the same variety has been noticeable and has caused more than passing notice for one can not help thinking that such varying rapidity of growth would be likely to cause equal variations in bearing. It would seem as if this must be caused by the variations in the stocks for the scions all come from the same tree. Inspection of seedling trees has shown that some grow much faster than others. If normal growth trees are considered, trees making less than half this are numerous and those making double are rather rare. Apparently we have in seedling stocks enough variations in vigor of growth to account for the variations in growth noticed in grafted trees of the same variety. Mr. Jones tells me that he expects to discard nearly 50% of his seedlings because not vigorous enough to bud or graft. Then there are some trees which seem incapable of taking grafts or buds. It would seem very desirable to select rapid growing stocks that will take buds and grafts readily and use those but this will mean working out means of propagating them by cuttings, layers, or some asexual method and these have not been well worked for nut trees, other than hazels, although some work has been done on it.

The above conclusions are largely from the limited observations I have made on my small place. None are very new for I believe I have heard all of them advanced before, but observing them myself has fixed them in my mind in a way that they could not have been otherwise. Many of them have been corroborated by others. For example, Mr. Jones has shown me walnut trees of the same size set out at the same time, some severely pruned and others not, where the severely pruned ones in two or three years had so far outstripped the others as to make it very noticeable and it seemed as if the difference in vigor would continue. On the other hand it is possible that there may be points where the experience of others differs from mine.

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THE PRESIDENT: There is one more address this morning. That is by Doctor
Morris, the subject being, "Pioneer Experience and Outlook."

DOCTOR MORRIS: Mr. Chairman, ladies and gentlemen:

Lord Byron said that the reason why he did not commit suicide was because he was so curious to know what was going to happen next. For any one to do pioneer work in almost any department of human activity there are two essentials: First, he must be more or less stupid and not read the handwriting on the wall; and in the second place he must be very obstinate and persistent. Given those qualities one may succeed in pioneer work in almost any department of life.

Something over twenty years ago I had the idea of putting upon my country place every kind of American tree that could be grown there. I planned to occupy a little time away from professional work and attend to this. As I began to acquire information the subject grew so rapidly that I found it would be necessary to give up my profession wholly and employ several assistants in order to carry out this idea. Consequently I cut down my ambition to include only coniferous and nut trees. This study in turn grew so rapidly that I found it necessary to cut out everything except nut trees, and then I found that one might devote his entire life to the subject of hickories alone to the exclusion of all other occupation.

In the beginning of the development of my nut trees there were failures continually and it became interesting. Lord Byron found it interesting to live in order to see what was going to happen next. My failures were so interesting that I was very curious to know what was going to happen next. I started in with a very large lot of shagbark hickory trees. I had them grafted for me in the South. I think I expended something like $250 for that lot. I had it grafted upon the common hickory stock of the South. They lived through the winter, the summer, and the next winter, but in the spring, following a few warm days and a freeze, the bark of every one of those common stocks exploded, fairly, and the entire lot was lost, not one tree lived.

A great many trees that I brought from farther south, from California and from the Pacific coast, all died. I learned then that the climate there will allow trees from western Europe to grow because they have the Japanese current furnishing similar conditions of climate; that trees from that part of the country would be mostly failures here in the East; and that trees for the East should come from northeast Asia where climatic conditions are similar.

I learned also that trees from a distance, not accustomed to our soil and climate, would not adapt themselves readily, and it would require long selection and breeding to acclimatize or adapt to our soil trees which were developed under differing conditions. Out of a large lot of things that I got from Chili, hoping that their altitude would correspond to our latitude, nothing grew. Consequently by elimination of things that would not live I gradually arrived at the conclusion that it is best for any locality to develop the species, or a like kind of tree, which belong to that locality. Well, they say, how about the prairies that are treeless? Of course we have there to deal with a question of fire that from time immemorial has swept the prairies covered with grass and has been halted only when it reached the regions of established forests; so that on the prairies I have no doubt we may have great groves of nut trees flourishing. In my locality the trees that are indigenous are the ones which do the best, and that is the line for perseverance.

Then I took up hybridization. I found there were many disappointments. It was difficult to be sure of securing reliable pollen and of getting it to the flowers at the right time and surely, so that we would have good hybrids instead of parthenogens which sometimes develop as the result of the female not making fusion with its mate.

On one occasion I remember I covered a lot of branches with large bags for pollenization, and going out a few days later to add pollen I found a wren's nest with two eggs in one of my bags. Now if a wren could lay two eggs in one of those bags the cross-pollenization was not likely to be a success. In this work, however, I find that we have a tremendous field opened up and one which might yield particularly to the ladies. It is very pretty work, it is nice work. It includes idealism, speculation, the idea of developing new trees, or trees that one has never seen before. After many failures in hybridizing I find now that by following rules it is simplified very much. Almost any one who is persistent enough may learn eventually to hybridize very easily.

The question of labeling trees and of keeping track of different specimens was one that gave me many disappointments. I would lose the labels, lose the records, so I was not able to tell truthfully about trees when visitors came to ask me about them. I know in one lot where I had a lot of hybrid trees, each one marked with a stake and number, the cow of a neighbor got over the fence into the field and the boy who came after that refractory cow found that to pull up those stakes gave him very convenient objects for throwing at the cow, and my labels were all hybridized.

This sort of thing was the kind of disappointments that I had in early experiences in growing nut trees. It is very essential, however, to keep good records and I find now that the best way is to use a galvanized iron rod with a metallic tag stamped with a machine and fastened on in such a way that it will not be injured by any sort of use. These galvanized rods, galvanized spring wire, are very durable if one is careful about placing them on the trees. That experience in keeping the labels was one which was very disappointing at first, but the question has now been finally settled.

The number of animals and birds that like a good thing is perfectly surprising, and in trying to raise my seedling nuts I have had great difficulty and have had to take up a new department of natural history in order to study the habits of rodents and of the birds. The crows have been, perhaps, the worst enemy, after the field mice, of the seedling nuts that were planted out in the field. But the crows may be kept away if we put up bean poles with a simple cotton string stretched between them at a distance of twenty-five or thirty feet. One of my friends who took my advice said that it didn't work, that he had not only put up the string but had fastened a piece of tin onto the string. That is just where he made a failure. The crows sized up the situation immediately. They sat on the fence and looked it over and made up their minds that those things were not meant for them, and then they went in and destroyed his grain. But a simple string between the poles will keep the crows guessing, and that alone will suffice to keep them out of the grain, nuts, or anything else.

These are a few rambling remarks which come to my mind, but still they belong to the experiences that we have in getting things under way in our experimental work. As to the outlook, there is no doubt whatsoever but that any man who is interested in the subject, who loves trees and loves plants, can manage all the problems. We shall eventually have horticulturists and amateur gardeners who will raise all of this great new food supply without difficulty.

We must now look for new food supplies. Wheat, grain, corn, and the other cereals are not going to supply this country indefinitely but the nut trees will. It is absolutely impossible to have over-population. It can't be done. Over-population as a social matter relates wholly to the habits acquired by people in using established kinds of food, but with the development of the nut trees, which furnish the appropriate starch, oils, and essentials of human diet, the danger of over-population becomes absolutely nil. We can not have over-population anyway, because nations of people reach cultural limitation, just as breeds of cattle run out, just as a breed of dogs runs out, just as a breed of any cultivated animal runs out. We are sure to do that. In all of our cultural periods we are sure to rise to a certain point, decline, and go out, and somebody else will follow, so that we never can really have over-population excepting as a matter of choice rather than one of necessity. On the question of food supply we may avert over-population by taking up something new to meet the conditions. That new thing right now is the development of the nut trees which furnish all of the food essentials and will take away any fear whatsoever of any over-crowding of the people of this country.

EVENING SESSION, SEPTEMBER 8th, 1922

The convention was called to order by the President at 8:30 o'clock P. M.

MR. SPENCER: Mr. President: I have an idea I would like to present on behalf of the ladies. Quite a number of years ago I was entertained at dinner on the plantation of Mr. John Todd, St. Mary's Parish, Louisiana. It is on the banks of a stream lined with live oaks at a point where Evangeline and the Arcadians passed on that trip to the next county which is known as Arcadia. The whole country round there is full of reminders of the Arcadians.

Mr. John Todd has several thousand acres in his plantation and four thousand acres are in sugar cane. When it came to the dessert a beautiful two-storied white cake was placed on the table. After eating it I turned to Mrs. Todd and said, "I dislike very much to comment on a lady's cooking but I hope you will excuse me if I ask you what this cake is made of. There is something peculiar about it that I do not recognize." "Well," she said, "while you and the other gentlemen were down inspecting the land that you came to see, I had the boys go out and rattle down some pecans. They cracked them, picked out the meats, and I put them in the oven and dried them. I knew that they would not dry out ordinarily in time for my meal. I then ran them through the meat chopper and chopped them as fine as I could and then I put them through a very fine sieve. The parts that were fine enough to go through I put in the flour of the cake, the rest I put in the filler between the two layers of cake and in the frosting." It was one of the most delicious cakes I ever tasted in my life. With that recipe you can make a white cake in about three minutes, fill your flour and your frosting with pecans and you certainly will have a feast for the gods.

DOCTOR MORRIS: Mr. President, the committee on resolutions has referred matters to the secretary for action.

THE SECRETARY: It was the duty of the committee on resolutions to prepare the resolutions on the deaths of Doctor Van Fleet and Colonel Sober, copies of which are to be sent to their families. The committee not having had time to meet that task has been assigned to the secretary, who will be very glad to carry it out to the best of his ability.

The other and more important task of that committee was to take action on the suggestions made by the president in his paper in regard to increasing the membership of the association. As it has been impossible to take such action in the committee I propose that we now take up consideration of that matter as a committee of the whole.

I would like at least to say that Mr. Jones has offered to the association five hundred nut trees to be given as premiums with new memberships. I think Mr. Jones said that they included Stabler walnut trees, Chinese walnuts, and what others, Mr. Jones?

MR. JONES: Chinese English walnuts, or Chinese Persian walnuts, Mayette & Franquette English walnuts and Stabler black walnut seedlings. I have an idea the Chinese walnuts would be the most attractive.

THE SECRETARY: They would all be seedling trees, of course?

MR. JONES: Yes; they would all be seedling trees. We would put them up and mail them out.

THE SECRETARY: Think of what an extraordinary, generous offer that is on the part of Mr. Jones, to contract to send out five hundred nut trees to as many new members, dig and pack and send them out!

MR. JONES: Well, growing the trees doesn't cost very much. Of course packing the single trees will cost more than the trees but we are glad to do that if it will help out.

THE SECRETARY: I know that to some members this premium offering for new members does not seem an advisable thing; to others it does seem a good thing to do. Perhaps that would be a good question to debate at the present time.

THE PRESIDENT: I think it is a good idea, Doctor, to the end of getting a thousand members this year?

MR. JONES: Set aside a thousand trees if you get a thousand members.

MR. OLCOTT: Mr. President, Mr. Jones said the cost of growing the tree isn't so much, but the packing and mailing is something. How would it do to offer the tree at cost of packing and mailing—fifty cents, or so? I suppose the value of that tree would be about a dollar, grown, packed and delivered. Suppose we made it twenty-five, thirty-five or fifty cents, something to cover the cost of packing? Would that not make it——

MR. JONES: (Interrupting.) We don't want anything for packing.

MR. O'CONNOR: Mr. President, If you make a bonus of that kind, which is very generous of Mr. Jones, I think it would be appreciated by some, but others would say, "Well, a thing which you get for nothing isn't worth much." This gentleman behind me here says, "Make it cost a little something, which would make it more attractive." How about putting the membership up a little, so as to cover the cost of mailing.

MR. JONES: I would say that the association was giving these trees because it wants them tried out for new varieties.

MR. SNYDER: The fact that our association offers these trees ought to be enough to establish their value. A new member would appreciate receiving something in this way. The largest horticultural society in our country is the Minnesota Horticultural Society. They have followed the practice for years of giving to each new member a tree of some kind, scions or plants of new fruits, and it has been a great success in building up their society. I doubt not that it will be here.

MR. SPENCER: I'm heart and soul in favor of the movement for better nut trees. I'm tired of having trees planted that produce nothing but litter, and for the small boy to keep breaking all the time instead of going fishing. As I said the other day through the committee on trees of the Bird and Tree Club of Decatur we have placed in that city a hundred and fourteen nut trees. I believe that I can go to the different purchasers and say that this association is anxious to increase the knowledge of the people as to the value of nut orchards and nut trees for food and shade and I can get them to become members. When those subscriptions are sent in send the names to Mr. Jones and have all the trees put in a little package and sent to me. Then I can deliver them and Mr. Jones will only have one package to do up.

I believe by a little effort among our friends a great deal of good can be accomplished. For instance I stated here that I was going to buy a subscription to the American Nut Journal and send it to the Maitland County Farm Bureau. Likewise, I hope I can get the Board of Education or the Public Library, which purchased twenty-eight different trees to put in the library grounds, to subscribe for the Nut Journal and take out membership. It won't be very hard, I should say, to get fifty or sixty new members in Decatur without going out and making myself a regular canvassing agent. I have got a great many friends there and I know that upon my representation they would be very glad to take out a membership and get a tree. Anybody can go and plant a Carolina poplar or a soft maple, or a basswood, or an elm, but his lot won't look different from any other. If all the ladies in town dressed in the same calico and the same cut you would not know whose wife was who. This idea of having all the yards, all the lots, all the places look alike, is wrong. You might as well have your home look distinctive and if you will take that idea, to have your place stand out as a place distinct in horticulture on your street, in your block, or in your city, you can appeal to civic pride. You must appeal to something besides dollars and cents. You must appeal to their public spirit, their civic pride. Then you can get them interested. A great many people are proud of their city and there are a great many people who can very easily say with Paul, "I am a citizen of no mean city."

Keep at it and take advantage of this offer of Mr. Jones. I believe by following those lines you can very easily go out and get five or ten members apiece.

MR. BIXBY: I don't want to throw cold water on any idea that is going to increase the membership but it seems to me that there are some objections to the proposed plan. In the first place the association has gone on record as favoring largely the planting of grafted trees. Now on the proposed plan the minute we get a new member in we have to send him a seedling tree. That does not seem to me the best thing to do. In the second place, I have had a good many years' experience in merchandising and it has always worked out with me that people do not much appreciate what they get for nothing. You can do this if a man is going to buy a certain kind of goods, by offering him an inducement, giving him something for nothing you can make him buy more than he would otherwise; but if a man who has never had a certain kind of goods, generally speaking you can't sell them to him by offering him a prize with them.

In the case suggested by Mr. Spencer, where a member working in a certain location could club with others and get several new members, why that hasn't the same objection. I do think that it would be a fine thing if the members in the different sections each agreed to get five or ten members, go after them and get them. I think that would be fine. And if they are willing to be responsible at the end of the year if they don't get them, and pay two dollars apiece for the ones they don't get, why that would help out the treasury.

MR. SMITH: Mr. Chairman, I am rather in favor of the premium plan. In this great state of New York there exists an organization at Geneva known as the New York State Fruit Testing Co-operative Association. In order to get members they offer premiums, a yearly premium. The year that I joined the association they sent me a new apple which had been tried out and found to be a very desirable fruit. They named it the "Tioga" variety. The next year they sent me as a premium twelve new raspberries that had been tested first by the Geneva Experiment Station, a branch of the agricultural college, and then by this association of fruit growers.

Now I don't know how it would operate with others but it was an inducement to me in the first place to get that new apple to experiment with, and the next year it was an inducement to get the twelve new raspberry bushes which are claimed to be the best raspberries grown.

The objection raised by Mr. Bixby seems to be, however, quite a valid one. The organization has put itself on record as opposed to seedling nut trees and it is a question whether we ought to encourage the distribution of seedlings. But in some way or other I'm in favor of the premium plan to attract new memberships.

THE PRESIDENT: Is it not better to plant seedlings than none at all? It is possible that some of the seedlings might be really worth while. Those that are not really worth while can be top worked.

MR. JONES: Mr. President, my idea about the Chinese walnuts and the Stabler walnuts was that if we want to get new varieties we have to get them from seedlings. My plan was to grow these and send them out as extras to people who had sent in orders for other trees. I thought that in that way we could introduce them to those who would take an interest in them. It would take a good deal of land and a good deal of money and a good deal of attention to care for several hundred or several thousand such trees, but you could send them out in that way one at a time and possibly get new varieties superior to anything we have. That was my idea in disposing of these trees. I thought that if the association felt that that would be an inducement for new members we could send them out in that way as premiums. The only difference in the cost to me would be the packing.

MR. SMITH: Would it be possible for the association to take out from this first year's dues sufficient to compensate Mr. Jones for the difference between the value of a seedling and some of the best nut trees, so we could say to a proposed member, "We are giving you something that years of experience have proved to be the very best thing up to date, and we want you to plant this and care for it"? I think he would be more interested if he knew he were getting a tested tree than if he were getting a seedling. The seedling may be a good thing and it may not.

MR. WEBER: Mr. President, we know that in the spring the dry goods stores distribute shade trees, and people carry them all day with the tops tied up and the roots uncovered. You might as well expect a fish to live out of water as to expect those trees to live. If we send the average person a tree he may make it grow but the chances are he will not, so why let him ruin a good grafted tree with his initial experiments in planting a nut tree. On the other hand you will emphasize the distinction between seedlings and grafted trees, because on his coming into the association you will present him with a seedling and explain to him in advance just the purpose for which it is being given. He will then plant that tree. If it grows he can see its performance along side of a later grafted tree which he will buy if he is interested in furthering his nut tree plantings. If he isn't, why, you get his membership fee and he centers his membership around that seedling which he thinks is the finest thing in the world.

Last summer I was talking nut trees to the wife of a rather prominent Detroit man. They have traveled around the world considerably. We were discussing some nut trees which had been sent out. I knew the size of the trees and I didn't laugh, or I sort of saved my face, when she asked me the question, "How many bushels of nuts could we get next year?" I just closed my jaws a while and looked out of the window. I didn't want to dampen her enthusiasm.

THE PRESIDENT: Mr. Tobin, I would like to have your views on the subject.

MR. TOBIN: This offer of Mr. Jones's is of great importance to this association. I have been interested in trees and forestry and plants of all kinds but until the present time I have not been so much interested in the cultivation of nuts. I wish to say that if there is any way I can help this association along in regard to an experimental station or in any way whatsoever, financially or otherwise, if the suggestion could be made I would be glad to hear it.

THE SECRETARY: Mr. President, this association is not opposed to the planting of seedling trees. One of our founders, the late John Craig, advocated the planting of seedling trees in great numbers, for only thus can we originate new varieties. The association is opposed to the dissemination of seedling trees as grafted trees. It does not advocate the planting of seedling trees for commercial purposes or for ordinary home use. It does not advise the purchase of seedling trees for growing nuts. In sending out these premium trees we should send with them a letter distinctly stating that the association does not advise the planting of seedling trees from a commercial point of view, but it does wish to disseminate these seedling trees which we offer as premiums for new members, for the purpose of testing and the possible discovery of new varieties of nuts. It would then be clearly understood. Certainly such seedling trees shouldn't be sent out to give members the idea that we advocate the planting of seedling trees for any other purpose than of possibly obtaining valuable new varieties.

MR. O'CONNOR: Mr. President, I'm a life member of the Wisconsin Horticultural Society which has offered a thousand dollars for an apple better than the Wealthy. We also offer premiums for new members every year. Sometimes it is a seedling apple tree. Among those premium trees may be a seedling which will win the prize. We do not know what the seedling nut tree will do. We may get something from a seedling which is far better than anything we have today on the table before us. Nature is something wonderful and no one can tell you what she will do. Only this last year has what is called the "O'Connor" come out. But we find this O'Connor nut is not hardy enough for certain sections of the country. This Persian walnut before you is a seedling, too you know, from nature.

So it is through seedlings that we are going to get better fruit. I believe that Mr. Jones's offer is a very good thing. But I suggest that we send these seedlings out with the understanding that they are seedlings and that we don't know what they will produce. If the new member will plant them and take care of them (and we should give a little instruction as to how they should be planted) in a few years, seven or eight if it is a pecan, he should see it coming into fruit.

I would like to say that if you will dynamite the hole with a one-half stick of twenty per cent. dynamite, or, if you are afraid to use the dynamite, dig a large hole so as to give these young roots a chance to spread, a grafted tree will come into bearing in three years. I have seen them do it down there with us in Maryland and I believe they will do the same thing anywhere else.

THE PRESIDENT: I would like to hear from Mr. Vollertsen on the subject.

MR. VOLLERTSEN: I haven't a great deal of confidence in seedlings. As a general thing we find all the nut trees are inclined to go back to their original type. If we take our filberts, even the best varieties, the chances are that they will go back to the European type that they originally came from. I have proven it time and again on the farm down there. I don't think it wise for this association to send out seedlings.

THE SECRETARY: Mr. President, in order to bring this question to a head, I move that Mr. Jones's offer be accepted and put in to practice if a suitable plan can be devised and carried out in the estimation of the executive committee.

Seconded and carried.

MR. OLCOTT: Mr. President, I wonder if the suggestion of Mr. O'Connor is clearly appreciated. It was barely suggested in his talk but he did not seem to clinch it at the end. As I understand his idea it was that this plan of furnishing a tree as a premium might well be accompanied by an offer of a prize for results, which would be an added inducement to membership.

THE SECRETARY: I will see that that point is considered by the executive committee.

I wish also to say that Mr. McGlennon, if I understand him aright, has offered to get one hundred members in the ensuing year if the others present will get ten each.

THE PRESIDENT: That's right, Doctor.

THE SECRETARY: I don't know just which comes first, whether Mr. McGlennon is to get one hundred members and then the rest of us to get ten each; or whether we are to get ten each and then Mr. McGlennon is to get the others!

THE PRESIDENT: Well, Mr. Secretary, I have associated with me the champion membership getter. When we can go out and get twenty or twenty-five in a month I think we can go out and get the others. We are all enthusiastic now and happy. We are glad we are here and we are going to do wonders this next year. But I'll wager inside of a week our ardour has materially cooled and it will be getting colder until about a month before the next convention. We are not going to get anywhere that way. We want to get busy immediately after this convention, and if we do there is no reason why we can't have a thousand members by the time of the 1923 convention. I repeat that my office will have a hundred members by the time of the next convention but it is with the understanding that the rest of you co-operate in this movement and that each of you here, and the other members who are not here, be informed and instructed what is expected of them, to get at least ten each.

MR. BIXBY: I don't believe you will ever succeed, Mr. President, in getting each of the other members to get ten members each. If the rest of the members get a hundred between them they would be doing more than we ever did before.

THE PRESIDENT: Yes, Mr. Bixby, but even if the members here get ten each I think if we follow them up closely and keep right after them we can increase this membership to a thousand.

MR. BIXBY: I will agree for one to get ten or else pay the amount in to the treasury.

MR. JONES: We can get ten.

MR. WEBER: Mr. President, I will get ten or kick in to the treasury the money they would have brought.

THE PRESIDENT: That's fine. That's thirty right there. How about you, Mr. Thorpe?

MR. THORPE: I think I can get it.

PROFESSOR NEILSON: I think I can get ten.

THE PRESIDENT: I think there is no doubt about it. Mr. Spencer will get ten won't you, Mr. Spencer?

MR. SPENCER: I will try to.

THE PRESIDENT: Well, you will, won't you, or else you will "kick in" with the money,—$20?

MR. SPENCER: Yes, I think so. Well, if I am going to put in $20 I want to say something more on the subject. If we send out this Chinese tree it would be very easy to put in a slip stating that the association is very anxious to know whether this is suitable for the receiver's particular part of the country. We should tell him that we don't know whether it will grow in Illinois or in Louisiana, and that it's an experiment on the part of the association to learn whether this tree, which is desirable in China, is suitable for his particular locality. We should ask him to please take care of it, watch results and report to the association. Make him sort of a partner in the discovery.

THE PRESIDENT: Pat, you will get ten, won't you?

MR. O'CONNOR: I will promise myself ten.

THE PRESIDENT: And Mr. Tobin, you will get ten, won't you? You said you were anxious to help this association.

MR. TOBIN: Yes.

THE PRESIDENT: Even financially?

MR. TOBIN: Yes, Mr. President.

THE PRESIDENT: Now, you will get us ten members during the year?

MR. TOBIN: Well, I would not promise you.

THE PRESIDENT: But if you don't, if you promise to help us financially, you would "kick in" with the money, wouldn't you?

MR. TOBIN: Yes.

THE PRESIDENT: Yes, sure. Then you get ten members.

MR. TUCKER: Mr. President, I want to ask about the vice presidents from the different states. Are those still in existence?

THE PRESIDENT: Yes. The secretary said yesterday they had been changed every year.

MR. TUCKER: Why can't memberships be also increased through the vice presidents? Put it up to them.

THE PRESIDENT: Joseph A. Smith, of Utah?

MR. SMITH: Mr. President, I will guarantee ten.

THE PRESIDENT: Now, that's an exceptionally fine offer of Mr. Smith, who comes to us from Utah. Just try and fix in your mind the map of the United States and realize where Utah is. Mr. Rawnsley, you will get ten, won't you?

MR. RAWNSLEY: Yes, I will get ten.

MR. TUCKER: I will get ten. Carrie and I together will get ten.

THE PRESIDENT: There are two hundred right there.

MR. TUCKER: My ten went with your hundred. I think we ought to do something through the vice presidents.

THE SECRETARY: The secretary will get up a letter and send it to each one of the vice presidents, stating what was done at this meeting in the way of pledging these new members and asking the vice presidents to do the same, to each guarantee ten members or to turn the money in themselves.

THE PRESIDENT: If we had more money so that some of the officers of this association could get about and confer with our state vice presidents there isn't any doubt but what we could stimulate their interest and get many new members. Of course, ladies and gentlemen, we have got to get new members, that's all there is to it.

MR. O'CONNOR: The more the merrier.

THE PRESIDENT: What is the use? Here we are meeting with a deficit every year. That's all wrong.

MR. OLCOTT: Mr. President, I am glad something is going to be done about the state vice presidents. I also have proposed that the state vice presidents be brought into line. Mr. Spencer has made a very good suggestion for them and that is to encourage friendly competition in dressing up yards, one section against another. If the state vice presidents would use that suggestion in getting new members I believe it would be a good thing. I believe also, as I have said many times that the state vice presidents should be the local directors of a state association subsidiary to this one, that Ohio, for instance, should have an association of Ohio nut growers. If they can't meet then let them correspond back and forth. Certainly the nut growers of Ohio should know each other and be brought in to correspondence. They could do that through an association of which our state vice president would be the chairman or the local president. I am a great believer in organization and I feel that the state vice presidents should amount to something. After the state organization is started by this association in that way, then the members of each association could elect their own chairman, if they wish, and report it to our secretary.

THE PRESIDENT: That is along the line of the suggestion offered by Professor Neilson this afternoon.

MR. OLCOTT: Yes. We could have a branch in Canada.

THE SECRETARY: The secretary will be glad to see that Mr. Olcott's suggestion is incorporated in the letters to the state vice presidents.

MR. JONES: We would be glad to make up a mailing list and turn it over to the secretary if he should want to circularize in making this offer or any other offer for memberships.

THE PRESIDENT: If we could get this thing where it ought to be it is possible that we might be able to induce the secretary to give his entire attention to the interests of the Northern Nut Growers Association. He would have to have a lucrative salary of course. That is one of my ambitions. I am frank to state it here right now.

Then the Northern Nut Growers Association would be the thing that it is supposed to be, the thing that it is not at the present time when we're meeting with a deficit every year. I hope and believe, in fact it must be, that this is the last time we are going to meet with a deficit. We are going to have a good surplus next year or what is the use of going on?

MR. SPENCER: The governors of three or four of the states met in Chicago not very long ago to consider the interests of the states that center around Chicago. The people in Illinois don't know that the Forest Reserve covers sixteen thousand acres and that it has English walnuts growing just as nicely as you have them here. That knowledge hasn't been spread. Also there are people who are propagating nut trees in Illinois and southern Indiana. Now if our vice presidents in Indiana, Illinois, Ohio and Missouri, which is the native home of most every kind of hickory, would get together and go to any one of the central cities of those particular states, call a meeting of their customers in that neighborhood, and spread a knowledge of this association I think that we could build up a local interest that would advertise this organization wonderfully.

You have got to advertise and you must show to the common people who are going to be your members, who are going to be interested in nut trees, that they are valuable; that an ordinary acre of nut trees is worth ten times the value of any crop of wheat raised in Illinois, and Illinois is the wheat country. Before the hard wheat was discovered in Minnesota the whole south half of Illinois was given to wheat. But now so far as white wheat is concerned, and spring wheat, it isn't wanted and the result is that you have got to get something else into that country. Now that wheat country of southern Illinois is a natural nut country. Pecans, persimmons, chinkapins, grow wild all over there, and there is no reason why that land, which can be bought for from ten and fifteen dollars an acre up to twenty-five, according to the improvements, if the oil rights are eliminated, can't be made to produce a hundred to five hundred dollars an acre. If that is so, why not do it?

Today, Illinois has over 11,000,000 bearing apple trees, and they raise just as good apples there as any where, but they haven't got the organization, they don't advertise, and we don't know it generally. If we can organize and distribute our information, get these vice presidents from two or three cities to join with the chambers of commerce and have a meeting down at Evansville, among the nut growers, for instance, the growers of Indiana pecans, and see what they grow, and what they are worth, why then you can get the people interested. You must have somebody that is interested in the propagation of a new idea. Don't get somebody who just comes here for a good time without any desire particularly of learning anything. If he doesn't want to learn we don't want him.

THE SECRETARY: I understand that, in view of the very generous offer of the president to get a hundred new members in the ensuing year, and of the pledge of ten other men to get ten more members, or turn in the necessary amount to the treasury, each of us goes forth from the meeting tonight with the understanding that he is morally under obligations to do what the other members have promised to do.

THE PRESIDENT: It would be a nice thing to give a Christmas gift of a membership in this association and a subscription to the American Nut Journal. A great many of us receive Christmas gifts which are appreciated when received, and maybe for a week or ten days, two weeks or a month, and then they're forgotten; but this membership and the American Nut Journal that one would receive every month, would be a constant reminder of the giver. What do you think of that, ladies and gentlemen?

THE SECRETARY: It is a fine idea, Mr. President, and I will see that it is also incorporated in the letters to the state vice president that each vice president give to at least one friend a subscription and membership in the association. I suggest also that those who can write for the magazines and the journals get up little articles for the horticultural papers about nut culture. There can't be too many of those in the periodicals.

THE PRESIDENT: Apropos of that suggestion, I believe Mr. Tucker has something to say in regard to a special edition of the Journal. Maybe Mr. Olcott would be good enough to make one of his—

MR. TUCKER: (Interrupting) To make one of his numbers a convention number.

THE PRESIDENT: Yes; one of the numbers in the near future devoted largely to the proceedings of this convention, that is, if he could see his way clear to do it.

MR. OLCOTT: You mean in the matter of—

THE PRESIDENT: (Interrupting) Of this convention. Sort of make it a northern nut growers issue. It is merely a suggestion, Mr. Olcott.

MR. OLCOTT: Yes.

THE PRESIDENT: So that it is practically all about this convention of the Northern Nut Growers Association.

MR. OLCOTT: Yes. Well, it is rather difficult to do that, Mr. President, to the exclusion of all other matter. Is that what you mean? How are we going to take care of the news? It is not a magazine of stories and fiction; it is a magazine of news, and the news of the period between August 15th and September 15th, for instance, will become stale if it is not used in the September 15th issue and runs over until the October 15th issue. It is the American Nut Journal. I think your idea can be carried out very fully by featuring the convention as the main thing, but not to use every last page for it.

MR. TUCKER: No. My idea wasn't to give the whole magazine up to that. But when you got up that magazine, to have the northern nut growers convention stick right out.

MR. OLCOTT: Sure.

THE PRESIDENT: Wasn't it your idea to have some of the pictures, too?

MR. OLCOTT: I see.

MR. TUCKER: Yes; run some of the pictures, and so forth.

THE PRESIDENT: Mr. Olcott, I am sure, is willing to give that issue just as soon as we can get more members and more money.

MR. OLCOTT: We are carrying the nut journal on its subscription list. There is no advertising to speak of in this pioneer industry. The nut nurserymen do not advertise; they should. People want to know where they can get nuts, butternuts and hickory nuts. The people in the South who grow pecans are doing a commercial business but they don't have to advertise; they can't furnish enough nuts to meet the demand. There is no occasion for them to ask for customers; the customers are flocking to their doors and standing in line. People want to know where to get black walnuts; they write in to me. I don't know where to send them. I don't suppose anybody has enough for his local trade and he doesn't have to advertise; he can sell all he has. There is no advertising to speak of. We are living on subscriptions. Now if you enlarge the Journal, use pictures which run up all the way from six to fifteen dollars apiece, you are soon using up your $1.50 per that is left out of the combination membership.

THE PRESIDENT: Yes.

MR. OLCOTT: After paying the tremendously high printer rates. A special edition can be gotten out at considerable additional cost. We have done it in the past and come out at the small end and it took several months to get even again. We can do it again for the sake of the association; but I am saying this to show why it is not done oftener.

THE PRESIDENT: Yes, I understand it. What do you think, then, of a little co-operation on the part of the association in the way of that extra expense for a special edition?

MR. OLCOTT: That's all right. In Mr. Linton's administration I furnished some very large and rather expensive half-tone engravings on the part of the association and they worked in very nicely. I don't know whether the association paid for them or whether he did. I think we divided the cost of them.

MR. BIXBY: I know he did. I have furnished some cuts myself.

THE PRESIDENT: Yes, I know, Mr. Bixby. You are very liberal.

THE SECRETARY: I suggested also that those who can give talks before their local horticultural societies should do so on the subject of nut culture, and if they wish to go in to it extensively slides could be obtained. I think that I could guarantee to obtain them from the Department of Agriculture for illustrated lectures. I have also another question which I would like to put before the association, and that is if we cannot use in some way our surplus back reports to gain new memberships. We have never been able to work out any method of doing so. We have printed each year an edition of a thousand numbers of the annual report. We send out two hundred and fifty or three hundred; consequently, we have about seven hundred annual reports accumulating on our hands every year. Now, what good are they going to be? Can't we use those in some way to increase our membership? Can't we use those as premiums, distribute them gratis some way or other, or distribute them for a small sum to educational institutions, newspapers and agricultural journals? Can't we do something with that annual surplus of about seven hundred nut reports to increase our membership?

MR. TUCKER: Why is there a thousand of them printed?

THE SECRETARY: Because you can get a thousand for just about the same price that you get five hundred, or a very little more.

MR. O'CONNOR: Mr. President, why couldn't some of those be sent to the different experiment stations; also to some of our libraries? We have a number of experiment stations that don't see anything of this kind and that don't know that such a thing exists as the Northern Nut Growers Association. It is only recently that within the state of Minnesota they knew there was such a thing. I have offered a prize in that state for nut culture work. This winter I am going to speak at the Maryland Horticultural meeting, which will be held in Baltimore, and wherever I can get a chance at any of those meetings I always put in a word for the nut. Over on the eastern shore of Maryland, I went into one of the largest apple orchards and nurseries, I believe, in the United States. There were a few northern pecans growing in the yard, and when I asked one of the young men what kind of pecans they were, he said, "Well, I don't know whether it is Indiana or just what it is; but I know it is a pecan." That was growing very beautifully right under the window, you might say, of their dwelling house. That was over at Berlin, at Mr. Harrison's. He likes to sell to nut tree owners, and yet has he come to his year's meeting? Is he a member of the association? For that reason I don't feel like helping him to sell a tree as long as he is not a member. But every chance I get I will put in a good word for the nut tree firm.

I think by sending out our literature to different magazines, to the different experiment stations and over into Canada we would be greatly benefited. We have got some good friends to the north of us. Why not send them some copies and have them help spread this good thing along?

THE SECRETARY: I would like to have Mr. Bixby state about the distribution of those reports outside of the membership. Is there any gratis distribution now?

MR. BIXBY: No, there isn't. There used to be and I made every one of them who received them gratis buy them of me.

THE SECRETARY: About how many institutions now buy the journal?

MR. BIXBY: I should say about half a dozen. That's the same number that had them free before. In nearly every instance when they would write in and request it I would tell them how the association was doing work the

Department of Agriculture ought to do, supporting itself with great difficulty, and we would be glad to have them as a member; that if not a member we would furnish a report for so much. In nearly every case we got them as members or they bought the report. As I said before I don't believe in giving things away; I believe in trying to get the people to see the advantage of buying them.

THE SECRETARY: It would be quite an expense to send out all the back numbers of the reports.

MR. BIXBY: I don't think they would appreciate them either. Although I have not been able to do it the most practicable thing to do seems to me to make an index, say of the first ten and bind them up in a booklet and then I think you could sell them. I hope to do this some time.

MR. TUCKER: What is the expense of mailing?

MR. BIXBY: I think it is about eight cents.

THE SECRETARY: It would be considerable labor, but I think it might be best to circularize different experiment stations, horticultural societies, etc., and ask them if they wouldn't like to have in their libraries a complete file of the reports of the Northern Nut Growers Association which can be obtained for a certain small number of dollars.

THE PRESIDENT: Professor Neilson, what would your attitude be toward a communication you would receive of that nature? Supposing that you were not the enthusiastic member that you are of our association?

PROFESSOR NEILSON: I believe it would be favorable. I believe that is general, and judging from the interest shown in our province I believe that a good many of those horticultural societies and other organizations would be glad to have the reports on file; they would be glad to purchase them at whatever figure was set upon them, if it were a reasonable figure. And I think that I could interest several of our agricultural representatives in having these on file in their office, and possibly in subscribing, or getting the departments of agriculture to subscribe to the northern nut growers journal. There are several county offices along the northern shore of Lake Ontario and in those counties nuts are produced. I think their representatives might be induced to persuade the department to subscribe to your journal.

PROFESSOR TAYLOR: Mr. Chairman: I want to speak on the suggestion made by Mr. Bixby. I may illustrate it in this way: we people in California are, of course, in a little different situation from those represented by the Northern Nut Growers Association. Over there west of the Rockies, or west of the Sierra Nevadas, we have an entirely different situation. By virtue of our peculiar climatic conditions we have already gone through our experimental period and we now have nuts that we are growing on a commercial basis just as they have in the South.

For several years I was connected with the University of California and I used to have to teach students, among other things, the various nuts. That was my particular line, the various nuts, especially those adaptable to California, but also along with that the nuts of the United States and the nuts of North America. I believe that Mr. Bixby will bear me out when I say that it was during my time that all of the back reports of the Northern Nut Growers Association were ordered. That was prior to 1919, was it not?

MR. BIXBY: Yes.

PROFESSOR TAYLOR: It was prior to 1919 that all the back numbers were ordered, and I hope they are still taking them.

MR. BIXBY: They are. They get them every two years.

PROFESSOR TAYLOR: They ought to and if they are not I will see that they do. But I found this difficulty, that there will very shortly be thirteen numbers and if it comes to a question of looking something up, we will find that the average man will not be enthusiastically interested because he won't know how quickly he can get at just exactly what he wants. Mr. Bixby suggested that ten of these volumes be taken together and indexed as a unit. That is one of the finest things that you can possibly ask for. I think the institutions will buy them in a way that they do not now because then they will not have to look through ten volumes to find a little idea they want.

I know it is an expensive proposition to index things of that kind; it takes time and a lot of patience. Not only that but it must be done by some one whose heart is in the work and who recognizes the problems that the man who is going to use that index is going to look up. But I do think that if it could be put in to a combined volume, and some sort of an effort made by the various vice presidents in the different sections to see the institutions in their own sections who would be interested, that something might be accomplished which would be of real worth. I believe this would be increasingly so in the future, because those people will want to look back ten, fifteen, twenty years, and see what the others went through. One of the biggest things that I think I did in our classes was to point out the problems that occurred in California ten, fifteen, twenty-five and thirty years ago, along the line of nut culture solely, and then point out where the nut growers succeeded.

And if I may just branch off here to one of the things I haven't spoken about before this evening, I am absolutely against planting seedling trees unless there is a very strong emphasis laid on the fact that they are not for commercial purposes and not for planting in orchards, but are simply and solely for the possibility of developing new varieties. I think that growers are going to want to go back over old reports in order to save covering the same ground twice. We have found our new people in California starting right in where people started fifty years ago because they didn't know what happened fifty years ago, because our reports out there were not properly indexed.

MR. WEBER: Mr. President, in order to bring the matter to a head, I move that the distribution of the old reports, by sale or otherwise, be left to the discretion of the executive committee.

(Seconded and carried.)

MR. OLCOTT: Mr. President, I would like to ask what the condition of the treasury is. I do so for this reason, that we have planned out a good deal to be done during the interim, from now to the next convention, and the secretary's office ought to be busy. We are planning upon making it so to keep up interest.

I think that the secretary shouldn't be handicapped by lack of funds for stationery and things of that kind. I think that with a deficit maybe he has been. Maybe more matter would go out if he had funds and to that end I am putting in my check for $20 for my subscription tonight in advance. If others will do that he will have funds to work with. (Applause)

THE PRESIDENT: In a discussion I had with the treasurer and the secretary before this evening's session we considered that point, Mr. Olcott, and I thought that we would go after the remaining deficit tonight and make it up, start off with a clean sheet. Mr. Bixby said that if we were going to enter into this new membership campaign in a really generous spirit, he felt that the matter of the remaining deficit should be taken care of.

MR. BIXBY: If we can get two hundred new members this year that will take care of it.

THE PRESIDENT: Two hundred are already pledged.

MR. BIXBY: If we get them that will take care of it.

MR. OLCOTT: It will take to the end of the year to get the returns.

MR. WEBER: I will send my check when I get home, because I don't want to go in to my pocketbook now.

THE PRESIDENT: What was the deficit, Mr. Bixby?

MR. BIXBY: The deficit was $176. There was pledged yesterday, $75, and there has been $10 more today. That's $85 of the $176. Then there is $20 of Mr. Olcott's. That would make it $105. Mr. Weber, when he gets home, will make it $125. We will clean it up one way or another.

THE SECRETARY: I think we should proceed now to the report of the nominating committee and the selection of the next place of meeting.

THE PRESIDENT: The hour is growing late and there is just one message I want to give you here. While it may savour some what of advertising our filbert enterprise, it was not with that idea in mind that we proceeded to get the information we have got. Our filberts have been distributed through the L. W. Hall Company, nurserymen of this city, who have exclusive sale of them at this time. They have been distributed during the past three years over a considerable area: Illinois, Idaho, Iowa, North Carolina, Massachusetts, Nebraska, New Hampshire, Ohio, Delaware, New York, Kentucky, West Virginia, Virginia, Georgia, District of Columbia, Pennsylvania and Kansas.

Some little time ago I conferred with Mr. Hall in regard to communicating with his customers to whom he had delivered filbert plants, the first in the spring of 1919. He has written them asking them how the plants have done, and particularly with regard to fruit bearing. I have the replies here and the gist of them is this: that the plants have done finely, have been entirely satisfactory in that respect. There has been a complaint that they have not borne; there are some instances of extreme pleasure expressed over the way they have borne. My own idea is, and I believe it is that of Mr. Vollertsen also, that they have not had quite time enough yet, since the spring of 1919.

MR. BIXEY: That is not time enough.

THE PRESIDENT: Furthermore, Mr. Hall, in offering them—against our advice as we endeavored to persuade him to offer them as improved European filberts, assorted varieties—thought that the people would be attracted to those unpronounceable names that we have them under. Maybe they were. He listed some six or eight varieties, I think, and those varieties were of our larger fruited kinds. We frankly confess that those varieties will not bear as abundantly as the smaller fruited varieties—not that they are very small they are quite a good sized nut. I believe if Mr. Hall had made a freer distribution of the so-called smaller fruited varieties that there might have been an even more favorable report in connection with fruiting. Another year or so will give us more definite information.

We have now cleaned up our program pretty well. You are going to find Doctor Kellogg's paper in the report, together with the secretary's. We have the papers here. That completes the program up to the present time with the exception of Senator Penny and Mr. Linton. We supposed Mr. Linton would be here. I had telephoned this morning as Mr. Penny promised to send a paper but he hasn't been able to do so. Those are the only two papers of the program that we haven't got.

There are two more things we have to take care of; one is the election of officers, and the other is the selection of a place for the next convention. I call for the report of the nominating committee.

MR. WEBER: The report of the nominating committee is as follows:

> *President:*—JAMES S. MCGLENNON. *Vice President:*—J. F. JONES. *Secretary:*—WILLIAM C. DEMING. *Treasurer:*—WILLARD G. BIXBY.

(Signed)

ROBERT T. MORRIS, G. H. CORSAN, HARRY R. WEBER,

Nominating Committee.

MR. O'CONNOR: I move the nominations be accepted.

THE PRESIDENT: Just one moment. Up to this evening I understood that the president was to be elected for a year. I do not know much about the condition prior to President Linton. He was elected at Battle Creek at the same time I was elected vice president. There were extenuating circumstances justifying the re-election of President Linton. I feel that similar conditions do not prevail justifying my re-election as president of this association. It is not going to make any difference to me whether I am president or just simply a soldier in the ranks. I want to see this association the success it ought to be and I feel, in view of the wonderful work that has been done in this association and for its best interests at all times by Mr. Jones, that it is due him that the presidency should be passed to him at this time. He is going to the next convention of the National Association of

Nurserymen, he and Doctor Morris, Mr. Olcott and Mr. Weber, to get before that convention of nurserymen something more of the history of this association and its ambitions and desires. I know he could appear before that convention in a much more advantageous way for the benefit of this association if he were president of it. I feel that Mr. Jones ought to be elected president of this association here tonight.

MR. OLCOTT: Mr. President, the presidents of this association have been elected for two years and I think it has become an established custom.

THE PRESIDENT: Mr. Bixby referred to that tonight. I didn't understand it that way. I supposed I was elected last year at Lancaster for one year.

MR. O'CONNOR: Mr. President, I took that matter up with Mr. Littlepage and he told me it was customary to elect the president for one year at a time and to re-elect him for the second year if he proved all right. So far I think every member of this association has been well satisfied with the service you have given us and we want you to continue on for another year.

THE PRESIDENT: Pat, I thought you were my friend.

MR. SPENCER: I move the secretary cast the ballot of the association in favor of the officers nominated by the committee on nominations.

MR. O'CONNOR: I second the motion.

(Upon the motion being put to a vote of the members, it was declared duly CARRIED, the secretary cast one ballot for the persons nominated, and they were declared duly elected.)

THE PRESIDENT: All right, ladies and gentlemen. Here we are with our coats off and sleeves rolled up for another year; but I want to give you all fair warning that if we don't have that thousand memberships at that next convention, this child is going to drop out. (Laughter.)

The next in the order of business is the selection of a place for the next convention. You heard the telegram this morning from Mr. Littlepage and the telegram from the Washington Chamber of Commerce. Personally, I would like to have you come to Rochester again because we sure enough have enjoyed the sessions with you all here this time. There is no finer city in the world than Rochester, N. Y., and we would like to have you come back here. I want you to come here.

MR. O'CONNOR: Mr. President, I feel it is quite an honor that we are asked to the capital city of the United States to hold our meeting. It shows we were appreciated there some few years ago. I move you we have the next meeting in Washington.

MR. OLCOTT: I second the motion.

(The motion being duly put to a vote of the members, it was carried.)

THE SECRETARY: Now, Mr. President, we should decide upon a date.

THE PRESIDENT: I think that is true. In Lancaster, last year, it was held later than this. I believe the ordinary time has been considerably later than this, about a month.

MR. O'CONNOR: Mr. Chairman, Mr. Littlepage asked me to say, after the convention city had been selected, that it would be best to make it about the last week in September as that would show the pecans and walnuts at about the right time.

MR. BIXBY: I move that the time of the next convention be fixed at September 26th, 27th and 28th, 1923.

MR. O'CONNOR: I second the motion.

(The motion being put to a vote of the members, it was declared CARRIED.)

There being no further business to come before the Convention, it thereupon adjourned.

PROCEEDINGS OF THE TREE PLANTING CEREMONIES AT HIGHLAND PARK, ROCHESTER, N. Y.

September 9th, 1922, 11 A. M.

PRESIDENT MCGLENNON: This occasion represents the custom of the association of planting a nut tree in one of the parks of the community, in which the annual convention is held. We had expected to have some black walnut seedlings grown from nuts presented to ex-President Linton by the superintendent at Mount Vernon, Washington's old home. I am not sure but I have quite a vivid remembrance that the trees from which these nuts were gathered were fruiting in Washington's time. However it would be a very delightful time if we could have such trees to plant in memory of that great character. But I am sorry to say that we have been disappointed in not receiving the trees from Mr. Linton. He expressed them from Saginaw the day before yesterday and we have made diligent effort to locate them in this city this morning but have been unable to get any trace of them. Anticipating such a happening Mrs. Ellwanger, who had on exhibition at the convention some Persian walnuts grown in pots, at our request very kindly consented to let us use one of those trees. If we had had a little more time to consider it undoubtedly Mr. Dunbar would have arranged to have this tree planted on the land that was given to the city by George Ellwanger, Mrs. Ellwanger's father-in-law, and Patrick Barry of the world famed nursery of Ellwanger & Barry. We are going to plant one of these Persian walnut trees here (the planting is now going on) and there is a greater likelihood that this tree will live than the black walnut, as that tree had to be dug and transported. We feel reasonably sure that this tree will live to commemorate our meeting in Rochester this year.

We are also going to plant an Arkansas hickory, that Mr. Dunbar has had dug from the park nursery, a short distance from where the walnut is planted. I think this, too, is an appropriate tree to plant because of the success of the hickory in this community. Mr. Dunbar tells me that practically all of the varieties of hickory of North America are planted on this park slope. We took great pleasure in driving through here the other day and listening to an explanation of their history by Mr. Dunbar.

We are honored today by the presence of the Dean of the New York State School of Forestry, Dean Mann, who has consented to address us. It gives me great pleasure to introduce to you Dean Mann.

DEAN MANN: President McGlennon, ladies and gentlemen:

I assure you it gives me great pleasure to be here because as a forester and tree lover by profession I am also a tree lover by nature. I can conceive of no more worthy, more beautiful nor attractive memorial than a tree dedicated to the Father of our Country, something which will grow in size, in beauty and in productivity as the years roll by. As foresters would remind you, ladies and gentlemen, the Father of our Country served his apprenticeship long before he became a land owner and patriarch on those broad Virginia acres. The Father of our Country started out in life as a forester and surveyor. You may remember that he piloted, or was to be one of the pilots of Braddock's expedition, having gained his knowledge of the woods through his early life as a young surveyor in the forests of Virginia.

There are in New York state approximately fourteen million acres better suited to tree crop production than to field crop production. Here in the northeastern corner of the United States, where our great centers of population are found, we have in the state of Maine seventy per cent suited to tree crop production but unsuited to tillage; we have similar conditions in Massachusetts and Connecticut. Throughout this northeastern section of the country we have a tree soil domain which will grow trees and which can't be plowed with profit. All who are interested in the production of trees for whatever purpose should realize that this nation cannot permanently prosper unless every acre of its land is put to its best permanent use.

I think that you will agree with me that it requires no prophetic eye to see the day not far distant when we will have, stretching from the Island of Manhattan up to where Albany now stands, one vast series of teeming cities with suburb touching suburb. The problem then will be how to feed this multitude. Developments in Russia show that, no matter how idealistic one's theory of government may be, food, in the last analysis, is the thing which makes or breaks a nation.

Those of you who have studied some of the interpreters of early Scripture will remember, perhaps, that the Garden of Eden was in reality an oasis of trees in the great valley of Mesopotamia, and even today "garden" in the oriental term means a group of trees. It has been proven by experience in these different tropical realms that where tree production is biggest and nuts and other products are grown under intensive cultivation, an acre will produce more food than where grazing is practiced. I spent a very pleasant year in California and saw some of the operations of the California nut growers, where they are growing English walnuts on a most extensive scale. I believe I will be making no false statement when I say that those areas in southern California which are growing nuts produce more in fats, proteins and calories for the maintenance of the health and strength of the human race than do the acres which are given up to the growing of animal crops.

So I applaud the idea of planting a tree in the memory of the Father of his Country. I believe I belong to your group, at least through interest, because I have been doing a little experimenting of my own in my back yard at Syracuse where I have an English walnut which I planted in 1915 which is this year producing for the first time. I am going to take those nuts and see what can be done with them in perpetuating that particular variety, because it is hardy, fast growing, and early to mature.

The New York State College of Forestry has a platform as broad as the entire state. We are interested in every kind of land which is not suited to agriculture, fish, game, recreation, conservation of water, and I pledge to you the sympathy and the support of the New York State College of Forestry. We have three experiment stations; one in Oneida county, one in Onandaga county, and another in Cattaraugus, with a fourth in St. Lawrence, if you wish to call it such. We would be delighted to receive from you any slip or any sort of fruit which you wish us to try out at these experiment stations. I believe that the time will come when some combined system of forestry and horticulture can be maintained which will aim at the production of food stuffs from trees, with lumber, perhaps, as a by-product. That works out in the old country and the day is not far off when it can be practiced here.

I congratulate the members of this association on having completed what was, from all accounts, a most successful meeting. I regret that I couldn't have been here earlier and met the other members of your body. I congratulate you; I wish you God speed, and I again tender the support of the College of Forestry.

* * * * *

PRESIDENT MCGLENNON: We certainly have received great encouragement from Dean Mann's remarks, which to me, and I believe to all present, were most interesting and instructive.

I want to hear just a few words from our esteemed friend, Mr. John Dunbar, Assistant Superintendent of Parks.

MR. DUNBAR: I think it is a very happy and fortunate circumstance that Mr. Mann is here this morning representing the College of Forestry of Syracuse. Every word that Mr. Mann has said is absolutely true. The forestry question of this country is indeed a very serious question. Every man, and every woman, should give most serious thought to it, and I hope the words Dean Mann has spoken to us here this morning will go in to all our hearts very deeply.

Of course the Park Department is studying trees from the ornamental and arboricultural point of view. We think, however, that arboriculture, horticulture and forestry, as the Dean said, are very, very closely allied and should surely work together. I think his idea is a very excellent one; that there should be a very close connection or union between forestry, horticulture, nut culture, and all kinds of fruit culture. I hope that day is not far distant.

PRESIDENT MCGLENNON: Ladies and gentlemen, the treasurer of our association is a man who is intensely interested in nut culture. He has done wonderful things for its advancement and especially for the advancement of the interests of the Northern Nut Growers Association.

MR. BIXBY: While Dean Mann was speaking the thought came to me, how could we better co-operate with the Department of Forestry? I think the work of the Nut Growers Association, which is particularly interested in the use of nut trees for orchards, and that of the Department of Forestry, which looks upon them particularly as producers of timber, could be very closely allied. The thought came to me, could not we right here work out some practical suggestion whereby we two could co-operate? I would like to ask Dean Mann what nut trees they are planting for forest purposes.

DEAN MANN: We have done very little. We have, at our experiment station at Chittenango, done some work with the English walnuts. This particularly hardy specimen that I have in my own back yard—I have two, one of them is growing very slowly—are from our experiment station. We have really had so much to do in the way of popular education in New York State in the timber products, that we are merely, as they say in the South, fixing to begin with other things. That is the only species with which we have made an actual start. There is this however: what can foresters, horticulturists and nut enthusiasts do to supply the place of the American chestnut? I really came here as a seeker after truth on this particular phase. You men probably know more about it than I. What can we produce? Is there any hybrid which can be introduced into this country which will take the place of the American chestnut?

MR. BIXBY: In reply to that I would say that I have hundreds of seedlings of the Chinese chestnut on which the blight has been working for years and has not destroyed them. I would be very glad to send them to the College of Forestry and let you try them.

DEAN MANN: They will be planted with extreme care and a barbed wire put around them.

MR. BIXBY: There is another thing, the rough shell Japanese walnut, so-called, which is really a butternut hybrid. I have planted it and it is growing at a tremendous rate, even faster than the Japanese walnut. I expect to get a lot of those nuts this year and I wondered how the College of Forestry would like to try some of them.

DEAN MANN: I would be delighted.

MR. BIXBY: Then there is one other nut the big shell bark hickory which is a native of the Mississippi Valley, which has been planted in Pennsylvania and up in Lockport, New York. It grows finely, it bears early, and I think that it might be worth trying.

DEAN MANN: We have adopted this platform: "Anything which will interest the people of New York State." We must, as a state institution, limit our horizon very largely to the state of New York. We do slip over occasionally, but anything which will interest the people of New York State in trees of any kind, for any purpose, is a step towards forest conservation. Take your city dweller in New York City, get him interested in a shade tree in front of his apartment house, or in a group of shade trees in the adjoining park, and you have converted that man along the line of King Forest. So we will be very glad to take any seeds you have and give them excellent care.

NUTS THE NATURAL AND ADEQUATE SOURCE OF PROTEIN AND FATS

By

JOHN HARVEY KELLOGG, M. D., F. A. C. S.,

Medical Director of the Battle Creek Sanitarium

In the writer's opinion, the most important thing which can be done to promote the nut growing industry is to make clear to men and women everywhere the necessity for returning to natural and biologic living. Since he left his primitive state, in his wanderings up and down the face of the earth to escape destruction by terrific terrestrial convulsions and cataclysmic changes in climate and temperatures, chilled during long glacial periods, parched and blistered by tropic heats, starved and wasted by drouth and famine, man has been driven by ages of hardships and emergencies to adopt every imaginable expedient to survive immediate destruction, and in so doing has acquired so great a number of unnatural tastes, appetites and habits, perversions and abnormalities in customs and modes of life, that it is the marvel of marvels that he still survives.

Man no longer seeks his food among the natural products of field and forest and prepares it at his own hearthstone, but finds it ready to eat, prepared in immense factories, slaughter-houses, mills, and bakeries and displayed in palatial emporiums. No longer led by a natural instinct, as were his remote forebears, in the selection of his foodstuffs, he finds his dietetic guidance in the advertising columns of the morning paper, and eats not what Nature prepared for his sustenance, but what his grocer, his butcher and his baker find most for their pecuniary interest to purvey to him. The average man no longer himself plants and tills and harvests the foods which enter into his bill of fare, that is, "earns his bread by the sweat of his brow," but accepts whatever is passed on to him by a long line of producers and purveyors who do his sweating for him, depriving him of the opportunity of earning both appetite and good digestion by honest toil. So he resorts to condiments and ragouts, palate-tickling and tongue-tickling sauces and nerve-rousing stimulants, as a means of securing the unearned felicity of gustatory enjoyment.

At the World's Eugenics Congress held in New York last fall, Professor Davenport expressed the opinion that the human race will ultimately perish, and Major Darwin, son of Charles Darwin, one of the world's leading economists, gave expression to similar views. We are evidently traveling a downhill road and the tide of degeneracy is rising so fast it will certainly sweep us on to race extinction unless we return to sane and biologic living. We are primates, not carnivores like the dog, nor omnivores like the hog. The primates are fruit and nut eaters in whatever part of the world they are found. All the primates adhere to the family bill of fare. The gorilla, reigning king of beasts in the forests of the Congo, his somewhat lesser relative, the chimpanzee, which tenants a wide area of the Dark Continent, the orang-utan of Borneo, and the gibbon of tropical Asia, diversified as they are in form and habitat, are all equally circumspect in their adherence to the diet of nuts and fruits, tender shoots and soft grains, foods which Nature has prescribed as the primate's bill of fare.

A return to natural eating would doubtless do, to say the least, as much as any one thing toward checking the downward race movement, and no one who has ever studied the economics of diet will question that the only way in which the earth's dense populations of the future can be fed will be by the elimination of the flesh-pots and a resumption of the natural dietary. This is clear when we recall the fact that the Agricultural Experiment Stations have demonstrated that 33 pounds of digestible foodstuffs are required to make one pound of beef. When an animal is fattened, the creature uses a large part of the food which it consumes for its own purposes. The eater of flesh does not get back the original corn and other foods given to the animal but only a small fraction of it; and hence dense populations can only indulge in beef eating by importing meats from other countries not yet fully occupied. Evidently, the present rapid increase of the earth's population will soon bring us to a point where this enormous waste must cease. Flesh eating will have to be abandoned for economic reasons. Even the milk supply will necessarily be limited, for we are compelled to feed the cow 5 pounds of digestible foodstuffs to obtain 1 pound of water-free food in the form of milk.

Those pessimistic economists who predict that by the year 2000 the American Continent will be so densely populated that means will have to be adopted to limit the increase of population because of the scarcity of foodstuffs, are evidently not aware of the activities of the Nut Growers Association and of the marvelous efficiency of nut trees as producers of protein and fats, the two elements of our foodstuffs which are most costly because hardest to produce.

I am creditably informed that one acre of land supporting 35 black walnut trees in full bearing, will produce not less than 350 pounds of walnut meats, each pound of which has a nutritive value in protein and fats fully four times that of an equal weight of beef or an equivalent of 1400 pounds of meat. To produce a steer weighing 1600 pounds, requires two acres and two years. Two acres and two years will produce 1400 pounds of nut meats, the equivalent of 5600 pounds of beef or more than 9 times the amount of nutritive material in the form of protein and fat produced by beef raising.

Of course, the question might be raised whether nuts as a source of food are equal in value to meats, which supply the same sort of food material, namely, protein and fats. If the anthropologists are right, this is a question which need not worry us, for, according to Professor Keith, the eminent English anatomist and a leading paleontologist, and Professor Elliot, of Oxford, nuts were the chief staple of our hardy ancestors of prehistoric times. Professor Elliot, indeed, tells us in his work, "Prehistoric Man," that the first representatives of the human race who appeared in the Eocene Period were fruit and nut eaters, and were abundantly supplied with this sort of nutriment. This eminent author says,—

"On the bushes by the rivers and along the shore there were all sorts of fruits and nuts. For the subsistence of our lemur-monkey-man in the early stages of evolution, what fruits would seem *a priori* most suitable?

"I think that one would select the banana and bread-fruit. Ancestral forms of both were flourishing in the Eocene. Many other fruits with which man has been afterwards continually (perhaps one might venture to say *most intimately*) associated, occur at this period. These are, most of them, found in so many places that one is apt to think they were then of world-wide distribution.

"In the temperate brushwood and on the river-sides, acorns, hazel-nut, hawthorne, sloe, cherry and plum might be found. Here and there, he might alight upon a walnut or an almond; figs also of one kind or another seem to have been common. Palm trees existed, and some of them were of enormous size."

If, in modern times, nuts have come to be used as a luxury rather than as a staple article of diet, it must be because we have neglected to cultivate this choicest of food products which Nature is ready to provide with a lavish hand when invited to do so by our co-operation. But as the public become better informed respecting the high food value of nuts and especially in view of the steadily rising cost of flesh meats, the nut is certain to gain higher appreciation, and the writer has no doubt that some time in the future nuts will become a leading constituent of the national bill of fare and will displace the flesh meats which today are held in high esteem but which in the broader light of the next century will be regarded as objectionable and inferior foods, and will give place to the products of the various varieties of nut trees which will be recognized as the choicest of all foods.

In nutritive value the nut far exceeds all other food substances; for example, the average number of food units per pound furnished by half a dozen of the more common varieties of nuts is 3231 calories while the average of the same number of varieties of cereals is 1654 calories, half the value of nuts. The average food value of the best vegetables is 300 calories per pound and of the best fresh fruits grown in this country, 278 calories. The average value of the six principal flesh foods is 810 calories per pound or one-fourth that of nuts.

Recent studies of the proteins of nuts by Osborne and Harris, Van Slyke, Johns and Cajori, have demonstrated that the proteins of nuts are at least equal to those of meat. This has been shown to be true of the almond, English walnut, black walnut, butternut, peanut, pecan, filbert, Brazil nut, pine nut, chestnut, hickory and cocoanut; that is, of practically all the nuts in common use.

Observations seem to show that, in general, the proteins of oily seeds are complete proteins.

Cajori's research has also shown the presence of growth-promoting vitamins in abundant quantity in the almond, English walnut, filbert, pine nut, hickory, chestnut and pecan.

That the nut is appreciated as a dainty is attested by the frequency with which it appears as a dessert and the extensive use of various nuts as confections. That nuts do not hold a more prominent place in the national bill of fare as food staples is due chiefly to two causes; first, the popular idea that nuts are highly indigestible, and second, the limited supply.

The notion that nuts are difficult of digestion has really no foundation in fact. The idea is probably the natural outgrowth of the custom of eating nuts at the close of a meal when an abundance, more likely a super-abundance, of highly nutritious foods has already been eaten and the equally injurious custom of eating nuts between meals.

Neglect of thorough mastication must also be mentioned as a common cause of indigestion following the use of nuts. Nuts are generally eaten dry and have a firm hard flesh which requires thorough use of the organs of mastication to prepare them for the action of the several digestive juices. It has been experimentally shown that nuts are not well digested unless reduced to a smooth paste in the mouth. Particles of nuts the size of small seeds may escape digestion. Nut paste or "butter" is easily digestible.

Delicious nut butters may be prepared from true nuts such as the almond, filbert and pine-nut, by blanching and crushing, without roasting. Peanuts require steam roasting. Over-roasting renders the nut difficult of digestion.

More than 50,000 tons of nut butters are produced in England every year. Peanut oil, palm kernel oil and coconut oil are the principal raw materials used. In face of vanishing meat supplies, it is most comforting to know that meats of all sorts may be safely replaced by nuts not only without loss, but with a decided gain. Nuts have several advantages over flesh foods which are well worth considering.

1. Nuts are free from waste products, uric acid, urea, and other tissue wastes which abound in meats.

2. Nuts are aseptic, free from putrefactive bacteria, and do not readily undergo decay either in the body or outside of it. Meats, on the other hand as found in the markets, are practically always in an advanced stage of putrefaction. Ordinary fresh, dried or salted meats contain from three million to ten times that number of bacteria per ounce, and such meats as Hamburger steak often contain more than a billion putrefactive organisms to the ounce. Nuts are clean and sterile.

3. Nuts are free from trichinae, tapeworm, and other parasites, as well as other infections due to specific organisms. Nuts are in good health when gathered and usually remain so until eaten.

In view of these facts, it is most interesting to know that in nuts, the most neglected of all well known food products, we find the assurance of an ample and complete food supply for all future time, even though necessity should compel the total abandonment of our present forms of animal industry.

Another of the great advantages of the nut is that with few exceptions, it may be eaten direct from the hand of nature without culinary preparation of any sort. Indeed, the common custom of offering nuts as dessert is an acknowledgment that in the nut the refined chemistry of Nature's laboratory permits of no improvement by the clumsy methods of the kitchen.

Every highway should be lined with trees. Many nut trees will grow on land unsuited to ordinary farm crops. The pinon flourishes on the bleak and barren peaks of the Rockies.

A few nut trees planted for each inhabitant would insure the country against any possibility of food shortage. A row of nut trees on each side of our 3,000,000 miles of country roads would provide half enough fat and protein for a population of 100,000,000.

If each one of the 6,000,000 farmers in the United States would plant and maintain an orchard of ten acres of black walnuts, the annual crop, with little or no attention, would yield not less than 3,000,000 tons of nut protein, the equivalent of more than 12,000,000 tons of meat, besides more than 6,000,000 tons of fat of the finest quality, sufficient to supply every one of 100,000,000 people with an ample amount of protein, and, in addition, the fat equivalent of 6-2/3 ounces of butter.

Nuts should be eaten every day and should be made a substantial part of the bill of fare. So long as the nut is regarded as a dainty, suitable only for dessert, the demand will be limited. But as its merits come to be appreciated, it will be in greater demand and the supply will rapidly grow in volume.

The Lime Content of Nuts

In proportion to their weight, nuts contain more lime than any other class of foodstuffs except legumes, the average being more than one-third grain to the ounce (.370 grs.). Certain nuts are surprisingly rich in lime. For example, the almond affords one and one-third grains of food lime to the ounce, while the hazel-nut or filbert affords one and three-quarters grains of lime to the ounce, or 11.3 per cent of a day's ration of lime. The pecan and the walnut are also fairly rich in lime, as is also the peanut.

An ounce and a half of each of almonds and hazel-nuts or filberts will supply one-third the total lime requirement for a day. In general, this addition to the ordinary bill of fare would be quite sufficient to insure against any serious deficiency of lime.

Meats of all sorts are poor in lime. The lime in animals is almost exclusively in the bones. One ounce of almonds, for instance, contains as much food lime as a pound of the choicest steak, and a quarter of a pound of black walnuts supplies as much food lime as nearly two pounds of average meats.

The Iron Content of Nuts

The almond, hazel-nut, chestnut, peanut, pecan and walnut, all contain a rich store of iron, the average iron content expressed as per cent. of the iron ration being 4.79, more than two and one-half times that of fruits (1.74), three times that of vegetables (1.46), greater than that of cereals and even superior to average meats. It is true that the extraordinarily high food value of nuts renders them less available than fruits as prime sources of iron, for one would have to eat 5,000 calories of chestnuts or walnuts or more than 4,000 calories of pecans or peanuts to get a day's ration of iron; but three-quarters of a pound of almonds or hazel-nuts would supply the needed quantum of iron with an energy intake of 2,500 calories, on account of their unusually rich store of iron.

It is worth while to know that vegetable milk prepared from almonds, by adding five parts of water to one part of blanched almonds made into a smooth paste, supplies two and a half times as much iron as does cow's milk in equal quantity, and furnishing the same amount of protein. It is worth noting, just here, also, that the protein of the almond is, like that of milk, a complete protein, that is, a protein out of which human tissues may be readily formed, which is by no means true of all vegetable proteins. Such a milk, however, would be somewhat deficient in lime, a lack which could be supplied by lentil soup.

A product commercially known as Malted Nuts, prepared from almonds or peanuts, has been found of very great service in meeting the needs of infants and some classes of invalids for an easily digestible liquid nourishment to take the place of milk when a substitute is needed.

The chief obstacle which at the present time stands in the way of making nuts a food staple is the meager supply. If the population of the United States should suddenly turn to nuts as the chief means of meeting their protein requirement, the total annual crop of nuts would be consumed in a day or two, or possibly less time. The American people readily change their eating habits. As nuts become more plentiful through the efforts of the Nut Growers Association, and the general enlightenment of the people concerning the superiority of this class of foodstuffs by a well conducted propaganda such as has been carried on in behalf of the raisin industry and such as the meat packers are now conducting in their effort to induce the American people to eat more meat, but of course on an honest, scientific basis rather than by means of untruthful and misleading statements, as the packers are doing, the intelligent people of this country could soon be brought to an appreciation of the great value of edible nuts and the important place which they should fill in the bill of fare.

Thirty years ago, the writer prepared a paste from peanuts which had been previously cooked by steaming or baking, and gave to the preparation the name of "Nut Butter." Little attention was paid to the product for two or three years, then it began rapidly to win favor and, according to a recent report by the Census Bureau, 56 establishments, in 1919, produced peanut butter to the value of nearly $6,000,000, and the peanut crop last year was

816,464,000 pounds. In 30 years, the peanut crop has increased from a few thousand acres to nearly 2,000,000 acres, and the peanut has come to occupy a place on the national bill of fare of considerable prominence. The peanut is not really a nut but a legume and is in flavor and other edible qualities greatly inferior to the products in which this Association is interested. Nevertheless, the fact that it is accessible has given it an opportunity to quickly gain popular favor. The writer feels very confident that if this association and other similar organizations will continue their efforts in behalf of nut growing, and will at the same time adopt measures to inform the public concerning the remarkable nutritive properties of these products which have been created expressly for the use of man and which are so wonderfully adapted to his sustenance, there will be a steady advance in their acceptance by the public and in the not far distant future, the raising of nuts will come to be as nearly universal among farmers as the production of apples or other fruit crops. If the uncultivated lands of this country not now occupied as farms were occupied by nut trees in good bearing, the annual crop of nut protein and fat would be amply sufficient, in connection with the corn, wheat and other crops harvested by our 6,000,000 farmers from our big billion acre farm to easily support a population of 1,000,000 persons. If the nut is given a chance, it will not only save the human race from perishing from starvation, but will give it a good boost upward in the direction of race betterment.

The Eat More Meat campaign which the packers are now conducting and for the support of which they at their recent convention in Kansas City, voted to raise a fund of $500,000, is being carried on by the grossest chicanery and misrepresentation. Pseudo-scientific men are being put before the public as great authorities in human nutrition and these men are sending out plausible but most misleading eulogies of meat as a foodstuff possessing essential qualities for the lack of which the American people are suffering. The only possible reason for these frantic appeals to the American people to consume more meat is the depletion of the packers' profits by the steady decrease in meat consumption which has been going on for a number of years and which begins to threaten the future development of their industry. The public will be damaged rather than benefited by an increase of meat consumption. A nation-wide campaign in behalf of the almond, the hazel-nut, the walnut, the pecan and other of our native nuts would unquestionably improve the health and vigor of the American people, provided the nut growers will supply the demand which would be created.

* * * * *

August 12th, 1922.

Dear Dr. Deming:

I have received your letters. I am sorry to answer you very late, because on March 28th my wife died. I have been again heart broken and delay everything for these few months.

I have not yet met Mr. Read, I went to the U. S. Consulate to find him, but no definite answer received yet.

The place Chuking is rather inconvenient to reach from Shanghai. I am gong to buy land near Shanghai i. e. one hour trip from business center. When I succeed that, I will remove all trees out.

I am sending you separate paper that you want for the convention.

The seeds that I sent you last year is Castanopsis sp. grows near Hangchow, 100 feet high and ever green.

Yours very sincerely

P. W. WANG.

CHINESE NUTS—WALNUT

P. W. WANG

Kinsan Arboretum, Chuking, Kiangsu Province, China.

Historic research by Berthold Laufer in his "Sino Iranica" published by Field Museum of Natural History of Chicago is very valuable. His conclusion is that China is not the original home of walnuts but imported from Persia via two routes, the earlier by Chinese Turkestan and little later by Tibet. I recommend every member to read this book. It contains many valuable historical informations about trees and vegetables in Asia.

According to the recent travel of late Mr. F. N. Meyer no grafting or budding of nut trees yet practiced in China. The walnuts varied from thinnest shell like peanut or hard shell with poor flavor. The Chinese walnut are proved to be hardier than Persian walnut in America.

There is no walnut in this province except a few in ornamental gardens. What we can get is through grocery stores. They imported them from Tientsin or Tsintao. The former is easy to crack with fine flavor and the kernel color is light. The latter is hard to crack, the internal partition has a peculiar construction that the kernel is very hard to take out even in broken pieces and the kernel has a brown color with the taste of bitterness and astringency. That shows that the walnut in Chili is far superior to that of

Shantung. I do not believe that the above difference is due to the latitude, because there is one walnut tree in a garden in Soochow, a big city 50 miles from Shanghai, the nut is very good.

The Chinese way of eating walnut is just like Americans. One thing that coincides with Dr. Kellogg's treatment to a Senator's daughter. In China there is no baby fed by cow's milk. When the mother lacks milk and the home is not rich enough to hire a milk nurse, walnut milk is substituted. The way of making walnut milk is rather crude here, they simply grind or knock the kernel into paste then mix with boil water. I wish to learn Dr. Kellogg's way of making walnut milk.

One tradition that believed by most Chinese even well educated Chinese for thousands years that if you eat walnut constantly, your life will be prolonged, and if you only eat fruits and nuts excluding all provisions other than produced from trees even rice and wheat your life will be eternal. I must recall the theory of Dr. Kellogg that may be the proof of the above tradition. "Beef fats is deposited in the tissue as beef fats without undergoing any chemical change whatever; mutton fat is deposited as mutton fat; lard as pig fat etc." Perhaps the influence of animal fat reduces the life as animals are generally short lived and nut fats increases the life as nut trees live for centuries.

Chinese walnuts are sometimes met with very good ones, moreover they are hardy and free from insect or fungus attack. They are really worth while to propagate. As I can not get propagate nor scions I am now planting seedling from best nuts. I wish you are doing the same work and finally we can supply the colder world with suitable walnut trees.

I suggest one plan that I know very big amount of walnuts of best quality are exported from Tientsin to the States. You can secure the best ones by selecting from the walnut importers for planting.

There is another walnut produced in the vicinity of Hangchow Carya Catheyensis, really a hickory, last year I sent to Mr. Jones for 50 lbs. The taste is far below that of Pecan, but just 3 months ago I ate at a friend's house. The hickory kernel was roasted with sugar syrup. It lost all bitterness and has a very good hickory taste with fine hickory flavor.

Pterocarya stenoptera grows best among any other trees in this region. It resists drought very well. I like to try to use it as stock for grafting.

I do not interest in chestnut yet. As far as I know the best chestnut is produced in Lian-Shang near Tientsin.

Castanopsis from Hangchow is very nice. They said the tree is over 100 feet high and is ever green.

Hazelnut is from Chili and North. They are not so good as yours.

Chinese almond is apricot kernel, the best one is from Peking or Tientsin.

Ginkgo nuts are never eaten afresh, we eat them sometimes roasted and most times cooked with meats. In which you will find both meats and nuts of good taste.

I like Torreya Grandis very much, I think Americans do not like it because they do not use the right way. Chinese roast the Torreya nut until all moisture gone then wait they are cold and eat them. They must be kept dry after roasted otherwise the taste is not so good and a second roast is necessary.

I hope you will try the above two kinds nuts by the above way, as Ginkgo can live over thousand years and Torreya in this country is also long lived, their nut fat would keep the human tissue less easy to decay.

The pine seeds kernels are sold here for Mex. $1.60 per pound. If your pine seeds kernel are cheap, it is possible to come over. The pine seeds are Pinus Bungeana and P. Massoniana.

RESOLUTIONS ON THE DEATH OF

DR. WALTER VAN FLEET

At the thirteenth annual convention of the Northern Nut Growers' Association, held at Rochester, N. Y., September 7, 8 and 9, 1922, a committee was appointed to express the sorrow of the association at the death of its honorary member, Dr. Walter Van Fleet, at the age of sixty-four, on January 26th 1922, and to inform Mrs. Van Fleet of its action.

Dr. Van Fleet, at one time the only honorary member of the association, was made so in recognition of his services to nut growing in breeding blight resistant chestnuts and chinkapins, and of his unfailing courtesy to the association whenever asked to present the results of his investigations.

Although incomplete his experiments had already produced results of great promise and shown the way that his successors must follow. Many of us knew him personally and had visited his home and experimental grounds at Bell, Maryland, some of us more than once. Few of us knew his varied and high attainments in many other fields than plant breeding, though a moment's thought would have made a discerning person see that his modesty, self-effacement, kindliness and sympathy were things that most often come to those whose experiences of life have been the widest. His accomplishments in plant breeding and other fields, a bibliography of his writings, and the events of his life, were fully and sympathetically related in a communication written by Mr. Mulford of the U. S. Dept. of Agriculture at the request of the association and read at the meeting.

The association feels that no one can ever quite take the place of Dr. Van Fleet in the field of his life work, in experimental nut breeding and in the hearts of the members of this association who had the privilege of knowing him, and it wishes to put on record its great sorrow at his untimely death in the very midst of his beneficent activity for the benefit of mankind.

RESOLUTION ON THE DEATH OF

COLEMAN K. SOBER

At the thirteenth annual convention of the Northern Nut Growers' Association, held at Rochester, N. Y., September 7, 8 and 9, 1922, a committee was appointed to express the feeling of the association at the death of one of its life members, Coleman K. Sober, at the age of seventy-nine, at his home in Lewisburg, Pa., in December 1921, and to inform his family of its action.

Colonel Sober, as he was most often called, was a frequent attendant at the meetings of the association in its early history. He was a pioneer in the culture of the chestnut in America and the grower and distributor of a variety which he called the Sober Paragon. He developed the production of this valuable variety, and its nursery stock, on a large scale and had demonstrated chestnut growing as the first of the established nut industries in the northeastern United States. He devised methods of grafting and cultivating the chestnut and invented means and machinery for harvesting and shelling the nuts, for which he found a ready market at good prices.

A man of strong personality, capable of large operations and unaccustomed to failure he found it hard to admit defeat of his deeply cherished purpose, and success already within his grasp, by that great national calamity the invasion of this country by the fatal chestnut blight.

Undoubtedly he foresaw, as did other advocates of nut culture, the great help and stimulus to the industry that would result from the commercial success of chestnut culture, and it was a bitter disappointment to him to find himself helpless before the irresistable progress of the blight. This failure came too late in life for him to recover and develop new fields in nut culture which, let us believe, he would have done if he had been younger, for we know that he was an advocate of the roadside planting of nut trees and a supporter of the efforts of those of us who are striving for the success of all forms of nut culture.

Nut growing and this association have lost an able and energetic worker.

An account of Col. Sober's life and works may be found in the August 1922 number of the American Nut Journal.

Telegram from Washington, D. C.

TO JAMES S. MCGLENNON:

Deeply regret my inability attend thirteenth annual meeting. Am sure it will be great success and all will enjoy trip to your beautiful city and surrounding country. The next few years will show fine results of efforts our Association, and nut culture in north will take on new life and result in planting thousands of acres trees. I hope Washington will be selected as place for next annual meeting.

T. P. LITTLEPAGE

* * * * *

Lincoln, Nebraska, September 5, 1922

My Dear McGlennon:

Nothing would give me greater pleasure than to be in your city this week. I have been through your city five times in three years. If I had known what you have there I should have stopped there three years ago. Since it is impossible for me to be there at this time I will save my coin to purchase trees and nuts for next year.

Dr. Deming's wonderful discovery of a monster pecan tree in Hartford, Conn., together with native pecans north of Burlington, Iowa also two Iowa pecan trees growing in this city for twenty-eight years, makes the field for pecan trees a very large one viz. from the Gulf to the forty-first parallel. Tell Dr. Deming we trust his wonderful discovery does not prove to be a pignut.

Our opportunities in the north for growing nut trees I think are wonderful.

The association with you will be a great success.

Sincerely,

W. A. THOMAS.

* * * * *

August 23, 1922.

MR. JAMES S. MCGLENNON,
Rochester, N. Y.
Dear Sir:—

I wish to thank you for your very kind letter of the eighteenth, and beg to assure you that it would afford me great pleasure to attend and meet you and others who are doing constructive work in the cause of nut culture. Unfortunately it will not be possible for me to do so. I have been on the sick list for the past few weeks which with my eighty-five years has left me so weak that I could not endure the fatigue connected with such an undertaking.

I would much like to see the results of your work with filberts, as I believe that is one branch of nut growing that can be made a success. Some years ago I planted out some filberts and they grew very well and tried to bear nuts. But unfortunately they had been planted near some woods that contained some squirrels who invariably ate all the nuts before the time they were half grown, so I grubbed them out. Recently I planted some more farther removed from woods and hope to see them fruit soon.

Some years ago I caused some filberts to be planted in ground used by the State Horticultural Society for testing new fruits. These are still living and bearing good crops.

I feel sure you will have a good meeting and am very sorry I can not be with you. Give my best regards to my nut growing friends, to all of whom a cordial invitation is extended to visit me and see what I am doing here with chestnuts.

Truly

E. A. RIEHL.

* * * * *

NEW YORK AGRICULTURAL EXPERIMENT STATION GENEVA, N. Y.

June 24, 1922.

Dear Dr. Deming:

It is kind indeed of you to ask me to help you out in your coming convention. Were I to be in the country I should be very glad to do anything I could to help out. I am leaving in a few days, however, to spend the summer in Europe and shall not be home at the time of your meeting.

You may be interested in knowing that we are growing some almonds on the Station grounds and that we have been trying to cross them with peaches. We think we have a cross but just what it will amount to I do not know. At any rate, we are living in hopes that sometime we may breed an almond for this part of the world. We are doing something with other nuts but not as much as I should like. We are always hoping that opportunity may offer to do more and possibly we shall be able to within a year or two.

Very truly yours,

U. P. HEDRICK.

* * * * *

The Battle Creek Sanitarium
Battle Creek, Michigan
September 5, 1922.

MR. JAMES S. McGLENNON,
Rochester, New York

Dear Sir:—

Enclosed you will find my paper.

I am very sorry, indeed, that I could not be with you, but an unexpected amount of surgical work compelled me to remain at home. I hope you will have a most successful convention. The Nut Growers Association, in my opinion, may prove one of the most important factors in the world movement for race betterment.

Sincerely yours,

JOHN HARVEY KELLOGG.

ATTENDANCE

Dr. Robert T. Morris, N. Y. City, Mr. and Mrs. J. S. McGlennon, Miss Norma McGlennon, Mrs. W. D. Ellwanger, Rochester, N. Y., Mr. and Mrs. W. G. Bixby, Baldwin, N. Y., J. F. Jones, Lancaster, Pa., Mr. and Mrs. J. M. Patterson, Putney, Ga., S. W. Snyder, Center Point, Iowa, Harry R. Weber, Cincinnati, Ohio, John Rick, Reading, Pa., Jas. A. Neilson, Guelph, Canada, Joseph A. Smith, Providence, Utah, Harry D. Whitner, Reading, Pa., Henry D. Spencer, Decatur, Ills., Mr. and Mrs. Samuel L. Smedley, Newtown Square, Pa., Mr. and Mrs. Geo. H. Corsan, Brooklyn, N. Y. Jacob E. Brown, Elmer, N. J., W. R. Fickes, Wooster, Ohio, W. J. Strong, Vineland Station, Ontario, Canada, P. H. O'Connor, Bowie, Maryland, Adelbert Thomson, East Avon, N. Y., A. C. Pomeroy, Lockport, N. Y., F. A. Bartlett, Stamford, Ct., Mr. and Mrs. S. H. Graham, Ithaca, N. Y., E. L. Wyckoff, Aurora, N. Y., M. G. Kains, Suffern, N. Y., Mrs. J. B. Comstock, Hollywood, Cal., Joseph Baker Comstock, III, Hollywood, Cal., Mr. and Mrs. E. A. Hoopes, Pa. John Dunbar, Rochester, N. Y., R. E. Horsey, John P. Lauth, Mr. and Mrs. J. B. Rawnsley, Geo. B. Tucker, Mrs. C. R. Nolan, D. D. Culver, M. L. Culver, C. A. Vick, Mrs. K. Dugan, W. J. Nolan, Mr. and Mrs. Fred Garrison, Mr. and Mrs. Clifford Spurr, Miss K. M. Pirrung, Miss Ida Schlegel, Alois Piehler, Miss Robena

Murdoch, John Herringler, Mr. and Mrs. Conrad Vollertsen, Elwood D. Haws, Mr. and Mrs. Ralph T. Olcott, Rochester, N. Y.

* * * * *

List of Nuts exhibited before the Northern Nut Growers Association September 7-8-9, 1922 at Rochester, N. Y., by Park Department.

Black Walnut, Juglans nigra, United States.
English Walnut, Juglans Regia, Europe and China.
Western Walnut, Juglans major, Western States.
Hybrid Walnut from Washington, D. C, supposed hybrid between Juglans rupestris and Juglans nigra.
Butternut, Juglans cinerea, North America.
Siebold's Butternut, Juglans Sieboldiana, Japan.
Juglans cathayensis, China.
Juglans coarctata, Japan.
Winged Chinese Walnut, Pterocarya stenoptera, China.
Winged Caucasian Walnut, Pterocarya fraxinifolia, West Asia.
King-Nut, Carya laciniosa, United States.
Shagbark, Carya ovata, North America.
Carya ovata ellipsoidalis, United States.
Ash-leaved Hickory, Carya ovata fraxinifolia, United States.
False Shagbark, Carya ovalis, United States.
Small Fruited Hickory, Carya ovalis odorata, North America.
Carya ovalis obovalis, North America.
Carya ovalis obcordata, United States.
Pignut, Carya glabra, North America.
Large Pignut, Carya glabra megacarpa, United States.
Bitternut, Carya cordiformis, North America.
Hybrid Hickory, X Carya Laneyi, Carya cordiformis X Carya ovata.

Hybrid Hickory, X Carya Dunbarii, Carya laciniosa X Carya ovata.
Beaked Hazel, Corylus rostrata, North America.
American Hazel, Corylus americana, North America.
European Hazel, Corylus Avellana, Eastern Hemisphere.
Constantinople Hazel, Corylus Colurna, South Europe.
Manchurian Hazel, Corylus mandshurica, Manchuria.
Sweet Chestnut, Castanea dentata, United States.
European Chestnut, Castanea sativa, Europe to China.
Japanese Chestnut, Castanea crenata, Japan, China.
Chinquapin, Castanea pumila, United States.

By the McGlennon-Vollertsen Filbert Nursery, twenty or more plates, of about a quart each, of named varieties of the European filbert grown in these Rochester nurseries, a very striking exhibit in demonstration of the commercial possibilities of this nut. By E. L. Wyckoff, Aurora, N. Y., a cluster of Indiana pecans, grown on a grafted tree at Aurora, of good size, apparently, other qualities not determined. A cluster of two small pecans grown on the great pecan tree in Hartford, Ct. One of these nuts was matured and filled. Brought by W. C. Deming who showed also chinkapins grown in Hartford and Redding, Conn. two strains of the Van Fleet hybrid chinkapins, Chinese chestnuts, C. mollissima, Japanese chestnuts, clusters of Kirtland and Griffin shagbarks from grafted trees, Ridenhauer almonds and several varieties of European and American filberts, all grown in Redding, Ct. filberts from the large trees at Bethel, Ct. and the large Sayre English walnut from Danbury, Ct. Illinois wild almonds were exhibited by Henry D. Spencer of Decatur, Ills. These have a fleshy covering like a thin peach. Mr. P. H. O'Connor showed specimens of the O'Connor hybrid walnut, J. regia X. J. nigra, and the Indiana hazel. Mr. A. C. Pomeroy had an exhibit of the Pomeroy English walnut. There were a number of other exhibits which have escaped record.

*** END OF THE PROJECT GUTENBERG EBOOK NORTHERN NUT GROWERS ASSOCIATION REPORT OF THE PROCEEDINGS AT THE 13TH ANNUAL MEETING ***

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